Bachelor Thesis

Proactive AI moderation for goal-oriented remote meetings: Design and acceptance of a context-sensitive system to support convergent decision-making processes
Bachelor thesis in the department of Computer Science by Oliver Hönig
Date of submission: 22.10.2026

1. Review: Prof. Dr. Jan Gugenheimer
2. Review: Yanni Mei
Darmstadt



TECHNISCHE UNIVERSITÄT DARMSTADT

Computer Science Department

HCI

Human Computer Interaction

Note on the Use of Artificial Intelligence

AI applications were used to edit the language of this work. The content, argumentation, and academic evaluation were developed independently by the author.

AI applications were used to assist with research in the preparation of this work. The academic review, selection, and classification of the content were carried out by the author.

AI-based programming tools were used to assist in the implementation of the prototype. This was done in consultation with the supervising examiners.

Since English is not the author's native language, DeepL [1] was used in some places to translate sections of German text written by the author into English.

Zusammenfassung

Abstract

Contents

1 Introduction **1**
- 1.1 General Problem and Relevance . . . 1
- 1.2 Research Objectives and Questions . . . 2
  - 1.2.1 RQ1: Facilitating strategies and challenges in goal-oriented remote meetings . . . 2
  - 1.2.2 RQ2: Design requirements for an AI-based facilitation system . . . 3
  - 1.2.3 RQ3: Perception of the meeting with a proactive AI system . . . 3

2 Theoretical Foundations **4**
- 2.1 Remote Meetings (RQ1) . . . 4
  - 2.1.1 Remote Meetings in Modern Work: Trends and challenges . . . 4
- 2.2 Facilitation and Convergent Group Decision-Making (RQ1) . . . 6
  - 2.2.1 Group Support Systems and collaboration tools . . . 6
  - 2.2.2 The facilitator role . . . 8
  - 2.2.3 Divergent and Convergent Thinking in Meetings . . . 8
  - 2.2.4 Biases in information exchange (Hidden profiles) . . . 9
- 2.3 AI in Meeting Support Systems (RQ2) . . . 10
  - 2.3.1 Overview of existing Meeting AI Tools . . . 10
  - 2.3.2 Related Research Approaches to AI-Supported Meeting-Systems . . . 11
  - 2.3.3 Proactive AI-Interactions: From Turn-Taking to intrinsic Motivation with the ”Inner Thoughts” Framework . . . 12
- 2.4 LLM-based AI-Systems and Agents in General (RQ2) . . . 13
  - 2.4.1 Agentic Systems vs. Programmatic Automation / Workflows . . . 13
  - 2.4.2 Prompt Engineering and Context Engineering . . . 14
  - 2.4.3 Building blocks for LLM-based Workflows and Agents . . . 15
  - 2.4.4 Evaluating and Testing of AI-Systems . . . 18

3 Empirical Analysis of Facilitating Strategies (RQ1 & RQ2) **20**
- 3.1 Methodology: Expert Interviews . . . 20
  - 3.1.1 Participants: Selection of 3 facilitating experts . . . 20
  - 3.1.2 Data Collection: Conduction of semi-structured interviews (30 min) . . . 21
  - 3.1.3 Analysis: Thematic coding . . . 22
- 3.2 Results: Facilitating Strategies and Challenges . . . 23
  - 3.2.1 What has to be true around the decision . . . 23
  - 3.2.2 What puts the decision at risk while the group is together . . . 25
  - 3.2.3 What the facilitators do in the convergent phase . . . 28
  - 3.2.4 Where an AI-System would help, and where it would not . . . 29

4 System Design and Implementation of a Scoped Prototype (RQ2) 34
4.1 Discussion of research results and decision to narrow down . . . 34
4.1.1 Retaining Proactive Facilitation Despite Empirical Concerns . . . 34
4.1.2 Focusing the Design Scope on Hidden Profile Szenarios . . . 35
4.2 Requirements Analysis and Scoped Requirements Catalog for the Prototype (User Stories) . . . 35
4.3 Evaluation-Driven Development: Formative Prototype Evaluation during Development . . . 37
4.3.1 Implementation Procedure . . . 37
4.3.2 Limitierungen des Vorgehens . . . 39
4.4 System Architecture: Adapted version of the “Inner Thoughts” framework . . . 40
4.4.1 Overview . . . 40
4.4.2 Components: Transcription, Analysis, and Intervention Modules . . . 44
4.4.3 Components: Transcription, Analysis, and Intervention Modules . . . 44
4.4.4 Components: Transcription, Analysis, and Intervention Modules . . . 44
4.4.5 Technology Stack: Facilitation Agent Runtime and Meeting Client . . . 50
4.4.6 UX und UI of Meeting Client . . . 50
4.5 Limitations of the Prototype . . . 51

5 Empirical Prototype Validation (RQ2 & RQ3) 53
5.1 Methodology: Hidden-Profile Scenario and Questionnaire Instrument . . . 53
5.1.1 Study Design . . . 53
5.1.2 Sample . . . 54
5.1.3 System einstellungen während der user studie und die überlegungen dahinter . . . 54
5.2 Results . . . 55
5.2.1 Observed Intervention Dynamics . . . 55
5.2.2 Perceived Meeting Quality (Davison Instrument) . . . 55
5.2.3 Participants’ Perception of the Proactive AI Moderator’s Interventions . . . 58
5.3 Discussion . . . 59
5.3.1 Identified Key Challenges and Potentials . . . 60
5.3.2 Lessons Learned: Implications for similar System Designs . . . 62

6 Conclusion 64
6.1 Summary of Findings . . . 64
6.2 Limitations . . . 64
6.3 Future Work . . . 64

7 Appendix 70
7.1 Semi-Structured Interview Guide for Expert Interviews with Facilitators . . . 71
7.2 Codebook for the Expert Interviews . . . 75
7.3 Hidden Profile Exercise . . . 79
7.4 FIM Scoring Prompt for Thought Evaluation . . . 98
7.5 Articulation Prompt for Utterance Formation . . . 106
7.6 Meeting-Success Questionnaire (Davison Instrument) . . . 109
7.7 Codebook for the Open Questionnaire Items . . . 113

List of Figures

1 Scope of the thesis: Primary focus and secondary focus, broken down by meeting phases – *Source: Own illustration, created with draw.io.* . . . 2

2 Venn-Diagram: Prompt Engineering as a subset of Context Engineering – *Source: Own illustration, created with draw.io.* . . . 14

3 AI-Building-Block:Prompt chaining: A complex task is decomposed into sequential LLM calls – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.* . . . 16

4 AI-Building-Block:Routing: A complex task is decomposed into subtasks and routed to specialized systems – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.* . . . 16

5 AI-Building-Block: Parallelization: Multiple tasks are processed in parallel and the results are aggregated programatically – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.* . . . 17

6 AI-Building-Block: Orchestrator-workers: A central LLM decomposes a main task dynamically into subtasks and delegates them to a variable number of worker LLMs – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.* . . . 17

7 AI-Building-Block: Evaluator-optimizer: A LLM generates answers, while another evaluates and optimizes them through feedback – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.* . . . 18

8 Evaluation-driven development loop: From exploratory testing to system adaptation – *Source: Own illustration, created with draw.io.* . . . 38

9 Langfuse: Screenshot of the Traceviewer in Langfuse that shows the trace of the AI-Facilitator processing a new conversation event. – *Source: Own screenshot, created with Langfuse.* . . . 38

10 System Architecture: Adapted version of the "Inner Thoughts" framework – *Source: Own illustration based on "Inner Thoughts" by Liu et al.[43], created with draw.io.* . . . 40

11 Thought Pipeline Sequence: Sequence of the thought pipeline – *Source: Own illustration based on "Inner Thoughts" by Liu et al.[43], created with draw.io.* . . . 41

12 From User Story to System Decision: Graphical overview of what User Stories motivated a system design decision. – *Source: Own illustration, created with draw.io.* . . . 44

13 Linear aging of the FIM score at three hold-times. Each panel shows the same scores $s_0 \in \{4.5, 4.7, 5.0\}$; the vertical line is t_{hold}, where a score-5 thought crosses τ. Within a panel, weaker thoughts expire sooner. Left to right, a shorter t_{hold} steepens every line. Parameters: $\tau = 4.3$, $s_{\text{hold}} = 5$. – *Source: Own illustration, created with LaTeX and TikZ* . . . 48

14 Cooldown factor after a thought is spoken, for $t_{\text{cool}} \in \{30, 60, 90\}$ s. Each panel compares the Hermite smoothstep with a linear ramp; the vertical line is t_{cool}, where recovery completes. The smoothstep stays flat at both ends; the linear ramp starts immediately and kinks at t_{cool}. – *Source: Own illustration, created with LaTeX and TikZ* 49
15 Weighted FIM score of a thought with $s_0 = 5$, never spoken versus spoken at $t = 12$ s. Aging uses $t_{\text{hold}} = 45$ s; cooldown uses $t_{\text{cool}} = 60$ s. Recovery restores the aged remainder; whether that remainder still exceeds τ depends on how long the thought has already lived. Parameters: $\tau = 4.3$, $s_{\text{hold}} = 5$. – *Source: Own illustration, created with LaTeX and TikZ* 49

16 Perceived communication (Davison items C1–C4). Each bar shows the share of responses on the five-point scale. – *Source: Own illustration.* 56
17 Perceived discussion quality (Davison items D1–D4). Each bar shows the share of responses on the five-point semantic differential. The marker above the scale names the unfavorable and favorable ends. Counts are illustrative. – *Source: Own illustration.* 56
18 Perceived status effects (Davison items S1–S4). Each bar shows the share of responses on the five-point scale. – *Source: Own illustration.* 57
19 Perceived teamwork (Davison items T1–T3). Each bar shows the share of responses on the five-point scale. – *Source: Own illustration.* 57
20 Estimated share of meeting time spent in serious discussion (Davison item E4). Each dot is one participant. The line spans the lowest and highest estimate in that group. The vertical mark is the group median, and the diamond is the group mean. – *Source: Own illustration.* 58
21 Perceived attributes of the proactive AI agent (items A1, A3, A6, A7, and A8). Each bar shows the share of responses on the five-point scale. Counts are illustrative. – *Source: Own illustration.* 58
22 Perceived disruption and intervention frequency of the proactive AI agent (items A2, A4a, and A4b). Each bar shows the share of responses on the five-point scale. Counts are illustrative. – *Source: Own illustration.* 59

List of Tables

3.1 Facilitation moves in the convergent phase. 28
3.2 Functions the experts described as useful for an AI in or around a convergent meeting. 30
3.3 Limits the experts placed on an AI that intervenes in a meeting. 31

4.1 Derived user stories for an AI facilitator, with references to the source chapters 36
4.2 Architectural design decisions of the adapted Inner Thoughts prototype, with rationale and pointers to the motivating chapters of this thesis. 46

7.1 Codebook for the expert interviews. Anchors are shortened from the transcript and keep the original German wording. 76
7.2 Mapping from the original Intrinsic Motivation factors to the FIM heuristic, with the warrant used in this thesis. 104
7.3 Mapping from the original Inner Thoughts articulation prompt to the facilitation version, with the warrant used in this thesis. 108
7.4 Codebook for the free-text answers to questionnaire items F1 and F2. Anchor examples are quoted in the original German. 113

1 Introduction

1.1 General Problem and Relevance

As early as the 1980s, research on Group Support Systems (GSS) identified key challenges that are still evident today in unproductive meetings: an inappropriate pace, poor meeting design, a lack of focus, a failure to reach decisions, or inefficient processes [2, p. 152].

Knowledge work now runs, to a remarkable degree, through remote meetings. The medium that made distributed collaboration routine did not, however, resolve the process inside the meeting.

Despite the long-standing awareness of these problems and the recognized importance of facilitation approaches, these insights still do not appear to be sufficiently implemented in practice. A negative assessment can be drawn from a study by Atlassian, in which 72% of meetings were rated as ineffective. The study identifies the same typical problems such as off-topic discussions, uneven participant engagement, and a lack of structure. [3]

One possible explanation for why these well-documented problems have nonetheless persisted, despite long-standing awareness, is the combination of inadequate training, lack of experience, and a natural resistance to change [4, p. 153].

In addition, the responsibilities of a facilitator are extremely broad and complex (see 2.2.2 and 3), and it is likely that not everyone is suited for such a role.

AI-powered meeting tools have the potential to democratize this facilitating knowledge.

Current meeting tools primarily support the preparation and follow-up of meetings. Features used during the meeting, which could be categorized under “facilitation,” are mainly reactiv or interactive. (see 2.3.1)

One reason for this is that literature suggests that AI systems in this context should primarily be designed to be reactive, and that proactive approaches are significantly more difficult to implement (see **??** and 2.3.3).

At the same time, problems arise in meetings that participants are often unaware of, such as cognitive biases, drifting off topic, or communication difficulties (see **??**). In such cases participants cannot actively ask an AI system for assistance because they are not aware of the issue at that moment. Even if the system itself sends a notification, for example, via chat, it may be ignored or rationalized away. [5, ”4.4.3 Intervene”]

In the field of Human-Computer Interaction (HCI), there are some preliminary studies on AI systems in this context (see 2.3.2). However, the vast majority of these studies remain theoretical or follow an interactive or reactive interaction model. Practice-oriented systems that proactively intervene and moderate meetings via voice have barely been noted in the literature.

This work therefore focuses on convergent decision-making processes in remote meetings, that is, on group processes in which existing information and options are evaluated and synthesized into a shared outcome.

This narrow focus reflects the current state of meeting support: preparation and especially follow-up are already partially covered in commercial meeting tools and for divergent practices (such as brainstorming), there are relatively well-established AI approaches. (see 2.3.1) In the convergent phase, key GSS problems such as a lack of focus, failure to reach a conclusion, and distorted information exchange persist. Since participants are often unaware of these issues, a purely interactive design falls short here. Therefore, this work investigates how an AI system can proactively facilitate during this phase to support the participants.

1.2 Research Objectives and Questions

This thesis investigates support during a goal-oriented remote meeting, and it narrows that investigation to the convergent phase: when a group must evaluate existing options, remain aligned with a stated objective, and move toward a joint decision. Within that scope, the work asks how a proactive, context-sensitive AI System can intervene while the meeting is still running, especially where participants may not necessarily be aware that something is not going ideally or is even counterproductive.

The research problem gets addressed with a structured approach that integrates three complementary research objectives. This bridges the gap between theoretical facilitation techniques and a practically applicable AI-supported prototype. This prototype integrates a system proposal that builds upon existing HCI research and concretizes it through features derived from domain-specific knowledge in the field of facilitation.

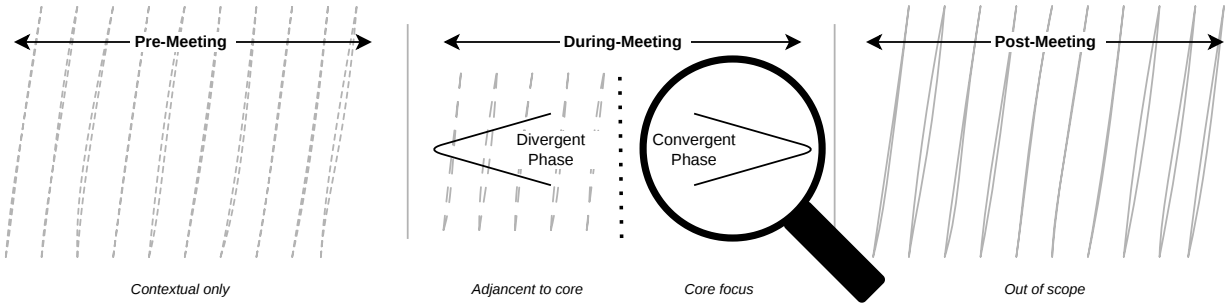


Figure 1: Scope of the thesis: Primary focus and secondary focus, broken down by meeting phases – *Source: Own illustration, created with draw.io.*

1.2.1 RQ1: Facilitating strategies and challenges in goal-oriented remote meetings

> *Which facilitating strategies are suitable for the goal-oriented management of remote meetings with convergent objectives (such as decision-making, prioritization, problem-solving)? What specific challenges typically arise in this context?*

The first research question aims to identify effective strategies for managing convergent remote meetings, such as decision-making, prioritization, and problem-solving, while also analyzing the specific challenges that arise. To achieve this the thesis conducts expert interviews. As preparation, a preliminary literature review was conducted on the topics of remote meetings and facilitation. Additionally, the literature review aims to identify existing findings and enrich them with insights from the expert interviews.

1.2.2 RQ2: Design requirements for an AI-based facilitation system

> *Which of the challenges identified in RQ1 can be addressed by an AI-based moderation system during a goal-oriented remote meeting with convergent objectives, and how should such a system be designed to intervene in a context-sensitive way?*

The second research question asks which of the challenges identified in RQ1 can be addressed by an AI-based moderation system during a goal-oriented remote meeting with convergent objectives, and how such a system should be designed. Rather than presupposing a single failure mode, the thesis first maps the problem space from the expert interviews and the literature, then assesses which challenges are both technically addressable and most promising for delivering added value.

1.2.3 RQ3: Perception of the meeting with a proactive AI system

> *How do knowledge workers perceive a goal-oriented remote meeting with convergent objectives in which a proactive AI system intervenes, with regard to communication, quality of discussion, status effects, teamwork, and efficiency, and how do they experience the AI system's contributions?*

The third research question investigates how knowledge workers experience a convergent remote meeting that is supported by the prototype. The evaluation uses a hidden-profile hiring simulation in which relevant information is distributed asymmetrically, so that a well-founded group decision depends on sharing unique information. Meeting outcomes are measured with an adapted version of the questionnaire instrument by Davison [6].

2 Theoretical Foundations

This chapter presents the theoretical foundations and the current state of research. These serve as the starting point for the analysis in the expert interviews (see Chapter 3) and provide indicators of what to focus on during the subsequent requirements gathering process (see Chapter 4.2).

2.1 Remote Meetings (RQ1)

Some studies define meetings as a goal- or outcome-oriented interaction between two or more people. [2, p. 148] One weakness of this definition is that some meetings would likely fail to meet the assumption of being goal-oriented.

Kayser [7], as cited in [8], offers a somewhat humorous definition that nevertheless captures the ineffectiveness of some meetings: ”A meeting is a gathering where people speak up, say nothing, and then all disagree.”

In their literature-based survey of meetings, Romano and Nunamaker derive the following summarizing definition. A meeting is ”a focused interaction of cognitive attention, planned or chance, where people agree to come together for a common purpose, whether at the same time and the same place, or at different times in different places.”[8, p.1]

This thesis uses the term “meetings” in this board sense of the summarizing definition.

In this thesis, “remote meetings” are defined as meetings that take place synchronously via an online video conferencing platform.

2.1.1 Remote Meetings in Modern Work: Trends and challenges

Although meetings are important for teams to achieve goals, solve problems, and make decisions, there are many signs that meetings are perceived as ineffective.

One of the most negative evaluations of meetings originates from a study by Atlassian, in which more than 72% of meetings were rated as ineffective by participants. The study identifies therefore problems such as off-topic discussions, uneven participant engagement, and a lack of structure. [3]

Earlier studies on managers’ self-assessments of meeting productivity ranged from 33% to 47%. [8, p. 1]

In a study by Geimer et al., of the more than 1,000 participants surveyed, fewer than half considered their typical meetings to be an effective use of time. [9, p. 2019] The most common negative statements were found in the area of “meeting organization.” [9, p. 2019]. Examples cited a lack of structure, the absence of an agenda, or an unclear objective. Generally poor preparation was also mentioned, as well as the fact that meeting participants had not been provided with all relevant information in advance. [9, p. 2019f]

So a clear issue with goal-oriented meetings is the absence or unclear communication of objectives.

An agenda was identified by many respondents (67%) as a key element for successful meetings in Doodle's State of Meetings Report 2019. Only the setting of clear meeting objectives (72%) was cited even more frequently. [10, p. 11]

While an agenda can serve as the foundation for such structure, it alone does not guarantee a successful meeting. Meeting goals are often dynamic and may need to be adjusted as the meeting progresses, due to the associated uncertainties [11, chap 4.1]. Although the agenda can outline a path to achieving the overall meeting goal [12, chap 2.1], without linked sub-goals, it guarantees neither the success of the meeting nor a timely conclusion. [11, chap 4.1] Discussions can nevertheless get out of hand if participants do no know what to priorities. [11, chap 4.1]

Furthermore, respondents to the study by Geimer et al. mentioned that meetings were often held with inappropriate participants. People were invited to meetings who had nothing to do with the projects or issues discussed during the meeting. [9, p. 2019f] Not having the right people in a meeting is one of the main reasons for unproductive meetings. [8, p.6]

Dysfunctional communication—such as complaining, criticizing, and digressing—reduces effectiveness. [13, p. 130]. It appears that dysfunctional communication has a stronger impact on the success of a meeting than functional communication. ("bad is stronger than good") [13, p. 148]

Chen et al. noted, as a secondary finding, that individuals in higher hierarchical positions tended to intervene more frequently and maintain focus on meeting objectives when a clear list of goals was provided. Conversely, those in lower hierarchical positions felt less obligated to intervene, even when they observed deviations from the main topic. [11, chap 4.3] Doodle's State of Meetings Report 2019 identified a correlation between salary and self-perceived participation in meetings. Respondents with higher salaries were more likely to describe themselves as active participants. [10, p. 6] This may stem from inexperience, hierarchical constraints, group dynamics, or a lack of error culture, where dissent risks loss of face. Low participation can narrow solution spaces, yield monotonous outcomes, and reduce meeting effectiveness.

Inclusion in meetings can increase satisfaction, performance, and creativity. [5, p.1]. In addition, this can counteract hidden profiles (see subsection 2.2.4).

Most of these insights apply to both remote and in-person meetings. Remote meetings, however, present additional challenges.

Thus, the main challenges of remote work are technology and communication issues. In addition, early-morning meetings and the use of video can be sources of stress for meeting participants. [**luebstorfDiggingZoomFatigue2023**] It was also observed that multitasking, that is, switching between different activities in a short period of time, is a critical stressor during video conferences. Multitasking occurred more frequently in larger meetings with cameras turned off. [**luebstorfDiggingZoomFatigue2023**]

Based on the expert interviews with facilitators conducted by Hughes and Roy, it can be concluded that in in-person meetings, facilitators find it easier to establish a shared "energetic presence," (including the perception of participants' nuances) which is more difficult in a remote setting. One facilitator described it as if he was almost blind online. [14, p.10]

Taken together, the studies trace the widespread perception of meetings as ineffective to the way the meeting is conducted: objectives stay unclear or shift without sub-goals that would tell participants what to prioritize, while unsuitable participants, dysfunctional communication, and hierarchically uneven participation restrict the knowledge that enters the discussion. For synchronous remote meetings, an agenda and a stated goal are

therefore necessary but not sufficient. Without a structure that can absorb changing objectives and inclusion that counters status differences, the meeting is unlikely to stay focused or to close on time.

2.2 Facilitation and Convergent Group Decision-Making (RQ1)

2.2.1 Group Support Systems and collaboration tools

Research on group support systems (GSS) began in the 1980s in response to the constant challenges posed by increasingly complex problems that often had to be solved in groups or teams within organizations. [15, p. 628]

According to Ackerman, the early days of this field of research were marked by significant fragmentation. Various research teams worked independently of one another and without coordination on overlapping topics. [15, p.628]

As a result, there was no standardized approach, and different terms for the same field of research became established, such as “Meeting Support Systems,” (MSS) “Computer-Supported Cooperative Work Systems,” (CSCW) “Group Support Systems,” (GSS) or “Group Decision Support Systems” (GDSS). [15, p.628f]

In some cases, the term “electronic meeting system” (EMS) is also used. This term was coined by Dennis et al. when they attempted to bring the two research areas, GDSS and CSCW under a single definition. [16, p. 592]

According to Dennis et al., GDSSs have focused more strongly on tasks such as decision-making, planning, and problem-solving, while CSCWs are geared more toward communication needs. However, the authors themselves note that there are only subtle differences between these areas and predict that the various system classes will eventually be merged into a single class . [16, p. 592]

The terms GSS, GDSS, and EMS can therefore now be considered synonymous. [17]

The overarching goal of the various lines of research is to collectively optimize meetings in terms of both the quality of their outcomes and the structure of their processes. [15, p.629]

Computer-aided tools are often used for this purpose. Huber defined it as follows [18, p. 195]:

> ”GDSS consists of a set of software, hardware, and language components and procedures that support a group of people engaged in a decision-related meeting.”

GSSs are thus a combination of methodological structures and processes that are documented in a way that makes them usable by groups, as well as technical tools designed to make the use of these methods practical during meetings.

Group Support Systems (GSS) can vary greatly in terms of their complexity and functionality

Some systems allow participants to enter text contributions simultaneously and anonymously in order to conduct electronic brainstorming, criteria analyses, or prioritization votes. One example of this is GroupSystems (later known as ThinkTank, presumably based on Plexsys) [15, p.634].

Plexsy is a framework presented by Dennis et al. that made various facilitation methods available for groups using supporting IT systems. It includes, for example, methods such as electronic brainstorming, voting procedures (prioritizing, Likert scales, multiple-choice questions, etc.), and several other approaches. [16,

p. 606f.] Depending on the phase of the meeting, the group can select an appropriate method and switch between them . [16, p. 607]

Group Explorer, which has evolved from its single-user predecessors COPE and Decision Explorer into today's web platform "Strategyfinder"[**InteractiveDecisionMakingPartner**], is a visual group support system that models complex group problems as interactive cause-and-effect networks (Cause Maps). In these maps, the causal relationships (in the sense of "A leads to B") expressed by the participants (in the sense of "A leads to B") are graphically linked together. The goal is to make complex relationships understandable and visual in order to facilitate joint decision-making. [15, p. 636f]

With the further development of GSS, the Thinklets Framework has become established as a way to encapsulate formalized facilitation techniques and proven procedures used by facilitators, in order to generate predictable and repeatable behavioral patterns in collaborative groups. They consist of small, standardized tools and scripts designed to enable groups to accomplish group tasks. [15, p. 635]

Modern tools such as Miro, Conceptboard, or Figma Jam, which can be classified as digital whiteboard solutions or CSCW, draw on some of these original methods (such as asynchronous information gathering, private information sharing, and voting (anonymous or non-anonymous)) and combine them with the capabilities offered by modern technologies that enable distributed collaboration.

Integrated AI-based features enhance these tools with smart functions such as the automatic grouping of ideas into topic clusters by AI or the enrichment of ideas through AI-generated suggestions (see subsection 2.3.1).

In addition, there are various modern collections of exercises, methods, and tools—including instructions for carrying them out (similar to the Thinklets concept), as well as guidance on what to look out for, which are suitable for various group situations and challenges in meetings, such as:

- Liberating Structures: A collection of methods, tools and templatesfor facilitation. [**LiberatingStructuresInnovati**
- 9 Spaces: A collection of methods, tools and templates for facilitation. [**9SpacesEntwickle**]
- SessionLab Library: SessionLab is a tool for creating workshop agendas. There is a community-maintained library where many methods are shared. [19]

In GSS research, there were approaches that assumed that GSS, due to the intuitly aimed interface design, could be operated independently by group members. However, as system and task complexity increased, it proved difficult in practice to simultaneously participate substantively in the discourse, operate the technical GSS software, and moderate the group process. [15, p. 631]

As a result, an additional, somewhat rigid role structure emerged at that time: In addition to the facilitator, who focused on content and social dynamics, a so-called "chauffeur", decribe by Whiteley and Garcia was as the "technical support person," responsible for the purely technical control of the GSS. [20, p. 31]

This historical necessity of providing two highly specialized roles—even though the role of the "Chofferus" has become obsolete today—underscores just how complex it is to conduct meetings effectively, even with support systems.

2.2.2 The facilitator role

Even in the early days of GSS (Group Support Systems) research, Beranek et al. pointed out that GSS alone likely cannot guarantee the effectiveness of a session. Rather, success depends largely on how these systems are used. [2, p. 147] In this context, the quality of the group session would depend largely on the facilitator. [2, p.146]

In the German-speaking context, the term “facilitator” is sometimes mistakenly translated as “moderator.” However, this equivalence is inadequate, as the roles have different focuses: While a moderator is primarily responsible for managing a meeting, a facilitator’s role is aimed at supporting the group in terms of both process and content. Although the role of the facilitator includes elements of moderation, it goes beyond that

According to Haney, researchers in the 1990s agreed that facilitation is a combination of “(1)managing relationships between people, tasks and technology, (2) structuring tasks, and (3) contributing to the effective accomplishment of the meeting outcomes.” [4, p. 73]

In his book ”The Skilled Facilitator”, regarded as a standard work in the field, organizational psychologist Roger Schwarz defines the term “Facilitator” as follows: ([21] as cited in [22] & [23, p.9]):

> ”Group facilitation is a process in which a person who is acceptable to all members of a group, substantively neutral, and has no decision making authority, intervenes to help a group improve the way it identifies and solves problems, and makes decisions, in order to increase the group’s effectiveness.”

Whiteley and Garcia attribute a number of key functions to a facilitator working in conjunction with a GSS. These include, process and agenda management, in which the facilitator guides the group through the problem-solving process and the agenda, but must adapt these flexibly and in real time when new problems arise. [20, p. 36] [20, p. 31] In addition, the facilitator is responsible for mood and relationship management (mood making) [20, p.33] , which includes assessing when the group’s mood needs to be lifted or lowered, and when to intervene, confront, calm, or reconcile. [20, p.31] Furthermore, the facilitator is responsible for creating a trusting work environment by setting guidelines and establishing a high-trust environment. [20, p.31] The facilitator interprets and perceives symbolic signals, uncovers myths and hidden interests, and refocuses the discussion on the actual core issues through framing. [20, p.32] In addition, the facilitator supports interest-based negotiations, consensus building, and the promotion of effective interpersonal behaviors to enable teamwork. [20, p.34]

In summary, according to Whiteley and Garcia, a facilitator needs skills in the following areas: problem-solving, decision-making, discrimination, reflection, imaging, intuiting, emotional, and being skills. These range from structured problem-solving and evaluation to causal, systemic, and lateral thinking, as well as visualization and pattern recognition. In addition, there are emotional skills such as humor and self-esteem, as well as “being” skills, which are characterized by resonance, balance, and autonomy within the group. [20, p.34]

2.2.3 Divergent and Convergent Thinking in Meetings

The concepts of divergent and convergent thinking can be traced back to J.P. Guilford. Divergent thinking describes the process of generating a large number of alternative, original, or unusual answers to an open-ended problem. Convergent thinking aims to find the single, conventional, and logically correct solution to a well-defined problem. This involves applying existing factual knowledge, logical-analytical rules, and predefined criteria. [**razumnikovaDivergentConvergentThinking2020**]

According to Anne Manning, generating new ideas and evaluating them are two fundamentally different processes. However, she says that in meetings, people often try to carry out both steps at the same time. [24, min 02:30] Immediately evaluating and weighing an idea hinders the creative process of generating ideas.

A predominance of divergent thinking without sufficient integration of convergent thought processes can lead to unreflective and excessive modifications. Corply further argues that both a lack and an excess of convergent or divergent thinking have adverse effects on creativity. [25]

In the context of facilitation, the terms ”Convergence Zone / Phase” and ”Divergence Zone / Phase” are widely recognized. A recurring framework in this domain is, developed by facilitator Sam Kaner inruduced in his book ”Facilitators Guide to Participatory Decision-Making”, which is commonly referred to as ”Diamond of Participatory Decision-Making” or ”Kaner’s Diamond”. [26]

Kaner’s Diamond model describes typical phases that groups go through in the decision-making process. The left side represents the Divergent Zone, where diverse perspectives are expressed and disagreements are explored. The right side embodies the Convergent Zone, where opinions align and the process moves toward closure. At the center of the Diamond lies the ”Groan Zone”, a transitional phase where the tension between divergence and convergence often manifests. [23, p. 32f.]

2.2.4 Biases in information exchange (Hidden profiles)

In meetings, participants typically have access to different types of information. Some of it is known to everyone (shared information), while other information is known only to certain individuals (unshared/private information). And since a foundation of information is an important factor in the decision-making process, this phenomenon is considered relevant within the scope of this thesis.

A hidden profile exists in a group when information about various decision alternatives is distributed among the individual members in such a way that an individual member, based on his or her partial knowledge, arrives at a different (suboptimal) preference than the group would if it were to combine all of its collectively available knowledge. [27, p. 257]

A common problem is that groups tend to be more likely to repeat the information they have been given than to discuss information they have not been given. [28, p. 619] (Also known as the “shared information bias.”)

Baker summarizes the factors identified in the literature regarding systematic bias in group decision-making processes involving shared information bias or hidden profiles as follows: Members often bring preconceived opinions, dismiss contradictory information, rate their own contributions higher than those of others, and consider information that can be validated by others (multiple members) to be more credible. Rapid consensus-building leads members to adopt these opinions rather than seek out relevant information. Majority opinions create social pressure on individual group members to align with the majority’s view. A majority decision disadvantages minorities who, although they possess important information, are unable to have it taken into account. In addition, shared information is mentioned more frequently simply for statistical reasons. (The more participants who have the information, the more likely it is to be mentioned or repeated.) [27, p. 261f.]

Confirmation bias and groupthink reinforce this blind spot because unique information disrupts the existing consensus [27, 29].

According to Verhein-Jarren et al., a prerequisite for effective communication is recognizing one’s own personality style, as well as understanding that there are different frames of reference within which the participants operate during the conversation. People tend to unconsciously interpret others according to their

own patterns. According to Verhein-Jarren et al., breaking through this automatic mode of perception is necessary to create a shared frame of reference. [30, p.9]

Within the same frame of reference, however, differences of opinion may still arise, which is not necessarily a bad thing. Schulz-Hardt et al. demonstrate empirically that dissent significantly increases the intensity of discussion and drastically improves the solution rate of the Hidden Profiles. [29, p. 1080]

Baker proposes the following solution strategies, summarized on the basis of the literature. Groups should structure their decision-making strategy in advance and maintain an open-minded, critical attitude that prioritizes a thorough evaluation of information over hasty consensus-building. In addition, the exchange of unbiased information is promoted by specifically encouraging divergent perspectives, visually documenting contributions for everyone to see, and employing facilitators to ensure the equal involvement of all participants. [27, p. 262f.]

2.3 AI in Meeting Support Systems (RQ2)

In addition to the fundamentals of Group Support Systems (GSS) outlined in subsection 2.2.1, modern approaches specifically designed for meeting support systems can be identified, some of which are supplemented by functions that are more or less supported by AI. This chapter analyzes these systems to identify potential gaps and to serve as a source of inspiration for the development of the prototype.

2.3.1 Overview of existing Meeting AI Tools

There is a wide variety of AI-based tools that provide support for meetings. Most of these tools focus on transcribing meeting content. Well-known examples include fireflies.ai [31], Otter.ai [32], and Jamie [33]. AI transcription features are also integrated into established meeting platforms such as Google Meet [34], Zoom [35], and Microsoft Teams. [36] More advanced features include the automatic extraction of action items. Depending on the tool, these features are available in real time or only after the meeting has ended. Microsoft Teams specifically offers a feature called “Facilitator,” [37] which provides additional capabilities beyond the features already mentioned (transcription, identification of action items): For example, it can edit the agenda if instructed to do so by a user with the appropriate permissions. It also displays a timeline to track topics and the flow of the meeting, and sends reminders to manage time. (Once halfway through the meeting and again shortly before it ends.) These interactions take place via the integrated chat. Google Meet, for example, also offers a live speech translation feature. [38]

Nevertheless, contemporary meeting platforms remain primarily technical infrastructures for conducting meetings. Their design does not systematically account for the professional knowledge, needs, and methodological repertoire of facilitators [14], nor are they conceived to support the range of tools, methods, practices, and techniques on which facilitation typically relies.

There are also platforms specialized for facilitation needs, such as Stormz [39], offer AI support across different phases of a meeting, including brainstorming. [40]

Platforms such as SessionLabs curate methods from the facilitation community and, with the help of an AI assistant, make it possible to find suitable methods and generate an agenda tailored to the meeting’s objectives.[19]

2.3.2 Related Research Approaches to AI-Supported Meeting-Systems

Several works in the literature pursue a comparable approach or related research questions . Among them, two papers proved particularly relevant to this thesis. Their key findings are summarized below to clarify how they inform the subsequent work.

Are We On Track?

In an empirical study involving 15 knowledge workers, Chen et al. examined how AI can support goal orientation and intentionality in meetings through passive visual cues and active, inquiry-based reflection prompts. [11, p. 1] To increase ecological validity, the authors used past meeting recordings provided by the study participants and applied a video-stimulated method in which the participants rewatched the same meeting situation, with the addition that it now included AI interventions. They evaluated counter-scenarios with simulated AI interventions. [11, p. 4] Technically, the prototypes developed were based primarily on GPT-4, which processed transcripts in simulated real time to passively visualize topic progression using D3.js or actively display context-related reflection questions. [11, p. 6] The results show that passive visualizations maintain the flow of conversation and promote individual understanding, whereas active AI questions, while triggering immediate adjustments, carry the risk of disruptions to the discourse. [11, p. 2] However, due to the passive designs, particularly in the visualization system,it was found that participants overlooked or ignored them. [11, p. 2] Based on the participants' responses, the authors conclude that it should be possible to dynamically adjust the intensity of the intervention to the subjective need for support in order to ensure a balance between democratic participation, efficiency, and user control. [11, p. 13]

One limitation of this study is that the system was not tested in the context of a live meeting .

Observe, Ask Intervene: designing AI Agents for More Inclusive Meetings

In the study by Houtti et al., the authors examined the design of an AI agent as a so-called "virtual co-host" to specifically promote inclusion and equal participation in video conferences.[5] First, ideation sessions were conducted in which participants were asked to design a virtual co-host that could intervene to balance out inequalities in participation. [5, p. 6]

This concept was then evaluated in an exploratory laboratory study involving 68 participants in 12 groups, using a structured decision-making task ("Lost at Sea") on Discord, followed by questionnaires on meeting quality and 18 qualitative interviews. [5, p. 14]

Technically, the prototype was implemented as a Discord bot in Python ('Discord.py'), which used Selenium and custom JavaScript to track the cumulative speaking times of all participants in the browser interface in real time. The bot communicated exclusively via private direct messages, generated individual bar charts showing speaking time, and always waited for a speaking pause of at least five seconds before sending messages to avoid interrupting the conversation. [5, p.6]

A tendency was observed to design the AI facilitator in a rather reactive manner. The participants wanted to retain a certain degree of control over the co-host. [5, p.6] Nevertheless, the authors concluded that this is not always feasible, because they suspected that a system that must always be triggered might remain unused. Based on this, the authors derive a framework they call "Observe, Ask, Intervene." The AI agent continuously observes the meeting but asks a human participant for permission before intervening. Nearly all interviewees stated that they would prefer to be asked before an intervention affecting them takes place. [5, p.15]

Specifically, this worked as follows: After certain thresholds for distributed speaking time were reached, the bot sent a private direct message with questions to people who had a low level of participation. Only after the person responded accordingly did the agent send the speaking time charts.

Although private feedback was perceived as socially acceptable[5, p.17], the bot did not exert sufficient social or normative pressure, which is why facilitators or dominant speakers largely ignored or rationalized the AI's critical feedback. [5, p.18]

It should also be noted that the system is a rule-based prototype that does not utilize a generative LLM pipeline.

2.3.3 Proactive AI-Interactions: From Turn-Taking to intrinsic Motivation with the "Inner Thoughts" Framework

So an AI-System can proactivly interact and therefor reduese the need for explicit prompts by the meeting participants it should be able to proactively recognize when it is its turn in an interaction sequence.

In dyadic human-AIinteractions, pauses or so-called "stop words" are often used for this purpose to identify a change in speaker. In multi-party scenarios (e.g., group conversations), however, this becomes significantly more complex and leads to the "multi-party turn-taking problem". [41]

Some approaches model the problem as a machine learning task by attempting to predict the next speaker based on previous speaker changes. [42] Bayer et al. employ traditional algorithms such as Maximum Likelihood Estimation (MLE) and Support Vector Machines (SVMs), as well as deep learning architectures including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks.

With these approaches, however, it remains unclear why an agent should interact at a specific point in time. Due to the design of the ML algorithms, the timing of the interaction is decoupled from its cause or is only implicitly included in the model, but cannot be explicitly explained.

A naive approach to turn-taking prediction would be to prompt an LLM to ask who is next.

Liu et al. investigated various language models (including GPT-3.5, GPT-4o, and fine-tuned variants) with regard to their ability to predict the next speaker (next-speaker prediction). They showed that these approaches work well when speaker identification (turn allocation) is explicit. [43, p.4]

This makes sense, since the current speaker explicitly assigns the next turn to a specific person, and the conversational context thus contains clear semantic cues, such as direct mentions of names—that an LLM should be able to process well.

However, in self-selection scenarios, where a participant chose themselves as the next speaker without being asked to do so, all models performed no better than chance. [43, p.4]

The authors conclude from this that the conversational context alone is insufficient for self-selection* scenarios, and that hidden, internal factors (such as interest, relevance, or motivation to engage) play a decisive role. [43, p.4] Based on a thinking-aloud study with 24 participants, they concluded that people are driven by intrinsic motivation to voice their thoughts. [43, p.2]

Building on this, they modeled a framework for chatbots that they call "Inner Thoughts." An LLM-based agent continuously generates internal thoughts in parallel with the conversation and evaluates them based on a defined heuristic (a fixed prompt in the LLM call). If the intrinsic motivation (i.e., the evaluation of a thought)

exceeds a configurable threshold, the agent decides whether to participate, taking into account the turn-taking type. [43]

In chat evaluations, this system achieved significantly higher performance in self-selection scenarios than conventional baselines, and human evaluators perceived the conversations as significantly more natural, intelligent, and coherent. [43, p.2]

The framework by Liu et al. was renamed “Thoughtful Agents” on GitHub.[1] In this paper, the original name “Inner Thoughts” is used, but it also explicitly refers to the renamed version.

2.4 LLM-based AI-Systems and Agents in General (RQ2)

The previous chapters examined remote meetings, facilitation, and existing GSS primarily from an application perspective. In the case of the last, technical approaches were also described. To address RQ2, typical patterns to llm based AI systems are also relevant, as well as the possibility of testing these assumptions. However, this chapter does not claim to be exhaustive but serves primarily to clarify what the author has focused on.

2.4.1 Agentic Systems vs. Programmatic Automation / Workflows

Huang explains that it is difficult to provide a correct and complete definition of the term “AI agent” because our understanding of it and the technologies behind it are changing rapidly. [44, p. 2]

In addition, there is a subjective impression that the term “AI agent” is used excessively and inconsistently in the literature, which leads to a diffuse understanding of the concept. To narrow down the definition of the term, this thesis attempts to approach the concept by distinguishing it from programmatic automation methods.

In this thesis, “programmatic process automation” refers to the rule-based automation of workflows, in which defined conditions and structured decision-making logic are used to execute processes efficiently, reliably, and reproducibly.

While traditional software executes predefined instructions, “AI agents possess the capacity to initiate actions, make decisions, and pursue goals without constant human intervention” [44, p. 2]

Thus, AI agents have a higher degree of autonomy in this context.

Anthropic uses the following definition for LLM-based workflows and agents:

> ”Workflows are systems where LLMs and tools are orchestrated through predefined code paths. Agents, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.” [45]

Existing papers, such as the study by Houtti et al. or by Liu et al. [43], also refer to an AI agent, for example, although it is questionable whether these can be classified as AI agents according to Antroplic’s definition or whether they are rather LLM-based workflows. The system developed by Houtti et al. does not use any LLMs. The authors themselves state that they used a rule-based prototype system to perform better testing. [5] Nevertheless, they refer to it as an AI agent.

[1] `https://github.com/xybruceliu/thoughtful-agents`

For the purposes of this thesis, it should therefore suffice for now to provide a rough classification of whether the developed system is more agent-based or workflow-based. The author therefore mostly uses the general term “AI system” for certain conclusions or system descriptions. When other systems are cited and the authors use the term “AI agent,” this term has been adopted.

A description of the actual system can be found in the section on system architecture.

2.4.2 Prompt Engineering and Context Engineering

In general, context engineering refers to the design, organization, and management of context-sensitive information, so that machines act in accordance with human intent. [46, p. 3]

Although context engineering already played a role in early human-computer interaction frameworks [46, p. 1], research has recently placed greater emphasis on this field due to the growing challenges in human-agent interaction (HAI).

Mei et al. (2025) haben in ihrer Meta-Studi [47] über 1.400 Forschungsarbeiten zum Thema Context Engineering für LLMs analysiert. Laut den Autoren haben sich LLMs von einfachen Systemen, die lediglich Anweisungen befolgen (sog. Prompt Engineering), zu zentralen Schlussfolgerungsmodulen entwickelt. Dabei habe sich die Gestaltung und Verwaltung von Informationsinhalten zu einer eigenständigen Disziplin weiterentwickelt (Context Engineering) [47]

For LLM-based systems, context engineering can be defined as follows::

> ”Context engineering is the practice of deliberately structuring and optimizing the information passed to an LLM or Agent to produce more accurate and relevant outputs.” [48, 00:34]

Context engineering thus goes beyond traditional prompt engineering and encompasses advanced methods such as retrieval-augmented generation (RAG), memory systems, tool-integrated reasoning, and multi-agent systems (see Mei et al., 2025). [47, p. 12-26]

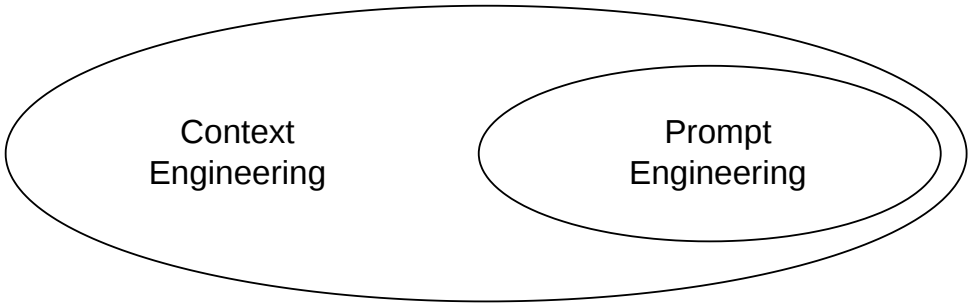


Figure 2: Venn-Diagram: Prompt Engineering as a subset of Context Engineering – *Source: Own illustration, created with draw.io.*

Although various methods of context engineering are discussed in the literature, Mei et al. critically point out that current practice operates without a unified theoretical foundation. Such a foundation would be necessary to link the various techniques together and derive principle-based design guidelines. This research gap is further exacerbated by the lack of mathematical frameworks that would make it possible to characterize

optimal design principles for context engineering in complex systems. This is related to the lack of methods for determining the optimal context composition. [47, p. 51]

2.4.3 Building blocks for LLM-based Workflows and Agents

In order to develop the AI agent described later, it was first necessary to examine the fundamental building blocks of AI agents. Anthropic has published a blog post on this topic, [45], which summarizes these fundamental building blocks from their perspective and serves as the primary basis for this chapter. We first examined all currently common patterns, even though not all of them were ultimately used in the actual system.

Prompt chaining

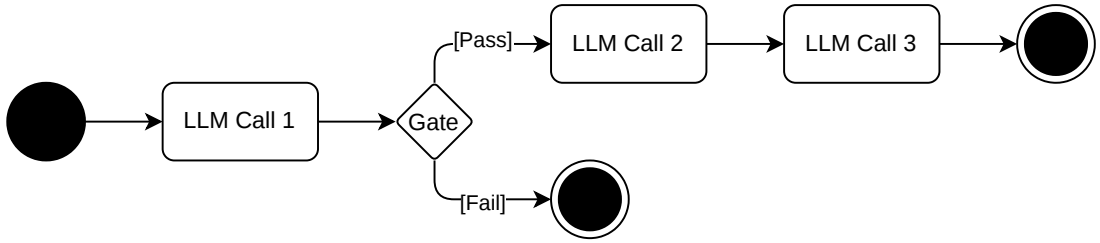


Figure 3: AI-Building-Block:Prompt chaining: A complex task is decomposed into sequential LLM calls – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.*

In prompt chaining, a complex task is broken down into a sequential series of substeps. Each step is described by a prompt and an LLM call, which uses the result of the previous step as context for the next step. Programmatic checks can be inserted in between. [45]

Routing

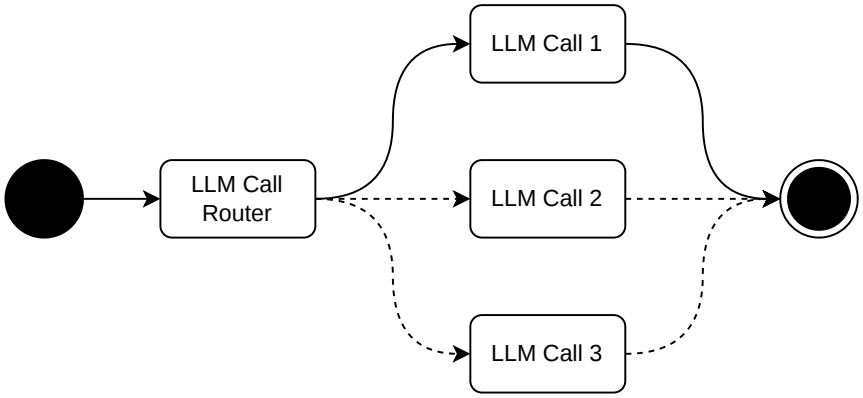


Figure 4: AI-Building-Block:Routing: A complex task is decomposed into subtasks and routed to specialized systems – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.*

During routing, tasks are classified and forwarded to subsystems that specialize in solving that category of tasks. This results in a clear separation of responsibilities, which is particularly well-suited when complex tasks can be divided into clearly definable categories and processed by specialized LLMs, other AI models, or algorithms. [45]

Parallelization

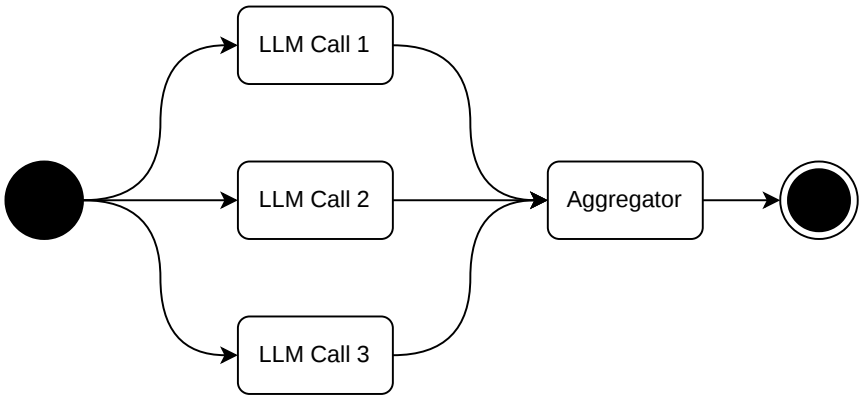


Figure 5: AI-Building-Block: Parallelization: Multiple tasks are processed in parallel and the results are aggregated programatically – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.*

In the parallelization workflow, multiple tasks are processed simultaneously, and the results are aggregated programmatically. Typical use cases arise when tasks can be broken down into independent subtasks or when multiple perspectives are required to achieve a more reliable result (e.g. voting). For complex tasks with multiple aspects, LLMs achieve better results when each aspect is addressed separately using a specialized prompt. [45]

Orchestrator-workers

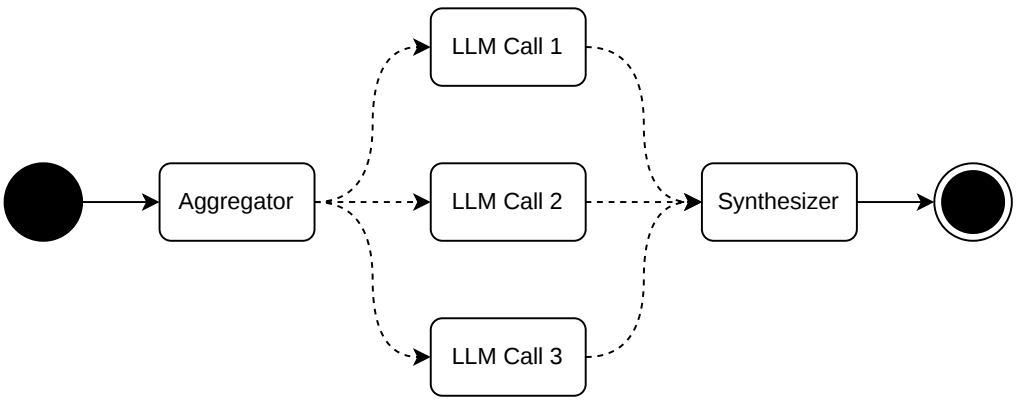


Figure 6: AI-Building-Block: Orchestrator-workers: A central LLM decomposes a main task dynamically into subtasks and delegates them to a variable number of worker LLMs – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.*

The Orchestrator-Worker workflow is similar to the parallelization workflow, but differs in that it is used when the required subtasks are not known in advance. A central LLM dynamically breaks down the main task into

subtasks and delegates them to any number of worker LLMs, whose results are then aggregated. [45]

Evaluator-optimizer

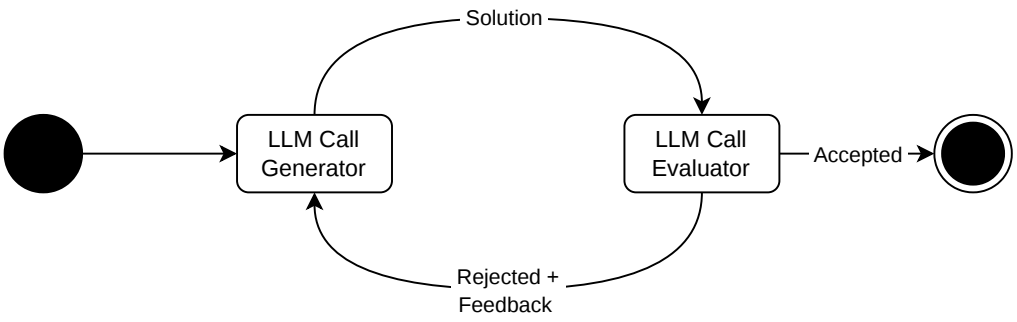


Figure 7: AI-Building-Block: Evaluator-optimizer: A LLM generates answers, while another evaluates and optimizes them through feedback – *Source: Own illustration based on "Building Effective AI Agents by Anthropic"[45], created with draw.io.*

The evaluator-optimizer workflow uses a feedback loop in which one LLM generates responses while another evaluates them and attempts to optimize them through feedback. This approach is particularly effective when clear evaluation criteria are in place and iterative improvements are measurable. [45]

2.4.4 Evaluating and Testing of AI-Systems

Due to the probabilistic nature of AI-based systems, testing AI-based systems differs fundamentally from conventional testing methods.

The International Software Testing Qualifications Board (ISTQB®) therefore suggests that this probabilistic nature would require statistical testing methods. [49, p. 39]

An additional challenge is that some AI systems are not only static but also adaptive, and can therefore continuously adjust themselves [49, p. 39].

These adjustments can occur on at least two levels::

- **Context-Based Adaptation:** Within the application, the AI system can dynamically adapt to changing conditions or the diverse possibilities offered by the context available at that moment.
- **Model-Based Adaptation:** Regardless of the specific context of the use case, the underlying AI models may change on their own, for example through the use of self-learning algorithms. When integrating external large language models (LLMs), the provider may make modifications to the model that are not immediately apparent to the user.

Defining a test oracle that is, specifying the expected outputs for specific input values for the system under test is therefore extremely difficult or even impossible in complex AI systems [49, p. 39].

In a live meeting scenario, context-based adaptation is particularly relevant. In any given meeting, there may be a variety of participants working on different topics, which makes the context in which the AI system operates highly variable.

Mohammadi et al. [50, p. 1] also point out that evaluating and testing LLM-based agents is a complex and still under-researched field, particularly because agents often operate in a more dynamic and interactive context. They therefore describe the testing of LLM agents as an interplay of the various perspectives of natural language processing (NLP), human-computer interaction (HCI), and software engineering.

Code-based methods aim to evaluate the performance of AI systems using explicit program code, defined rules, and test cases with assertions. This involves verifying whether the actions performed exactly meet specified criteria. [50, p. 7]

For this purpose, for example, possible states that the system goes through can be modeled and analyzed using trajectory comparisons. However, existing benchmarking methods such as AgentBoard [51] are often general in nature and only partially suitable for domain-specific use cases.

Code-based methods reach their limits, particularly with open-ended or qualitative responses, as well as with creative problems where correctness can be evaluated subjectively [50, p. 7].

The "LLM-as-a-Judge" approach leverages the reasoning capabilities of LLMs to evaluate agents' responses based on qualitative criteria. In this process, the responses are analyzed based on the evaluation metrics specified in the instructions. [50, p. 7].

The authors Zheng et al. [52, p. 4] propose three variants of the LLM-as-a-Judge approach, which can be applied individually or in combination:

- Pairwise Comparison: An LLM compares two answers to a question and decides which is better or whether there is a tie.
- Single Answer Grading: An LLM evaluates a single answer directly using a scale or a set of criteria.
- Reference-Guided Grading: An LLM evaluates an answer by comparing it to a given reference solution, e.g., for math problems.

The appendix to the paper "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena" contains the corresponding prompts that can be used for the various variants of the LLM-as-a-Judge approach [52].

According to Mohammadi et al., however, human-in-the-loop evaluation is still considered the gold standard, in which people (such as in user studies, expert audits, or through crowdworkers) annotate or evaluate the outputs [50, p. 7]. However, this approach is often difficult to scale and involves significant effort and costs.

A widely used approach is Evaluation-Driven Development (EDD), in which AI-based systems and agents are continuously evaluated during the development phase to detect deviations and identify new use cases [50, p. 6].

3 Empirical Analysis of Facilitating Strategies (RQ1 & RQ2)

3.1 Methodology: Expert Interviews

As described in the theoretical foundations, interaction in meetings encompasses several levels at once: psychological effects, personality differences, and communication among the participants.

It remains to be seen how this will be handled in practice, which factors will be the focus, how facilitators will conduct themselves in meetings involving convergent decision-making, and which practices can be translated into the actual behavior of an AI-powered meeting system.

The interviews were intended to facilitate precisely this transition. Their purpose was, on the one hand, to move from the general descriptions of meetings and facilitation in the literature to the concrete experiences of practicing facilitators, in order to identify a smaller set of characteristics relevant to an AI agent, as well as to uncover additional insights, which may not yet have been captured by the literature review, regarding what an AI agent should pay attention to and where its limitations might be.

Semi-structured expert interviews were chosen as the method. This type of interview allows for in-depth insights into the respondents' perspectives through open-ended questions and flexible follow-ups, while at the same time ensuring a thematic focus through a guide.

Expert interviews are particularly suitable because experts, through their specialized knowledge and practical experience, provide efficient access to specific areas of knowledge [53, p. 877]. The definition of an expert is context-dependent and must be determined on a case-by-case basis depending on the research topic [53, p. 877]. In this study, only individuals with verifiable professional experience in the field of facilitation and who are professionally active in this context were interviewed. Therefore, the interviews conducted can be classified as semi-structured expert interviews.

3.1.1 Participants: Selection of 3 facilitating experts

Three experts were interviewed for this study. Their recruitment involved a longer process that relied on personal outreach. At the beginning of the study, local events and online meetups in the facilitation industry were attended. This made it possible, on the one hand, to establish direct contacts with practicing facilitators and, on the other hand, to gain in-depth insight into the facilitation industry as well as into consulting services in the context of organizational development.

Key events included a two-day conference on cooperation and New Work in Darmstadt (June), attended by facilitators and organizational designers from all over Germany, as well as regular participation in a

monthly, two-hour online meetup for facilitators (since April), where the future prospects of facilitation and, in particular, the potential of AI in the field of facilitation are discussed.

The deliberate establishment of personal contacts served two purposes: First, it allowed for an initial assessment of which experts could provide relevant input on the research question. Second, the pre-existing personal acquaintance helped foster an open and trusting atmosphere for the discussions. No details about this study were shared with the selected experts prior to the interviews in order to avoid biasing them.

The following criteria were considered when selecting the experts:

- Active professional practice in the field of facilitation with at least five years of professional experience
- Regular work in remote-based formats
- Regular facilitation of groups in decision-making meetings
- Occasional use of AI to support their work as facilitators

Beyond these common criteria, the three experts were specifically selected so that their areas of focus differ slightly and each covers an aspect relevant to the research question.

Expert 1 is an external facilitator and the managing director of a consulting firm. Her focus is on facilitation and AI-supported facilitation, as well as the future possibilities that arise from them. She has already conducted several experiments with larger groups in which voice agents were used as a supportive element.

Expert 2 works in organizational development. He facilitates internal meetings within his own organization as well as client workshops in which groups reach decisions.

Expert 3 serves as a facilitator for large-scale projects and executive teams, both online and offline, often in change initiatives, such as those involving communication, stakeholder engagement, and collaboration within project teams. Additionally, one of his areas of focus is mediation, that is conflict resolution and consensus-building within groups. For decision-making processes, how and when to intervene in the group is particularly relevant.

3.1.2 Data Collection: Conduction of semi-structured interviews (30 min)

Both the order of the various sections and the questions within the sections follow a structure that progresses from the general to the specific. The questions are open-ended in order to avoid dictating too much or merely receiving confirmation of what is already known from the literature review. According to Helfferich in Baur et al., providing too much guidance is one of the biggest mistakes in conducting qualitative interviews. [53, p. 878]

On the other hand, Helfferich also notes that it is in the interest of research to limit openness. [53, p. 878f.] For this reason, the scope of the study (see Figure 1) was explained to the interviewees both at the time of the initial request and prior to the interviews.

The interview guide (see section 7.1) is divided into five sections. In the first section, the interviewees were asked to describe their professional context. This served to further validate their expertise and also to provide a simple way to begin the conversation. This was followed by a section focusing on challenges in targeted remote meetings involving decision-making, which led into a section where participants were asked about specific facilitation strategies. The final section focused on requirements for an AI-supported meeting facilitation system. A conscious decision was also made to include questions about how facilitators would envision such

an AI-supported system. One could argue that facilitators are not experts in this area and that this could lead to speculative, uninformed answers, since they are not aware of the technical capabilities and limitations of AI-supported systems. However, the questions are phrased in such a way that facilitators can describe the behaviors and characteristics of this system, which is within their area of expertise, without needing detailed knowledge of the technical implementation. Additionally, in Houtti et al., a co-host design was developed with participants in a meeting to determine how meeting participants would like such a system to function. Here, the facilitator's perspective is included.

Before the actual interview, a practice interview was conducted with another student to assess the general clarity of the questions and the approximate time frame. However, since the subject matter was specialized, another run-through was simulated using ChatGPT in voice mode as a practice partner.

The actual interviews were conducted via Zoom.

Es wurde immer direkte eine stunde mit den Interviewn vereinbart. Der interview leitfaden war initial für 40min konzipert, um noch puffer zu haben.

The interviews were scheduled to last approximately 40–50 minutes in order to allow for a time buffer in case of unexpectedly detailed responses. The very first interview confirmed the need for this buffer, as some of the participants' responses were longer than expected. The first interview significantly exceeded the planned duration. The actual interview durations ranged from 50 minutes (Interview 3) to 75 minutes (Interview 1), with Interview 2 lasting 65 minutes.

No more than one interview was conducted per day. After an interview was completed, notes were taken immediately to capture initial thoughts while the impressions from the interview were still fresh.

3.1.3 Analysis: Thematic coding

The interviews were analyzed by inductive thematic coding. The codebook was created after the interviews had been transcribed. The three transcripts were read and codes were formulated from those claims. While the transcripts were worked through, the codebook was extended when a passage required a new distinction, and definitions were tightened where neighboring codes overlapped. Passages already coded were then checked against the revised scheme.

The final Codebook with Rules, definitions, and shortened anchor quotations in the original German wording are documented in section 7.2. A passage can get coded with multiple codes.

Fifteen codes are grouped into four categories. Decision conditions (D) describe what has to be true around a decision, including work before and after the meeting: decision scope, process and rights (D1), preparation and follow-up (D2), and structural limits the meeting itself cannot remove (D3). Meeting dynamics (M) describe what puts goal orientation or decision quality at risk during the session: focus, time, and a decision that is not yet ripe (M1), voices, hierarchy, and dissent (M2), and remote conditions, including ambiguous signals (M3). A claim can carry both a D code and an M code: D is what the meeting depends on or cannot remove, and M is what that does while people are together, so the clock is not the cut. Facilitation practice (F) describes what the experts do in response and is the source of the requirements: process structure (F1), a visible state of the discussion (F2), bringing in a missing voice or fact (F3), the judgment of when and whom to address (F4), and neutrality, so that the decision stays with the group or a named owner (F5). AI co-facilitator (A) records the system: useful functions (A1), role, trigger, and acceptance (A2), risks and non-goals (A3), and context that lies outside the transcript (A4).

The unit of coding is one coherent claim. It may run across consecutive turns of the same speaker and may receive more than one code, for instance when it states both a problem and a response. Interviewer turns, consent, scheduling, and biographical self-description were left uncoded. One coder carried out the coding.

3.2 Results: Facilitating Strategies and Challenges

This section reports the thematic analysis of the three transcribed expert interviews. E1 (interview 1) works in organizational development, strategy, and multi-stakeholder processes, and combines facilitation with consulting *[e1#15, e1#16, e1#17]*. E2 (interview 2) maintains internal decision routines and facilitates commissioned workshops *[e2#6, e2#8]*. E3 (interview 3) facilitates large change projects and leadership teams, online and in person *[e3#8, e3#10]*. Names of persons and organizations that appear in the transcripts are not repeated here. The interviews were coded with the codebook in section 7.2, developed inductively from the transcripts as described in subsection 3.1.3. Codes were applied to claims by the interviewee. One claim could receive several codes. Interviewer turns, consent, and biographical self-description were left uncoded. One coder did the coding. No second coder was used, and nothing in this section is offered as a frequency beyond these three accounts. Consecutive rows often continue a single thought. Those rows are read as one line of argument. A marker such as *[e2#4]* locates row 4 in the structured transcript of interview 2. Where a conclusion rests on more than one passage, the marker lists those rows. The markers are locators in the interview material, not literature citations.

The interviews were asked to serve two purposes. They describe how a remote group stays with a joint decision, and they are a source of requirements for an AI moderator. Both purposes are reported here. Findings that later fall outside the prototype are included, because they bound what an in-meeting system can be asked to do. Design consequences are not settled in this section.

3.2.1 What has to be true around the decision

This subsection reports conditions the live discussion depends on. Each condition is stated once. Where the same condition disturbs the hour, subsection 3.2.2 reports only that disturbance.

The first condition is a shared information base, and a goal that has been said (D2). E1 calls lack of preparation the largest failure of remote meetings: people enter without having read the document, the group has no basis for a decision, and the meeting is postponed *[e1#24, e1#26, e1#27]*.

> "That happened to me again just this week. We waited four weeks for a meeting, […] so that five people could coordinate, and then three of them walk in and say, 'Yeah, oh, I'm afraid I didn't read the document.'" *[e1#25]*

She treats the meeting as an investment of organizational time, and she treats unprepared attendance as a waste of that investment *[e1#26, e1#28]*. For her, what happens before the meeting is the central part of the work, and she does it with the people who carry responsibility for the meeting *[e1#122]*. E2 makes the same observation. People will not enter a workshop unprepared, and they regularly enter a meeting unprepared *[e2#13]*.

> "[…] people usually go into a meeting unprepared." *[e2#13]*

The invitation often contains no information, so participants cannot tell what is to be decided, what their role is, or why they were invited *[e2#28, e2#30]*. E2 estimates that at least half of the work is preparation: the concrete problem, who must be present, whether the decision owner is in the room, and which decision procedure will be used *[e2#90]*. E3 puts the same weight on the briefing with the decision-maker *[e3#47, e3#84]*.

> "The most important thing isn't what you do during the meeting itself. The most important thing is what you clarify with your manager beforehand. What is the real goal?" *[e3#84]*

Goals and non-goals have to be stated in that preparation *[e3#47, e3#86, e3#88]*. Leaders, in E3's account, often have the goal in mind and still fail to say it *[e3#86, e3#88]*. For a short slot with senior decision makers, E3 prepares a decision memorandum: the dilemma, options including the option of doing nothing, and the costs and benefits of each *[e3#47, e3#48, e3#49]*.

The second condition is that the decision is specified before the discussion starts (D1). The specification has three parts. The group has to know what is being decided, and of what kind. It has to know who has the right to decide, including who decides if the group does not agree, and that person has to be in the meeting. It has to know the procedure, and the scope that procedure actually leaves to the people in the room.

(1) What is being decided takes more than one form. E1 distinguishes documented formal decisions from informal agreements that rest on trust, and she says that many of the decisions in her workshops are decisions about next steps rather than a final choice *[e1#20, e1#21, e1#73, e1#86]*. The point to be decided, and the level at which it sits, has to be named in advance. A choice that looks technical still carries choices about processes, roles, and tasks *[e1#128, e1#129]*. E2 separates reversible decisions, which can be tried under a "safe enough to try" rule, from irreversible ones, which justify a longer and more careful process *[e2#40, e2#80, e2#94]*. Not deciding is, for E2, itself a decision, because delay has a cost *[e2#80, e2#168]*.

(2) Who may decide is stated most directly by E3. E3 asks who has the authority, including budget authority, and who is responsible if the meeting ends without a decision *[e3#51, e3#53]*. If the team does not agree, E3 puts that responsibility on the leader, who then has to decide alone *[e3#53]*. For a team that is meant to decide together, the same briefing asks how far the team is actually allowed to go *[e3#49, e3#50]*. E2 makes presence part of the same condition. The first question for a decision meeting is whether the person who can decide is in the room. If not, the group needs an explicit way to handle that absence, or it will leave without a decision *[e2#75, e2#90]*. People who will be affected, and who are not in the room, are a further gap. E2's example is a shift decided by the people in the meeting and handed to the person who has to work it.

> "They are not sitting in the meeting at all. […] I come back from the meeting, and, Oliver, your new shift is 1 p.m. to 5 p.m. You now work from 1 p.m. to 9 p.m. You were not heard. […] A week later your resignation is on the table." *[e2#122]*

Auf der anderen seite können auch zu viele irellevante Perosnen eingeladen sein.

When the exchange does not happen, the people who only listen will notice it.

> "One person speaks and the others listen, and they ask themselves, well, I have not even said anything. Why was I actually invited?" *[e2#103]*

(3) The procedure is the point E2 develops furthest. In E2's account, organizations usually know two procedures, a decision by authority and a decision by majority *[e2#52, e2#53]*. Majority is described as a poor fit for most organizational choices, because it produces winners and losers and can exclude a minority whose cooperation the result needs *[e2#54, e2#119]*. E2 prefers consensus for fundamental questions, systemic consensus when several options are on the table, and a consultative individual decision when one or two people hold the expertise: they must hear the others, and they then decide and carry the responsibility *[e2#119, e2#120, e2#121, e2#148]*. "Disagree and commit" is named as a practice that keeps dissent from turning into a

refusal to implement *[e2#40, e2#41, e2#117]*. The procedure is empty when the scope behind it is only apparent. E2's example is a meeting that formalizes a decision already taken *[e2#26, e2#28]*. E3 describes leaders who tell a team that the decision is theirs when it has already been made, and who want the team only to nod it through *[e3#50]*. E3's advice is to say so in advance *[e3#50]*. E1 describes hierarchies that have decided and do not want to hear where the decision does not fit *[e1#45, e1#46]*. In all three accounts, a meeting that asks for commitment without a real scope of decision damages the result *[e1#45, e2#26, e3#50]*.

The third condition is a limit the meeting cannot repair (D3). Organizational structure shapes the hour from outside it. A structural problem can disturb the decision, and it can be the reason a meeting exists. Such a structure is hard to remove inside the meeting. E2 states it most sharply. Many meetings exist because decision rights are missing: small choices wait for the next regular meeting, people who could decide are not allowed to, and the meeting becomes the symptom of that structure *[e2#9, e2#10, e2#65, e2#73]*. Better moderation, on this account, can support a meeting and still leave the structural problem in place *[e2#74]*. The absence of the person who can decide, already stated as part of who may decide, is for E2 also the sign of that structure *[e2#75]*.

Manchmal ist das Meeting ansich bereits nicht das korrekte foramt. E2 also treats meetings that only broadcast information as a misuse of synchronous time *[e2#102, e2#103]*.

> "Far too many meetings are misused simply to pass on information." *[e2#103]*

E1's version of the same limit is the guest list. Standard meetings contain people who cannot contribute and who cannot decide *[e1#29, e1#30, e1#32]*. Hearing something is not, for her, a reason to attend, because a short protocol can do that afterwards *[e1#31]*. She also says that an external facilitator can invite dissent for the duration of a workshop but cannot change a fear-based culture *[e1#56, e1#57]*. These limits are felt while the group is together. The speaking pattern they produce is reported in subsection 3.2.2.

Laut E2 ist der synchrnoe austausch der Hauptgrnd für meetings. das wird aber nciht benötigt um infach nur informationen weiter zugeben.

The same outside work continues after the meeting closes (D2). A consequential decision has a before and an after: options have to be clarified, and someone has to implement the result by a stated time *[e3#26]*. The decision is written into the system the organization already uses, so that people who were not in the room can act on it *[e3#27, e3#28]*. Ownership belongs with that handoff. A result that the leadership team does not take as its own is, in E1's words, waste *[e1#137]*. Ownership comes from having worked on the result and from hearing one's own voice in it. It does not come from receiving a finished proposal from above, or from leaving the content with a consultant *[e1#135, e1#136, e1#139]*. E2 states the operational form: who does what by when, and where the group checks that it was done *[e2#133, e2#134, e2#135]*. A decision that nobody follows is, for E2, the worst outcome *[e2#133]*. E3 separates that ownership from the facilitator's role. That boundary is taken up in subsection 3.2.3 *[e3#89, e3#90]*.

3.2.2 What puts the decision at risk while the group is together

Given the conditions in subsection 3.2.1, three things still happen while the group is together: loss of the question, suppression of relevant dissent, and signals that are hard to read at a distance, somethimes caues by the remote setting *[e1#37, e2#114, e3#30]*. Some are the in-room face of a condition already stated. This subsection reports that disturbance. The condition it comes from stays in the preceding subsection.

Loss of the question, and the time in which the group can stay with it (M1), takes several forms. A side track because the decision is not yet ripe is the in-room face of a missing question or of missing preparation. The claim here is the drift, and the close that comes too early. E3 names the side track directly.

> "What can of course happen is that side tracks keep getting opened. Perhaps because the decision simply is not ripe yet, even if it was prepared." *[e3#30]*

In a group of ten or twelve with no structure, the same loss shows up as attention. Half of them start writing emails or answering messages and are not listening at all. E3 calls that a main problem, and calls managing the participants' attention essential *[e3#31]*. E2 locates the same loss in the day around the meeting.

> "The meeting is embedded in twenty other meetings in the day, so there is no pause from one topic to the next. My head is not free." *[e2#31]*

The previous meeting overruns, and the switch from one context to the next cannot be processed *[e2#32]*. E1 describes the same side-work, writing emails while half listening, as the absence of a contribution *[e1#41]*. For her the loss matters because attention is what carries the process. When people are only functionally present, nothing happens *[e1#95, e1#98]*. E2 describes the drift in the same terms: the group is no longer talking about the actual problem and the decision. In the course of that divergence the group can notice that it has been working on the wrong question, and that it no longer makes sense to hold the original goal *[e2#131, e2#132]*. What follows is a judgment: go back to the start, or, for the moment, work with the question the group has now arrived at *[e2#132]*. The interviews asked about the convergent phase. The experts keep tying that phase to the open phase and to other meeting types. M1 is therefore a passage between phases, as well as a loss of the decision itself. E1 describes the moment when new information has just come back from small groups. That return is, for her, usually the most critical point, because the group is again in the open phase of Kaner's diamond and has to find its way toward a decision *[e1#72, e1#73]*. E2 names what sits between opening and closing.

> "In between is the groan zone, this groan zone that is so painful. […] Now we argue, we discuss, and so on, and as a facilitator I have to endure that. I must not close it or force it. And of course there is time pressure. […] If I only have 45 minutes, but a complex topic, I spend too little time in that zone, and yet we now have to come to a decision quickly." *[e2#113, e2#114]*

So it is important to find the right balance to get every body heard in the given time frame. E1 makes the same observation: once the content starts, the hour is already gone, and the alternative is fewer people and more time *[e1#105, e1#106]*. E3 adds a timing rule at the other end. Reopening a large controversy in the last third of a workshop is dangerous. The move then is to stop discussing the substance and to discuss only how a decision will be reached *[e3#45]*.

Dissent and hierarchy (M2) are described as conditions of decision quality, not as a problem of politeness. The fear culture that one session cannot change, and the hierarchy that has already decided, were stated in subsection 3.2.1. Both act inside the hour, on who speaks and on what follows when a needed voice does not. E1 says that few organizations treat contradiction as useful, because a contradiction shows where something does not fit *[e1#44]*. An objection spoken in the room is, for her, what the meeting is there to hear, and she names invited criticism as a success factor of a good meeting *[e1#43, e1#156]*. What is not invited does not disappear.

> "Then it stays at the bottom, it bubbles away down there, and at some point it splashes up and turns into a conflict very quickly, which is unnecessary." *[e1#46]*

The same failure has a form she treats as a mark of quality. If the meeting simply starts, and only one or two people speak, the voices that are always heard take the floor *[e1#68]*.

> "Then the ones speak up who are extroverted anyway, and for whom it comes easily. Those are the voices one always hears. And the others stay nicely silent in the background. You know that this is often exactly how it goes." *[e1#69]*

Trust is, for her, the condition of being able to say that something is wrong, and it is harder when the people have never met, as in a remote setting *[e1#43]*. She also describes the dissent that does not help: chronic objection and the habit of taking a substantive criticism as a personal attack *[e1#51, e1#53]*. For E2's critical moments are participation failures.

> "People do not take part. They have no way to take part: they do not get to speak, or the method does not bring them in. Or they do not want to take part. So participation is a problem." *[e2#51]*

He ties that to the reason for being in the room together. The point of meeting at the same time is the exchange, and hearing everyone. A message can carry information. It cannot replace that exchange *[e2#102]*.

The procedure can produce the same failure. A majority, already described as a procedure that produces losers, leaves people who then have to live with a decision they lost *[e2#54]*. Where a dissent is not carried, E2 describes the meeting as a fight that makes winners and losers.

> "Otherwise decisions become winner-and-loser fights, culturally, and then you say in your team: yes, I saw it differently in the meeting. That was not my opinion, but the others outvoted me." *[e2#41]*

E3 defines the relevant voices that should get heard by what would happen afterwards. Where people notice, during the meeting, that they are only there to nod a decision through, he says the team registers it quickly and the result is discontent *[e3#50]*.

> "The people who should get to speak are those who, if they have a criticism of the decision, could credibly cast doubt on it. […] If a decision is taken that does not suit them, they would afterwards say, I'll do it differently anyway, even if they do not do so openly." *[e3#68]*

The hierarchy shapes the order in which agreement appears. In some settings a lower rank will not agree until the senior person has agreed, while in others the senior person stays noncommittal until everyone else has spoken *[e3#77, e3#79]*.

The remote setting changes which of these signs can be seen (M3).

E1 prefers to do complex content in person *[e1#23]*. What she loses online is the reading she has in the room.

> "Especially when I am sitting there with 16 tiles, or more, and facilitating them, a certain feel for this group and for the people is lost to me as a facilitator. […] I can perceive mood, resistance, agreement, and so on much better when I am in a room with them." *[e1#37]*

Camera-off participation and parallel work are, for her, forms of absence *[e1#40, e1#41, e1#99]*. A black tile is something she does not accept, because the group then cannot tell whether a person is present *[e1#40, e1#41]*. Language adds a further gap. A meeting in a shared foreign language, with more than one cultural habit of what may be said, makes a common picture harder to reach *[e1#34, e1#35, e1#36]*. E3 describes a repair that a face-to-face break still allows and an online break drops.

> "If I take a break online, people are disconnected at first, and then the smaller points may not get clarified. If people disagree, or, 'hey, what did you mean earlier?' Those things do not come up." *[e3#19]*

The agreement itself becomes harder to read. Especially online, E3 finds it very difficult to judge whether people are really in favor *[e3#69]*. The sign is a thinner "yes": people grow quieter, or less enthusiastic, as the decision approaches, and a "yes" that already names the next action is a different commitment from a flat one *[e3#70, e3#72, e3#76]*.

> "A 'yeah' and a 'yes' are two very different things." *[e3#74]*

E3 refuses a single reading of that sign. The same word means different things for different people *[e3#76, e3#79]*. Shared visual tools are, for E3, a condition of a complex decision online, and so is the time required for people who do not yet know the tool *[e3#22, e3#24, e3#25]*. Technical failure across organizations, including the ordinary failure of a microphone, is named as a further remote obstacle *[e3#20]*. E2 is more sparing on the remote medium. The point that does appear is parallel work while half listening, which a workshop makes harder than a meeting *[e2#21]*. Some methods, in the same account, also depend on a shared board. Without one, E2 cannot run them *[e2#36]*. The remote-specific material is therefore strongest in E1 and E3. The material is a list of cues and repairs that become unreliable when the group is not in one room *[e1#37, e3#19, e3#74]*.

3.2.3 What the facilitators do in the convergent phase

The practices are reported here as practices. They are the material from which requirements can be taken. They are not yet requirements.

Although the focus was on the convergent phase of the meeting, the experts repeatedly drew connections to other phases of the meeting. This suggests that these phases are more closely interlinked than previously assumed. When connections were made to areas that did not strictly pertain exclusively to the convergent phase, these were nevertheless mentioned here to illustrate the interplay.

Table 3.1: Facilitation moves in the convergent phase.

Facilitation action	What the facilitator attends to
Open before closing (F1)	Whether the point is understood before evaluation
Make the state visible (F2)	Whether everyone can see where the group stands
Bring in a voice or a fact (F3)	Which missing voice or fact would change the result
Judge when to move (F4)	Whether this moment can carry a decision
Leave the decision with the group (F5)	Whether the result becomes the group's own

The shared shape of decision meeting (F1) is to open before closing, and to keep clarification separate from debate *[e1#72, e2#33, e3#45]*. E2 states the sequence as information, then a check that people mean the same thing, then perspectives, and only then evaluation *[e2#33, e2#44, e2#46]*. Jumping from a document into debate is named as a source of decisions that people would have made differently if they had understood the point *[e2#33, e2#34]*. E1 uses a check-in so that every person has spoken once before the extroverted voices take the floor, and small groups so that people can think by speaking *[e1#65, e1#66, e1#69, e1#70]*. She describes the overall shape as a diamond: build the complexity, then bring it back to a decision or to next steps *[e1#72, e1#73]*. E3 follows a prepared plan and leaves it when the discussion becomes controversial in a way the plan did not expect *[e3#61]*. Early or mid-way, the two sides may go into separate groups to sharpen what they do not understand in the other position *[e3#44]*. Near the end, the structure changes level. The group stops reworking the controversy and agrees the path to a decision: which queries, who fetches which information by when, and when they meet again *[e3#45]*.

Making the state visible (F2) is the convergent move all three describe in concrete terms. E1 clusters results on a shared board and then summarizes: given what has been heard, can the group agree these next steps *[e1#82, e1#85]*. The summary is offered as a reading of the group's material *[e1#84, e1#85]*. E2 has people write pros and cons, or the consequences of a candidate decision, so that the option space is outside any one person's memory *[e2#36, e2#38]*. With more than a few people, writing or a small group is also a way to keep the contributions in view *[e2#86, e2#88]*. E3's criterion is that everyone should see, at one glance, where everyone stands *[e3#56, e3#57]*. A serial round in which each person says an opinion is, in a larger group, a practice E3

rejects, because people stop listening *[e3#56, e3#57]*. Scales, a fist-to-five that includes an explicit veto, a matrix of effort against value, and a poll are ways to put that picture in front of the group *[e3#40, e3#42, e3#55, e3#57]*. E3 also paraphrases near the close, sometimes slightly past what was said, in order to test whether something is still unsaid *[e3#81]*. The summary and the display are checks. They are not the decision *[e1#85, e3#42, e3#90]*.

Bringing a voice or a fact in (F3) is selective. E1 invites people who have not yet spoken, and she prepares questions that are meant to produce a new angle on the topic *[e1#71, e1#88]*. She also asks leadership, or a product owner, to say what they will do with what they have heard *[e1#76, e1#83]*. The invitation of a quiet person is, for her, easier in the room than online, where it can feel intrusive, and it can stigmatize, especially across culture and age *[e1#88, e1#89, e1#90]*. Naming introversion as the reason is something she would do cautiously *[e1#89, e1#90]*. E2's corresponding rules are to hear lower rank before senior rank, so that the senior view does not set the room, and to let everyone speak before anyone evaluates *[e2#46, e2#111, e2#112]*. A role, such as a skeptic's, is a way to make a critical view sayable *[e2#45]*. If a needed role is absent, E2 will ask the group what that role would be likely to say, and will mark the answer as a hypothesis *[e2#77, e2#78]*. E3 will not invite every silent person. An intern who says nothing is left. A co-founder who has been silent on a topic that matters to him is asked, by name, after a short restatement of what has been said *[e3#35, e3#36]*. The people who are asked are those whose later resistance would matter *[e3#68]*. Alongside speech, E1 and E2 both use writing and chat, because some people think before they can speak and others think by speaking *[e1#70, e1#102, e2#86, e2#87]*. E3's minimum, when a speech is not needed, is still a visible degree of confidence *[e3#69]*.

The judgment of when to move (F4) is described as restraint as much as intervention. E2 has to endure the groan zone and not force a close *[e2#114, e2#115]*. E3 waits, counts about seven seconds, and only then asks in another way or repeats the paraphrase *[e3#81, e3#82]*. A non-answer is sometimes the information. E3 stops pushing when further questioning would be delicate *[e3#119]*. E1 watches whether the group is only functionally present or also engaged, and she treats absence of energy as a reason the process will stall *[e1#94, e1#95, e1#97]*. She also weighs chronic objection rather than inviting it without limit *[e1#51]*. E2 asks, before an irreversible decision, whether the moment is even a fit one. The example is late on a Friday, with no decision capacity left *[e2#93, e2#94]*. The same expert uses the state of the people, including hunger, as a reason to distrust the moment *[e2#98]*. None of the three describes a cue that would justify the same intervention every time *[e1#90, e2#129, e3#76, e3#79]*.

The boundary of the role (F5) is drawn most sharply by E3. Leaders sometimes want the facilitator to produce the decision and to make people accept it. E3 refuses that delegation *[e3#89]*. The facilitator's work is to structure the conversation so that there are real options and so that people can stand behind a result. Saying yes or no remains the leader's work, because the facilitator is not in that role *[e3#90]*. E2 draws the boundary at the other end of the system: the decision should not be taken out of human hands, particularly where the solution is not yet known *[e2#166]*. E1 requires the group to take ownership, and she also describes a practice that crosses the line E3 draws. Because her facilitation sits inside consulting, she prepares substantive proposals for how to continue, and she describes the art of the work as helping to make the decision as well as advising on it *[e1#107, e1#130, e1#131, e1#135]*. The three accounts agree that the result has to become the group's *[e1#137, e2#166, e3#90]*. They differ on how far the facilitator may put content into that result. E3 forbids it *[e3#90]*. E1 treats a consulting proposal she has prepared as material on which the group then decides *[e1#131, e1#135]*.

3.2.4 Where an AI-System would help, and where it would not

The experts were not asked to design a technical architecture. They were asked which behavior would be useful, how a system should enter the meeting, and what it should not do. Table 3.2 and Table 3.3 record

which of them stated which point. An empty cell means that this expert did not state it. It does not mean that they rejected it.

Table 3.2: Functions the experts described as useful for an AI in or around a convergent meeting.

ID	Function	E1	E2	E3
F1	Fast summary, clustering, or a visual restatement of options	yes		yes
F2	Timekeeping, agenda position, and a prompt to come to the point	yes		
F3	Walking the group through a decision: question, options, doing nothing, pros and cons, or a named procedure		yes	yes
F4	Pointing at a decision the group has not noticed, or at agreement that is thinner than it appears			yes
F5	A cue that a named bias may be in play		yes	
F6	A persona that speaks for an absent stakeholder or a missing expertise	yes	yes	
F7	Documentation into a knowledge base, a task system, or an approved summary	yes		yes
F8	Help with preparation, or with stating the goal at the start	yes	yes	yes

A fast summary, clustering, or a visual restatement of options (F1) is the use for which E1 considers an AI well suited. Transcription and summary are already available, including the summary a meeting platform already offers. What she would add is faster sensemaking of complex material *[e1#151, e1#157]*.

> "AI can come in in a really exciting way if it manages to make these summaries faster. It helps to make sense of complex data more quickly. It can do that perfectly. And that is what it is highly suited to." *[e1#157]*

The example is a board that would otherwise remain a jumble of notes. Clustering it by hand is what used to be required. The board's AI now returns a summary, including across groups, within a minute *[e1#158, e1#159]*. At the end she would also have the system restate which decisions it has understood, and ask whether it has understood them correctly. She calls that a summary function, and she would have the facilitator call it *[e1#175]*. E3's restatement is visual. Once an option has been taken, the system summarizes it and shows the advantages and disadvantages *[e3#99, e3#101]*.

Timekeeping, agenda position, and a prompt to come to the point (F2) is the other function E1 singles out. Interrupting a person who talks too long is a task human moderators dislike, and she would have the system take it on, as when someone has been speaking for three minutes *[e1#172, e1#173]*.

> "Or it says, hey, I've listened to you, now come to the point. What is this actually about for you?" *[e1#173]*

An agenda signal belongs with it. If the system knew the agenda, it could show whether the group is still in time or already out of it, and it could say afterwards that the agenda was badly planned *[e1#174]*.

Walking the group through a decision (F3) is, for E2, a named procedure once several options are visible. Systemic consensus, a resistance score from 0 to 10 on each option, is the procedure he names *[e2#147, e2#148]*.

> "I notice there are several options now. I propose we do systemic consensus." *[e2#147]*

Enriching the data behind the option space, clarification, and a prompt to "disagree and commit" or to "safe enough to try" are further impulses *[e2#164]*. So is the cost of delay: not deciding is itself a decision, and he would have the system say what the delay costs *[e2#168]*. E3 can imagine a decision mode that the group switches on. First the decision question, including what happens if nothing is done, then one option at a time, summarized and shown, including advantages and disadvantages *[e3#99, e3#100, e3#101]*. She finds it hard to imagine putting the value of a skilled facilitator into such a system *[e3#97]*. The guided decision is a part she can imagine *[e3#100]*.

Pointing at a decision the group has not noticed, or at agreement that is thinner than it appears (F4), is the function E3 calls directing attention. The system reads the transcript and points at a place where a decision seems to be required, or where the group is less agreed than it thinks *[e3#101, e3#102]*.

> "You may need a decision here, because you are not as agreed as you think. Talk about that again." *[e3#102]*

A cue that a named bias may be in play (F5) is a warning E2 would have the system give while it listens.

> "Look, careful, careful, halo effect, or this or that bias. That would be extremely helpful." *[e2#147]*

A persona that speaks for an absent stakeholder or a missing expertise (F6) is an invitation E1 would make. She would invite an AI persona as a customer into a regular meeting, where that customer would not be invited, or as a missing expert, such as sales *[e1#164, e1#165, e1#166]*.

> "Come in with your voice and set out what you have just heard, and what moves you." *[e1#166]*

She treats the move as hybrid work, in which agents hold roles, and as a different layer from human facilitation *[e1#167, e1#168, e1#169]*. E2 names the same device for a role that is absent. An AI persona can play it so that the space is opened again. His example is a works council that is not in the room, and what it would be likely to think, anticipated from earlier meetings *[e2#78]*.

Documentation into a knowledge base, a task system, or an approved summary (F7) is, for E1, what should happen to the transcript afterwards. She would build the transcripts into a knowledge base, through a workflow that files them, so that the record does not disappear into a single email *[e1#188, e1#190]*. The form she prefers for that filing is a summary a moderator has approved.

> "[…] only protocols that a moderator has approved are then filed, not as a transcript but as summaries, and anonymized." *[e1#205]*

Faster protocols, and building onward from them, are part of the same advantage *[e1#163]*. E3 considers a result protocol certainly useful, and she would integrate it with the task system the team already uses, such as a kanban board *[e3#97, e3#98]*.

Help with preparation, or with stating the goal at the start (F8), is where E1 locates a main advantage, alongside those protocols *[e1#163]*. E2 says the questions that belong there are the ones people apparently cannot ask themselves: whether the decision owner is present, and what is needed so that people arrive prepared. Those questions are easier to clarify than in a real-time conversation. A system that listens to the live meeting and then reacts to its complexity he marks as only a thought *[e2#154, e2#158]*.

> "Is the decision maker in the room? What do we need so that people come into the meeting well prepared?" *[e2#154]*

E3 can imagine the first ten minutes being used for the system to help establish what the goal of today's decision is *[e3#117]*.

Table 3.3: Limits the experts placed on an AI that intervenes in a meeting.

ID	Limit	E1	E2	E3
L1	Sentiment or attendance monitoring becomes surveillance	yes		
L2	A record chills dissent, or leaves the dissenter marked	yes	yes	
L3	Calling on a quiet person can shame, or can misread the silence	yes		yes
L4	People will not yield the floor to a machine, especially a senior person	yes	yes	yes
L5	The system must not decide, and must not impose a structure nobody asked for		yes	yes
L6	A transcript is not enough context for a sound intervention	yes	yes	yes
L7	The group has to agree beforehand that interruption is allowed	yes		

The limits constrain the functions.

Sentiment or attendance monitoring becomes surveillance (L1) for E1. She would be very cautious about it. A report that a person has not spoken in the last ten meetings, or a reading of mood and engagement from the voice, is for her a shift from contact to control, and it raises data protection *[e1#152, e1#153, e1#155, e1#160]*.

> "That can very quickly turn into control. […] the AI noticed that in the last ten meetings you did not speak up once." *[e1#152]*

A record chills dissent, or leaves the dissenter marked (L2). E2 says people become more careful in the phase where reservations are most useful, because the meeting may be recorded and kept, and the reservation then stays attached to the person who raised it *[e2#161, e2#162]*.

> "Jacob had raised reservations at one point, and somewhere it sticks: Jacob has raised reservations again." *[e2#161]*

E1's formulation is the later use. Critical comments, which she treats as a condition of a good meeting, can be turned against the speaker, and that threatens the safe space *[e1#156]*.

Calling on a quiet person can shame, or can misread the silence (L3). E3 says silence is often not a trait of the person. Naming the one who has not spoken can shame them for not having found a way in. An intervention can also arrive while they are about to be persuaded, and then leave them in the view they were leaving. She does not rule the move out, and she calls it unlikely *[e3#119, e3#120, e3#122, e3#125, e3#127, e3#131]*.

> "It can be entirely good that the person is saying nothing. Perhaps she is about to be convinced by the others. If the AI then barges in, she stays with her view." *[e3#125, e3#127]*

E1, asked about a system that invites a person who has not yet spoken, says she would be very careful about an AI bringing quiet voices in *[e1#159]*. In her own practice, that invitation can feel intrusive online, and naming introversion can stigmatize *[e1#88, e1#90]*.

People will not yield the floor to a machine, especially a senior person (L4). E3 expects an AI to fail most sharply at interrupting people who want to speak. The inhibition is higher with a human, including a beginner *[e3#104, e3#106]*. E1 expects a chat to be talked over, because people would rather talk to each other, and she expects a video agent that tells a senior person they have spoken too long to be rejected *[e1#178, e1#183, e1#184]*.

> "Sorry, you have already been speaking too long. Who are you? […] Especially if I were the manager, I would think: forget it." *[e1#183, e1#184]*

E2 doubts that a system will be obeyed where it has no accepted authority *[e2#160]*.

The system must not decide, and must not impose a structure nobody asked for (L5). E2 says the decision should not be taken out of human hands, particularly where the solution is not yet known and the group has to come to a judgment *[e2#166]*. E3 would have the participants trigger the decision mode, or have the system speak afterwards. A light that says a decision still seems open, which the group can choose to follow, is the most proactive form she will consider, and even that might damage the conversation *[e3#101, e3#108, e3#109]*.

> "An AI that barges in on its own and imposes a structure the participants did not ask for. I find that very hard to imagine." *[e3#111]*

A transcript is not enough context for a sound intervention (L6). E3 says a sound reading would require the decision brief, video, tone, facial expression, gesture, the history of this group, and the meetings before and after this one. A transcript does not supply them *[e3#123, e3#133]*.

> "A great deal more has to be clear than just the meeting transcript." *[e3#133]*

E1's case is a person who is only half present because a child is ill. The facilitator knows that because she was told. She would not have told the system, and it will still produce a reading of how the person is in the meeting *[e1#161, e1#162]*. E2 says that treating the question at a general level is difficult. A closer answer would require the kind of organization, and whether the work is remote or hybrid *[e2#172, e2#173]*.

The group has to agree beforehand that interruption is allowed (L7). E1 requires that the system be introduced, and that the whole group agree that it may interrupt, together with an explicit decision about what is stored *[e1#199, e1#200, e1#204]*. The timekeeping interruption is, for her, acceptable under that agreement *[e1#172, e1#199]*.

> "It has to be introduced. And it has to have an agreement of the whole group that it may do that." *[e1#199]*

4 System Design and Implementation of a Scoped Prototype (RQ2)

Accordingly, this chapter does not claim to present a system that resolves every problem arising in the divergent phase of a meeting. Instead, the prototype targets one specific sub-problem that was identified as both high-impact and measurable in the theoretical foundation and the expert interviews: the shared-information bias underlying hidden profiles and the closely related issue of unequal participation.

4.1 Discussion of research results and decision to narrow down

4.1.1 Retaining Proactive Facilitation Despite Empirical Concerns

This thesis is based on a specific premise: an AI system that proactively steps in to moderate a goal-oriented remote meeting while the meeting is in progress.

Since facilitators also intervene proactively in meetings and facilitation is part of their role, RQ2 does not ask whether facilitation would be useful in principle. It asks which of the challenges described in RQ1 such a system could actually address during an ongoing meeting, and what form the intervention must take.

One could argue that this assumption was too narrow. The literature has already identified a tendency for meeting facilitators to lean toward systems that intervene less proactively. (See 2.3.2)

The expert interviews also reveal some presumed limitations of proactive, interventionist AI meeting systems, with experts declaring precisely those proactive approaches, which this study aims to investigate in depth, to be unpromising.

At the same time, the literature and interviews indicate that proactive intervention is necessary in some decision-focused meetings with a convergent nature and is common practice among human facilitators. However, the research approaches examined by the author tend to take a more cautious stance when it comes to AI systems, which has sometimes resulted, for example, in the AI's contributions being ignored. (see 2.32)

Furthermore, the limitations implied by the expert interviews, althougt they seem plausible, hypothetical.

The question remains as to how participants in goal-oriented remote meetings would actually perceive a proactive intervention by an AI system. However, in his literature review, the author did not find any systems that had actually attempted to intervene proactively in a live meeting situation, which is why research in this area, with concrete implementation proposals, is limited.

Therefore, the author decided to stay with the original premise and attempt to develop an initial AI prototype that could actually be evaluated in a live meeting situation. This was not because the concerns or limitations

raised were deemed implausible, but rather based on the assumption that an initial attempt to design such a proactive system would reveal further insights, challenges, and opportunities that would otherwise have received little attention. It is hoped that patterns can be derived from these learnings to further develop such proactive moderation systems.

4.1.2 Focusing the Design Scope on Hidden Profile Szenarios

The previous chapters have opened up a broad range of issues. Goal-oriented remote meetings often fail due to a lack of pace, focus, and decision-making, as well as uneven participation and a skewed exchange of information (section 2.2, subsection 2.2.4).

The expert interviews identify several facilitating behaviors that are applicable depending on the meeting situation. Developing a system capable of responding to all of these behaviors would be too complex and would not answer RQ2. Any observed effect could not be attributed to any single design decision.

Some of the experts' suggestions regarding the potential of AI-supported meeting systems fall outside the scope. Preparation and documentation (F8, F7) take place before and after the meeting. Summarization, clustering, and time reminders (F1, F2) are already being addressed by existing tools (subsection 2.3.1). If a role is missing, facilitators ask the group what that role would say. Simulating this as in (F6) is a different approach that does not fall under proactive moderation. A decision-making process (F3) can be initiated by the group as soon as options become apparent. It does not require a system that intervenes in the ongoing conversation without being asked.

Therefore, the bias indicator (F5) was selected as a promising approach, as it is most closely aligned with the research question. At the same time, despite the presumed limitations, it remains justified to pursue this more proactive approach, as this is where the benefits of a proactive system are greatest. Since the group may not experience the bias themselves, those who would need to actively provide help might not even be prompted to do so in the first place, and overly passive approaches could potentially be ignored.

The bias associated with a core task of facilitators is the shared-information bias (subsection 2.2.4). In this practice, integrating participants means contributing the opinion or fact that would change the outcome (subsection 3.2.3). This is particularly essential for decision-making. In the hidden profile, the relevant information is already available, but distributed across the group. Based on their partial knowledge, each person forms a preference that is inferior to the one the group would form based on the shared situation.

The Inner-Thoughts Framework provides a technical foundation that can be adapted to this use case.

Consequently, this focus was deemed both promising in the context of the insights gained so far and technically feasible.

4.2 Requirements Analysis and Scoped Requirements Catalog for the Prototype (User Stories)

Die Anforderungen werden in Form von User Stories festgehaltens Laut Ott können diese einen in Alltagssprache formulierten Software-Anforderungskatalog darstellen [54, p. 18]. Die Formulierungen orientieren sich an diesem Muster:

> "Als <Rolle>, ich möchte <Ziel/Wunsch>, um <Nutzen>." [54, p. 18]

Die Anforderungen wurden aus der Literatur und den Experteninterviews extrahiert und der in 4.1.2 beschreibenen fokusiereung. Die Begründungen verweisen auf die Abschnitte dieser Arbeit, in denen die jeweilige Herleitung bereits ausgeführt ist, weshlab die quellen hier nicht wiederholt.

Die User Stories sind aus der Perspektive eines AI-Facilitators geschrieben, der mit modreine elmente proativ in das meeting eingreifen möchte und beim aufdecken von hidden profiles hilft. Table 4.1 bündelt sechs zentrale Stories mit stabilen IDs (US1–US6), damit spätere Design- und Scope-Entscheidungen eindeutig darauf verweisen können.

Table 4.1: Derived user stories for an AI facilitator, with references to the source chapters

ID	User Story	Reasoning / References
US1	Als AI-Facilitator möchte ich Ablauf und Ziel des Meetings klar machen können, damit die Gruppe einen gemeinsamen Bezugsrahmen hat und zu einer tragfähigen Entscheidung kommt.	Structuring tasks and communicating the meeting goal are core facilitation duties and a documented failure mode of goal-oriented meetings (see subsection 2.2.2; **??**). The expert interviews confirm that unclear purpose undermines facilitation quality (see chapter 3).
US2	Als AI-Facilitator möchte ich Personen, die noch wenig gesagt haben, auf rücksichtsvolle Weise in die Diskussion einladen, damit alle relevanten Stimmen gehört werden, ohne Schweigen zu stigmatisieren.	Unequal participation is a key meeting challenge and is linked to hierarchy and status (see **??**). Expert facilitators invite quiet members, but warn that doing so online can stigmatize or feel like control (see chapter 3).
US3	Als AI-Facilitator möchte ich gezielt nach noch nicht genannten oder nicht ins Bild passenden Informationen fragen, damit ungeteiltes Wissen in die Entscheidung einfließen kann.	Groups systematically under-discuss unshared information; asking for information that does not yet fit the picture is a documented counter-strategy (see subsection 2.2.4). The interviews describe the same practice of checking whether all relevant voices and facts are on the table (see chapter 3).
US4	Als AI-Facilitator möchte ich Meinungsverschiedenheiten nicht vorschnell auflösen, damit Dissent den Informationsaustausch vertieft und bessere Alternativen sichtbar bleiben.	Dissent improves hidden-profile solution rates; rushing through the Groan Zone closes that information search too early (see subsection 2.2.4; subsection 2.2.3). Interviewed facilitators likewise open first, then converge (see chapter 3).
US5	Als AI-Facilitator möchte ich nur sparsam und mit klarem Prozessnutzen eingreifen, damit die menschliche Diskussion nicht übersteuert wird.	A facilitator is substantively neutral and must not dominate the group (see subsection 2.2.2). Fully automatic or overly frequent intervention produces mistrust (see subsection 2.3.2; **??**). Experts warn against over-facilitation and a missing sense of presence (see chapter 3).
US6	Als AI-Facilitator möchte ich inhaltlich neutral bleiben und selbst keine Entscheidung treffen, damit die Gruppe die Verantwortung für das Ergebnis behält.	The facilitator is acceptable to the group, substantively neutral, and has no decision-making authority (see subsection 2.2.2). Interviewed facilitators invite leadership to interpret, but do not decide in the group's place (see chapter 3).

4.3 Evaluation-Driven Development: Formative Prototype Evaluation during Development

4.3.1 Implementation Procedure

Die Entwicklung des Prototyps folgte dem Evaluation-Driven Development (EDD) Ansatz (vgl. subsection 4.3.1). Dabei kamen bereits während der Implementierung automatisierte Prüfverfahren zum Einsatz, um das System kontinuierlich zu evaluieren.

Nutzerstudien, Human-in-the-Loop-Evaluierungen oder Expertenbewertungen waren in dieser frühen Prototyping- und Entwicklungsphase nicht praktikabel, da solche Verfahren ohne erheblichen Aufwand nicht skalierbar sind. Stattdessen erfolgte die Weiterentwicklung des Systems in kurzen Zyklen aus Implementierung, Prüfung und Anpassung. Dieses iterative Vorgehen ist in der Softwareentwicklung etabliert (vgl. [54, p. 8]) und besonders bei EDD von KI-basierter Systeme relevant, da aufgrund der vielfältigen Nutzungsmöglichkeiten hier kein vollständiges Testorakel vorausgesetzt werden kann [49, p. 39].

Für Komponenten, die nicht auf probabilistische KI-Modellausgaben angewiesen waren, sowie Unit- und Integrationstests, kamen klassische Softwaretests zum Einsatz, da sich für diese ein eindeutiges und deterministisch prüfbares Testorakel definieren ließ. Da der überwiegende Teil des zu testenden Systems jedoch auf LLM-basierte Komponenten angewiesen und somit probabilistisch ist, war dies in den meisten Fällen nicht möglich.

Deshalb wurde ein von der Methode „LLM-as-a-Judge“ inspirierter Ansatz gewählt (vgl. **??**). In der hier verwendeten Version bewertete das LLM den Output nicht in Echtzeit, sondern wurde genutzt um im Voraus Szenarien (meist mögliche Konversationen) für spezifische Situationen zu generieren, um ein lexikalisches Testorakel zu erstellen. Dieses wurde im Rahmen codebasierter (vgl. **??**) automatisierter Test, in Kombination mit regulären Ausdrücke verwendet. Das vom LLM-Judge genereierte Orakel operationalisierte, welche Formulierungen eine fachlich angemessene Facilitator-Äußerung in dem jeweiligen Szenario aufweisen muss und welche sie nicht aufweisen darf. Der Output der einzelnen Komponenten wurde überwacht und analysiert, um das Verhalten der Komponenten in den verschiedenen Szenarien besser verstehen und bewerten zu können.

Während der Entwicklung wurde das System kontinuierlich in unstrukturierten, explorativen Tests, sowohl einzeln als auch unregelmäßig mit mehreren Personen in verschiedenen, nicht weiter spezifizierten Szenarien überprüft. Ziel war es, bereits vor den eigentlichen Nutzerstudien (vgl. chapter 5) ein Gefühl für das Systemverhalten zu entwickeln. Dadurch sollten offensichtliche Fehler sowie kritische Schwachstellen, an denen das System vom intendierten Verhalten abweicht, frühzeitig erkannt werden, um das System anzupassen.

Dieses Vorgehen kann in einem Loop wie folgt dargestellt werden:

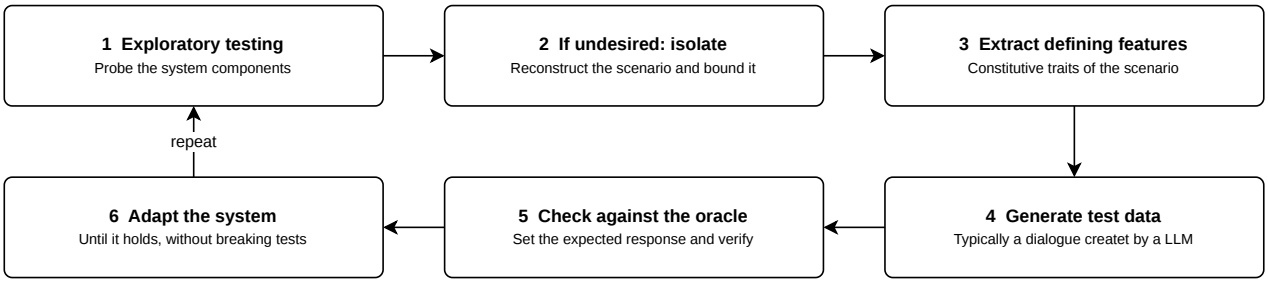


Figure 8: Evaluation-driven development loop: From exploratory testing to system adaptation – *Source: Own illustration, created with draw.io.*

Das sukzessive erweiterte Testorakel diente zugleich als Regressionstest, um zu erkennen ob eine Systemänderung zuvor gesichertes Verhalten nicht beeinträchtigt.

Zum genaueren monitoren des Systems wurde zusätzlich zu den automatisierten Tests Langfuse genutzt. Langfuse ist eine Open-Source-Plattform zur Observability und Evaluation von LLM-Anwendungen, die Trace-Level-Daten zu Modellaufrufen, Prompts und Nutzerinteraktionen erfasst und deren Analyse, Bewertung sowie kontinuierliche Optimierung ermöglicht. [55]

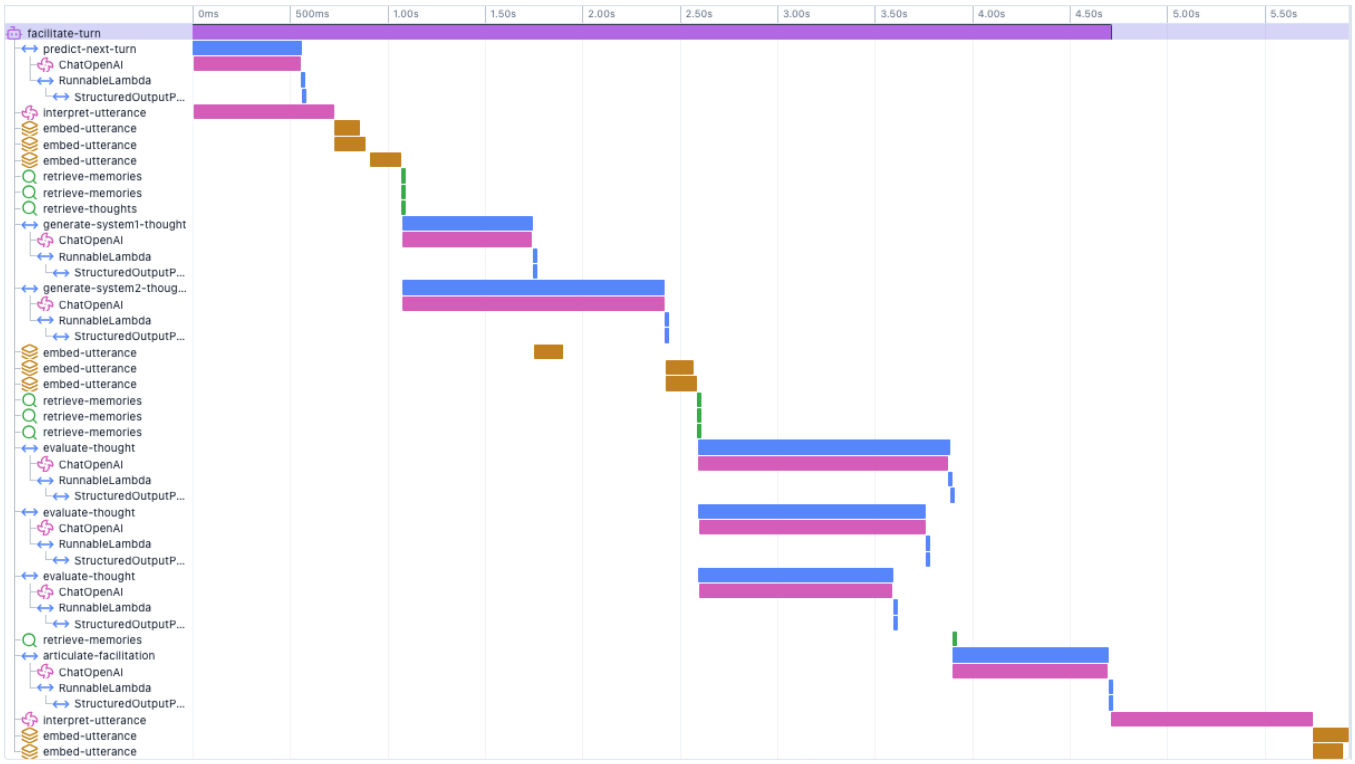


Figure 9: Langfuse: Screenshot of the Traceviewer in Langfuse that shows the trace of the AI-Facilitator processing a new conversation event. – *Source: Own screenshot, created with Langfuse.*

4.3.2 Limitierungen des Vorgehens

Ein lexikalisches Orakel ist unvollständig und die qualität hängt in diesem Fall von dem LLM-Judge ab. Eine gültige Facilitator-Äußerung kann das intendierte Verhalten zeigen, ohne die vorgesehenen Wortgruppen zu enthalten; umgekehrt kann ein Treffer formal korrekt und situativ unpassend sein. Isolierte Dialoge blenden zudem die Dynamik eines echten Meetings aus: Turn-Taking, Timing und die Reaktion der Gruppe auf eine Intervention bleiben unberücksichtigt. Welche Szenarien überhaupt als Testdaten aufgenommen wurden, hängt außerdem von den explorativ beobachteten Fehlern ab und ist damit autorenseitig selektiert.

Dennoch entspricht diese Kombination, von kontinuierliche, orakelgestützte Prüfung während der Implementierung und unabhängige Human-in-the-Loop-Evaluation danach (druch die User Studien), einem valider Ansatz zum prüfen von llm basierten agentischen Systemen. [50, pp. 6–7].

4.4 System Architecture: Adapted version of the “Inner Thoughts” framework

4.4.1 Overview

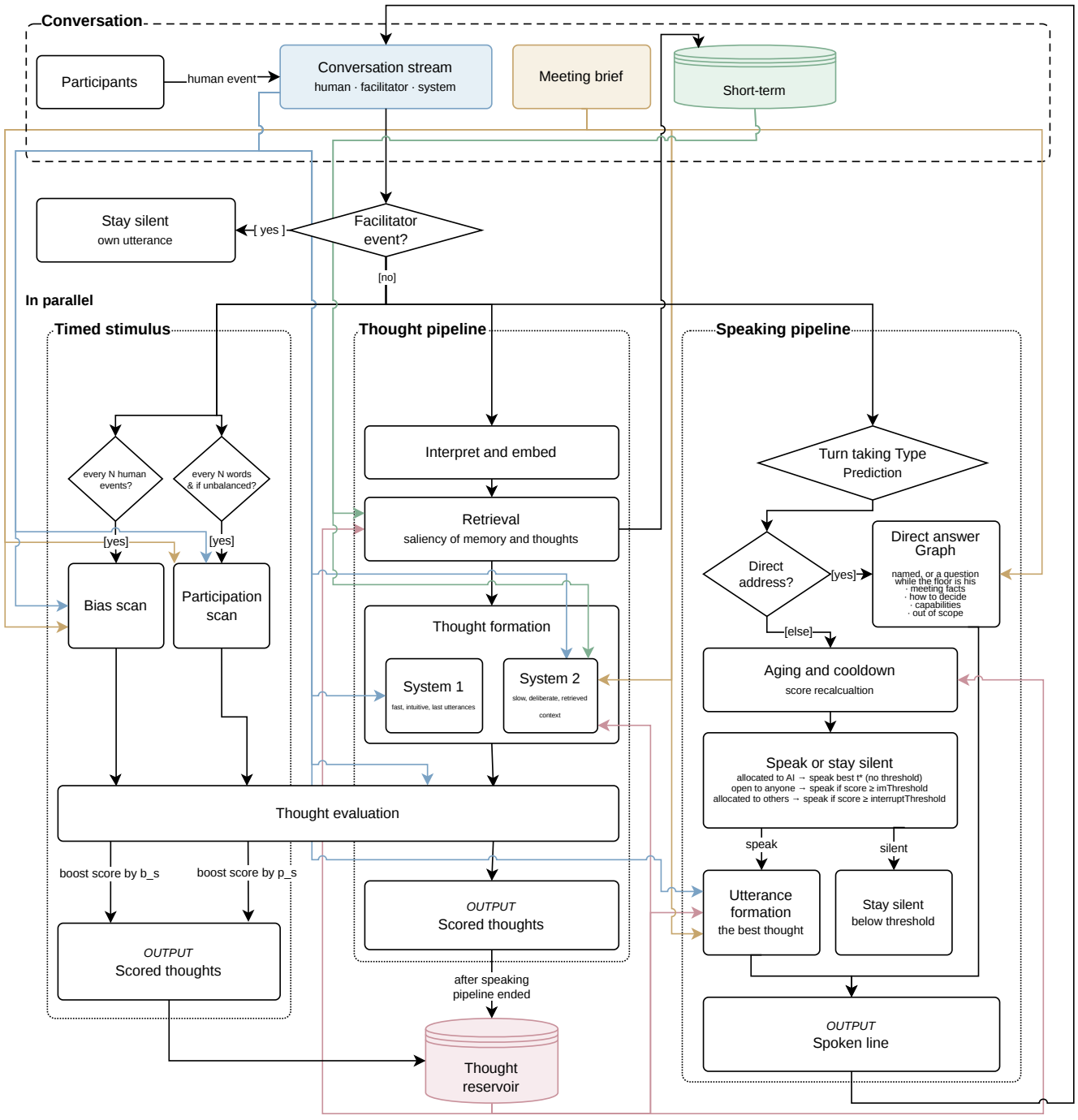


Figure 10: System Architecture: Adapted version of the “Inner Thoughts” framework – *Source: Own illustration based on ”Inner Thoughts” by Liu et al.[43], created with draw.io.*

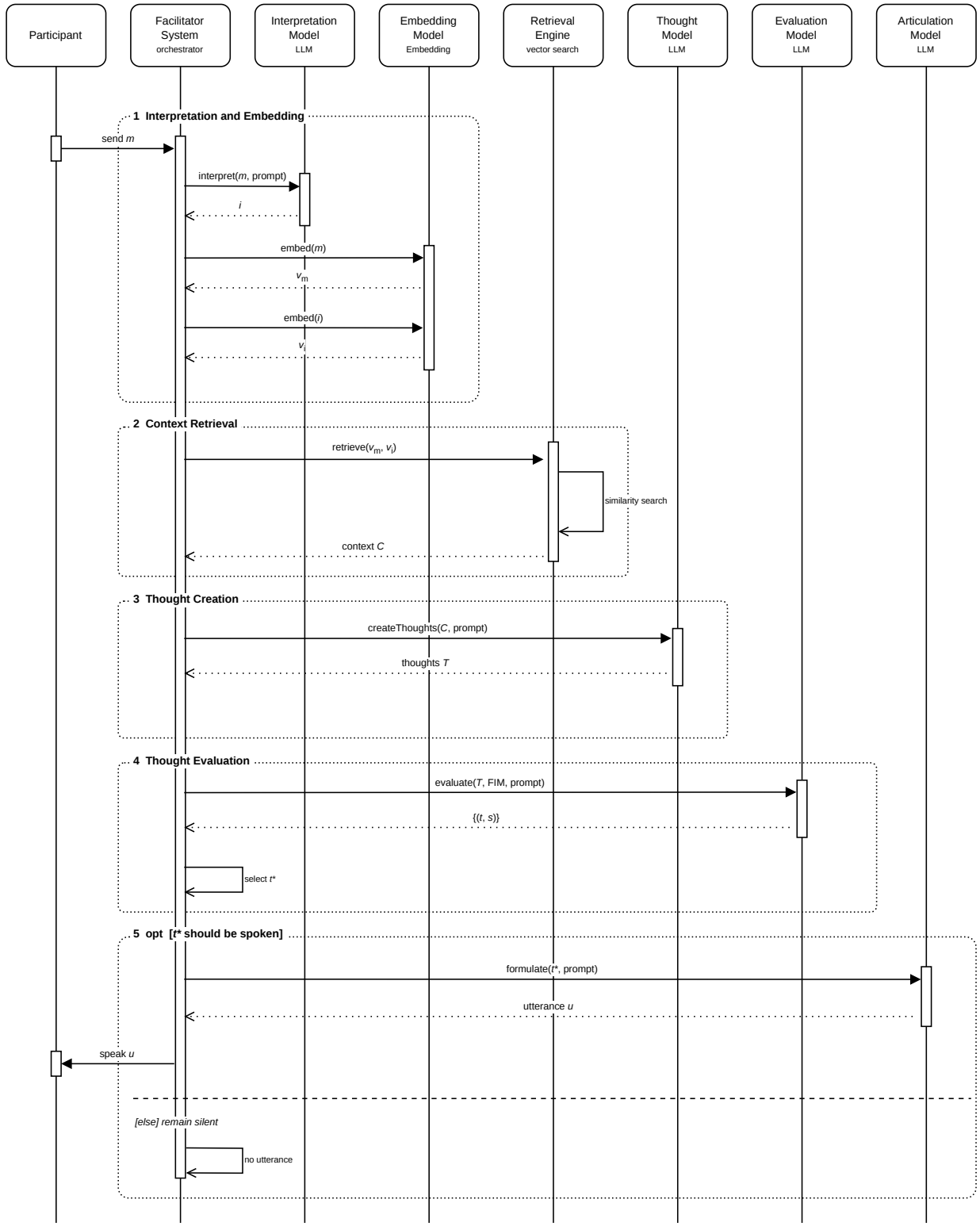


Figure 11: Thought Pipeline Sequence: Sequence of the thought pipeline – *Source: Own illustration based on "Inner Thoughts" by Liu et al.[43], created with draw.io.*

Wie im kaptiel ”practice” ai interaction besprochen, sind ansätze für turn prediction besondere schlcht in sitauion mit selfalocation. dies ist allerdigns genau das was ein faciliatato machen würde. insbesondere in scenarien in denen davon auszugehen ist das die gruppe nicht selbst erkennt und nahc hilfe fragen kann, wie es z.B. bei dem shared information bias in hidden profile szenairen der fall ist.

Das Inner Thougs freame work scheint hier einen vielversprechneden Ansatz zu bieten.

Warum wrude sich für das Inner Thougs framework etnschieden? weshlba nicht eher ein passivers System? (teilnehmenden ingoreren es, rationaliseren es weg) der fokus sollte gezielt darauf liegen, wie meeting teilnehmden in einer live siuation tasächlich reagleiren. Es war bewust, dass diese entscheidung gegen eingie vermutet limitrungen spricht. Teile der literatur schlißen eien so aktiven KI-Agent in meetings tendezell aus (vgl. sektion mit andern ki meeting systemn). dem entsprechend gibt es allerdings kaum arbeiten die tatsächlich versuchen ein solches system zu desingen und es dann auch in einem live meeting evaluieren.

Das Inner Thoughts- Freamwork von Liu et al. ist in deren version eher für interaktionen ausgelegt die als soziales Plaudern kotegorisertet werden können. Dies ist nicht was ein Faciliator machen würde. Die Idee ein KI-System anhand einer intrsisichen motivation zu modellieren, bevor jemand etwas zu einer Gruppen discussion beiträgt, ist dennoch auch für diese ARbeit relevant. Insbesondere weil dieser Ansatz darauf hindeutet eine erhöhte Genauigkeit bei self-trun-allocation zu erzeugen. Aus den Experten ging hervor ds eine Geanken gänge, welche Facilitatoren haben auf abstakter inutionen basieren andere wiederum druchdacht kombiniert mit experten wissen etc. Dies legt nach, dass legt nahe das sowhl system 1 als auch system 2 bei dem prozess involoviert sind, was eben falls in dem Innter Thougs Framewokr berücksichtigt wird.

Die Autoren wiesen in der Diskussion der Arbeit [43, ”8.2 Apllying Inner Thougs Beyond Casual Conversations”] selbst darauf hin, dass sich das System durch anpassungen der Bewertungskriterien auf aufgabenorietnerte Kontexte anpassen lässt und zielorientiertes praktives Handeln ermöglichen kann. Sie schalgen vor für weitere arbieten das system mit domain-specifische heuristiken anzupassen, um die interaktionen an real-world tasks zu orientieren.

Im ursprünglichen Modell von Liu et al. werden generierte Gedanken primär danach bewertet, wie stark sie die ”persönliche Relevanz” die individuellen Interessen oder die einzigartigen persönlichen Kenntnisse des Agenten widerspiegeln (TODO: prüfen bzw. konkretisieren wie die heuristik tatsächlich ist). Dies mag für generische Conversationen geeigent sein, da dieser Ansatz Authentizität und Sympathie in infromellen, sozialen Unterhaltungen erhöht, ist für einen AI-Moderator in zielorientierne Meetings die durch Hidden Profile geprägt sind, allerdings ungeeiengt. Wenn die Gedanken des AI-Moderator primär danach bewertet würden, wie stark sie seinen ”persönlichen Interssen” wiederspieglen, verfällt er in dassaelber fehlerhafte Verahltensmuster, das zu verzeruung in hidden profile situationen führen kann. Er bringt eingen Meinungen oder Präferzene ein statt die diskussion neutral zu strukturieren. Strukturelle prozeduelle Gedanken würde in der aktuellen Heurisik voraussichtlich mit zu niederiger Relevanz bewertet werden.

Bei der Anpassung der Heurisitken und prompts wurden sowohl die Literatur als auch die erkentnisse aus den Interviews berücksichtigt. Die Evaluierungs-Heursitik zu bewertung der Gedanken wurde so angepasst, dass sie das kognitionspyschologsche Fundament des Innter Thoughts-Modelles (basierend auf grice, Sacks, Goffman und Berly) mit en organisationspsychologischen Prinzipien zur Kompenstation des Shared Information bias (Stasser & Titus; Brodbeck et al.) und den HCI-Richtlinen für inklusive Moderation (Houttie et al; Davison) kombiniert.

In der original Modellering des Inner Thougs-Frameworks wird ide participation in einer kovversation ausschlielich basierend der innern gedanken abgeleitet. Die Authoern mekren an, dass exteren Cues nicht im ausschleiieß zu ihrer Methode stehen sondern complentär mit exteren cues kombiniert zu einem robustern system führen könnten. [43, 8.1 Proactive Agents via Intrinsic Motivation vs. Extrinsic Cues]

Eine kernproblematik bei hidden profies ist, das private inforamtione nicht getielt werden. Ein indikator dafür könnte sein, dass eine person sich an der converstation nicht beteiligt oder ander Teilnehmden die Diskussion domieren, so dass andere nicht zu wort kommen können. wie aus den interviews hervorging, würde ein facilitator hier typischer weise versuchen diese Person auf höfliche art zu integrieen, um deren Perspektive zu hören. Hierfür wäre extern Cues wie die Sprrecheranteile relevant. Um diese exteren Cues abzubilden wurde das Inner Thoughs Framework, um die Komponete "External Stimulus Injection" erweitert. Ist einer dieser externe Trigger erfüllt wird ein vordefinierter Gedanke in der Gedanken generierung von System 2 berügcksichtig und die Evaluations herusitik bewsust in richtung diesen extern Triggers gebaiased, um zu erzielen dass diese externe cue im innerthoughs modeel gewichtet wird.

Folgt man der Definition von Anthropic, enthält das System überwiegend Eigenschaften, die es als Workflow kennzeichnen: LLMs und Werkzeuge werden weitgehend über vordefinierte Codepfade orchestriert. Gleichzeitig entscheidet das System selbst, wann und wie es interagiert, und trifft dabei fortlaufend Entscheidungen. Aufgrund dieser wahrgenommenen Autonomie beschreibt der Autor das System als KI-System mit programmatischen Elementen, die agentische Eigenschaften induzieren. An den meisten Stellen der Arbeit wird die Bezeichnung daher bewusst offen gehalten und von einem KI-System oder einem KI-Agenten gesprochen.

Baseirend auf der Literatur und den Intervies wurde versucht dieses System entpsrechend zu adaptieren.

Ein prblem ist die tendenz von KI zu überkompensieren und zu viel zu sagen. Die authoren merken dies auch an und sind optimistischen, dasss ihr conversational agent hier besser performt. Alelrdigns handelt es sich um eine plausch. Ein facilitator würde nicht so oft in die Diskussion eingreifen. Das risiko für overcompensationg mit diesem system ist hoch. mit dem imThreshold kann gesteuert werden wie schwerwwiegend ein Gedanke sein muss damit die KI ihn überhaupt aussproche.

In dem Paper wurd ein Selective Participant modeliert, welche foglende Parameter hatte:

- System1 wurde auf 0 gesetzt so das die KI also niemals äußert niemals spontane, ungeplante System-1-Floskeln äußert. Jeder Beitrag muss einen teifen, kognitiven Nutzen haben. -imThreshold wurde auf 4.3 gesetzt. Ein strategeischer Gedanke muss auf einer Skala von 1-5 einen extrem hohe Bewertung erreichen, um die schwelle zu reißen. Ein einfacher Gedanke wie "ICh könnte Bob mal wieder fragen" reicht dafür nicht aus. Es bedarf eines Phase 1 Trigger:

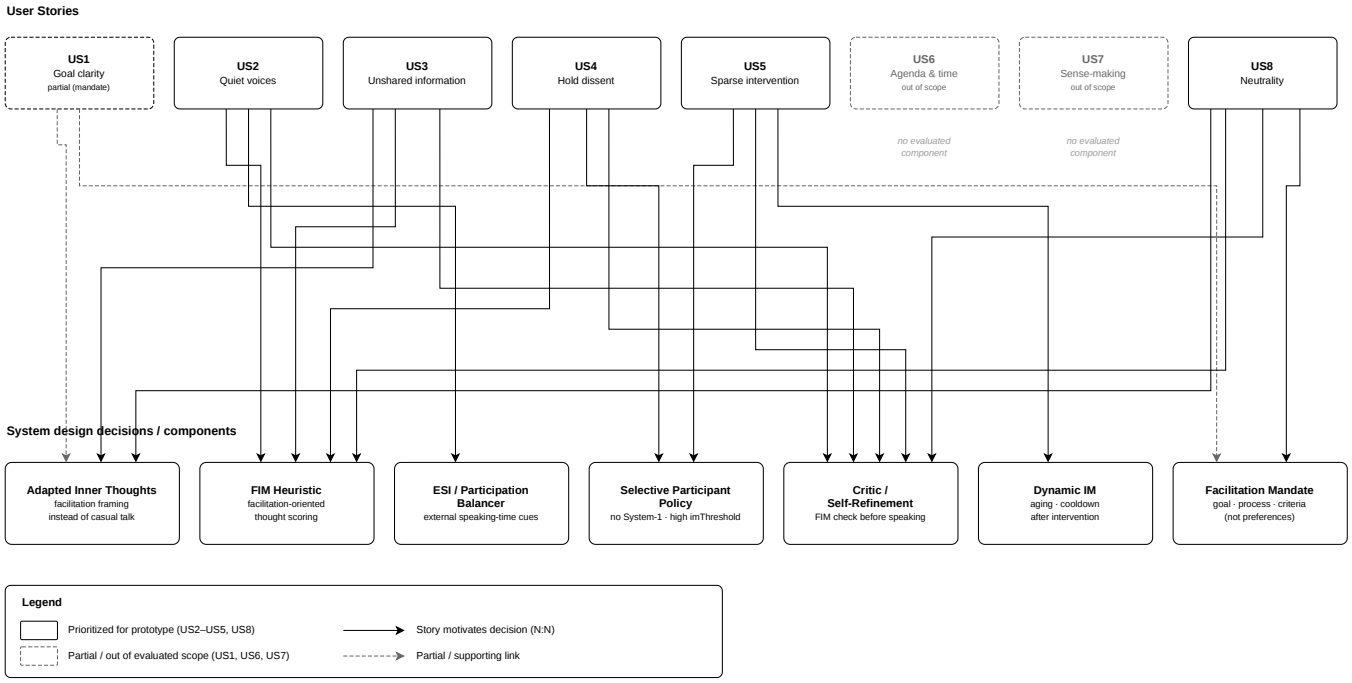


Figure 12: From User Story to System Decision: Graphical overview of what User Stories motivated a system design decision. – *Source: Own illustration, created with draw.io.*

4.4.2 Components: Transcription, Analysis, and Intervention Modules

4.4.3 Components: Transcription, Analysis, and Intervention Modules

4.4.4 Components: Transcription, Analysis, and Intervention Modules

Abgleeite aus dem Paper Keeper. Der Agent könnte auch eine visualle Warteschlage steuern. anstatt aktiv zu unterbrechen sezt ersich auf die warteschalge bzw. hebt die hand, um sich anzukündigen.

FIM-Heuristik

Thought-Evaluation

The full FIM scoring prompt, with a unit-by-unit provenance relative to the original Inner Thoughts evaluation protocol, is documented in section 7.4.

Angelengt an die Thoguht Evaluation Phase von XX (procativ Conversation Agents) wurde auch in diesem sich dem contextual self-refinement bedient. Self-Refinement und Reflexion könnnen bei LLMs zu drastischen ELsitugssteirgunen führen, wenn eine kognitive Feedbackschleife druchaufen wird. [47, p. 18] Ein Gedanke wird nicht direkt ausgesprochen, sondern druch einen Kritiker basierend auf der für diese arbeit entwickelte FIM-Heurisitk geprüft. The evaluation protocol follows the G-Eval prompt structure [56, p. 2512] similarly also used for Thoughtful Agents. Probability-weighted scoring as proposed in the original method was omitted as a pragmatic decision, because the APIs used here do not return output-token probabilities.

In practice every utterance is appended to a shared pending buffer of messages that have not yet been thought about. The turn-taking prediction only decides *when* that buffer is flushed (immediately if the floor is allocated

to the AI; otherwise after X seconds as one batch). Both cases continue to the same next step: the flushed pending set is assembled and the thought pipeline runs. External Stimulus Injection is not a later stage. Its monitors run continuously, but an ESI policy joins the pipeline only at that flush, and only if a trigger is active; otherwise the pipeline runs without it. A selected thought t^* is not spoken raw. An Utterance Formation agent formulates it into a facilitation act, with tools for conversation context, facilitation best practices, speaking-time shares, sending a chat message, or speaking aloud. The articulation prompt and the mapping from the original Inner Thoughts wording are documented in section 7.5.

Table 4.2 maps each architectural choice to a single warrant and to the chapters that already establish it, so the implementation can be traced without restating those arguments.

Table 4.2: Architectural design decisions of the adapted Inner Thoughts prototype, with rationale and pointers to the motivating chapters of this thesis.

Design decision	Rationale	Evidence
Adapt Inner Thoughts from casual talk to facilitation	The original model times utterances by intrinsic motivation, which improves self-turn-allocation in multi-party talk. Liu et al. already invite domain-specific evaluation criteria for goal-directed settings; without that shift, a facilitator would be scored as a social conversationalist.	subsection 2.3.3
Replace personal-relevance scoring with a facilitation-oriented FIM heuristic	Ranking thoughts by the agent's own interests would privilege preference talk, undervalue procedural structure, and reproduce the shared-information bias the moderator is meant to counteract.	subsection 2.2.4; chapter 3
Inject external participation cues into thought generation	Unique information stays private when speakers remain silent or are crowded out. Observable speaking-time imbalance is a complementary cue, not a rival, to intrinsic motivation; expert facilitators report inviting quiet members rather than waiting for self-selection.	subsection 2.2.4; **??**; chapter 3
Selective-participant policy: no System-1 speech, high *imThreshold*	Facilitation needs sparse, high-utility interventions. Unconstrained generation over-talks; disabling System-1 filler and raising the utterance threshold admits only thoughts with clear process value.	subsection 2.2.2; **??**
Critic-based self-refinement before any utterance	A thought is spoken only after a critic applies the FIM heuristic. Contextual self-refinement implements an evaluator–optimizer loop and is associated with substantial quality gains in LLM pipelines.	subsection 2.4.2; subsection 2.4.3
Cheap temporal reweighting instead of LLM re-scoring	Static IM speaks stale thoughts. Re-prompting every thought after each event is too slow and costly for a live meeting. Linear aging and a post-speech smoothstep keep the original score but discount age and repetition.	subsection 4.4.1

Dynamic intrinsic motivation

In the original Inner Thoughts model, intrinsic motivation is static: a thought is scored once at generation and that score never changes. Early tests during development showed that the system spoke thoughts that the subsequent discussion had already made obsolete, and that would, if they would have been evaluated again, probably no longer would have received a high FIM score given how the meeting had evolved.

Re-scoring every pending thought with an LLM after each event would restore recency, but the added latency and cost would make the system unusable in a live meeting. The prototype therefore keeps the original FIM score and applies a cheap, deterministic reweighting under two assumptions:

- A facilitation thought becomes a poorer description of the current state and process need as the conversation moves on, because later information was not available when it was scored. As the meeting continues, the group may already have closed that gap, a quiet participant may have spoken, unique information may have surfaced, or the discussion may have moved on. Therefore its priority should fall with age.

- Once a facilitation intervention has been spoken, repeating it immediately could result in over-talking and crowd the discussion. The motivation to raise the same process issue is therefore initially very low, but may recover if the group still does not take it up.

Let $S = [1, 5]$ be the FIM scale, $T = [0, \infty)$ elapsed time in seconds, and $T_\infty = [0, \infty]$ time since last speech, where ∞ means never spoken. To each triple $(s_0, t, t_{\text{spk}}) \in S \times T \times T_\infty$ the mapping s_{weighted} assigns one weighted score. Aging and cooldown address the two assumptions (see above), so they are modeled as a product:

$$s_{\text{weighted}} : \begin{cases} S \times T \times T_\infty \to [0, 5] \\ (s_0,\, t,\, t_{\text{spk}}) \mapsto \min\big(5,\, \max(0,\, s_{\text{aged}}(s_0, t) \cdot c(t_{\text{spk}}))\big). \end{cases} \tag{4.1}$$

The session-wide parameters τ, $(s_{\text{hold}}, t_{\text{hold}})$ and t_{cool} determine the kernels s_{aged} and c below.

Aging s_{aged}. Later information was not available when the thought was scored, so priority falls with age. Linear decay hits zero at a finite horizon H:

$$s_{\text{aged}} : \begin{cases} S \times T \to [0, 5] \\ (s_0,\, t) \mapsto \max\big(0,\, s_0(1 - t/H)\big). \end{cases} \tag{4.2}$$

Linear decay is a practical modelling assumption: it is simple and reaches zero in finite time (Figure 13). (4.2) is the straight line from $(0, s_0)$ to $(H, 0)$, hence of slope $-s_0/H$.

The horizon H is not chosen directly. A thought of score s_{hold} should remain above τ for t_{hold} seconds, so that line must pass through (t_{hold}, τ). Equating the two expressions for the slope and solving for H gives

$$\begin{aligned} -\frac{s_{\text{hold}}}{H} &= \frac{\tau - s_{\text{hold}}}{t_{\text{hold}}}, \\ H &= \frac{t_{\text{hold}}}{1 - \tau/s_{\text{hold}}}. \end{aligned} \tag{4.3}$$

All thoughts share this H. A weaker thought therefore crosses τ sooner. The pair $(s_{\text{hold}}, t_{\text{hold}})$ is the only design knob for temporal sensitivity: a shorter t_{hold} steepens every line (Figure 13).

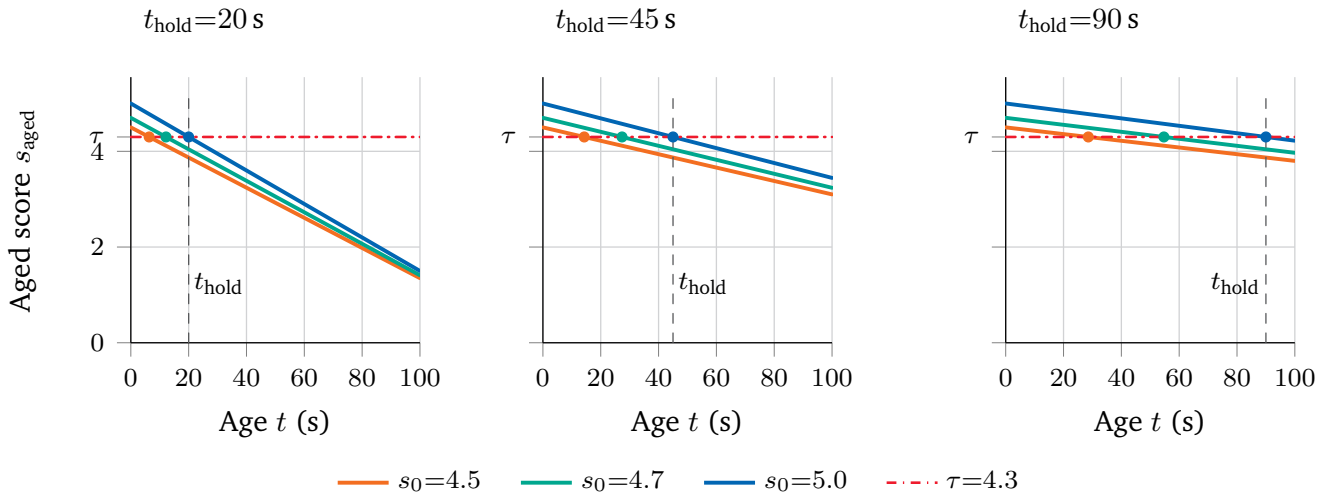


Figure 13: Linear aging of the FIM score at three hold-times. Each panel shows the same scores $s_0 \in \{4.5, 4.7, 5.0\}$; the vertical line is t_{hold}, where a score-5 thought crosses τ. Within a panel, weaker thoughts expire sooner. Left to right, a shorter t_{hold} steepens every line. Parameters: $\tau = 4.3$, $s_{\text{hold}} = 5$. – *Source: Own illustration, created with LaTeX and TikZ*

Cooldown c. Once spoken, the motivation to repeat a facilitation intervention is initially very low, but may recover if the group still does not take the process issue up.

$$c : \begin{cases} T_\infty \to [0,1] \\ t_{\text{spk}} \mapsto \text{smoothstep}(t_{\text{spk}}/t_{\text{cool}}), \end{cases} \qquad \text{smoothstep}(x) = \begin{cases} 0 & x \le 0, \\ 3x^2 - 2x^3 & 0 < x < 1, \\ 1 & x \ge 1. \end{cases} \tag{4.4}$$

For $t_{\text{spk}} = \infty$ the quotient is ∞, hence $c(\infty) = 1$.

The cubic $3x^2 - 2x^3$ is the Hermite interpolant on $[0, 1]$ with vanishing endpoint derivatives, chosen so that recovery pauses after speech and returns to 1 without a hard knee (Figure 14).

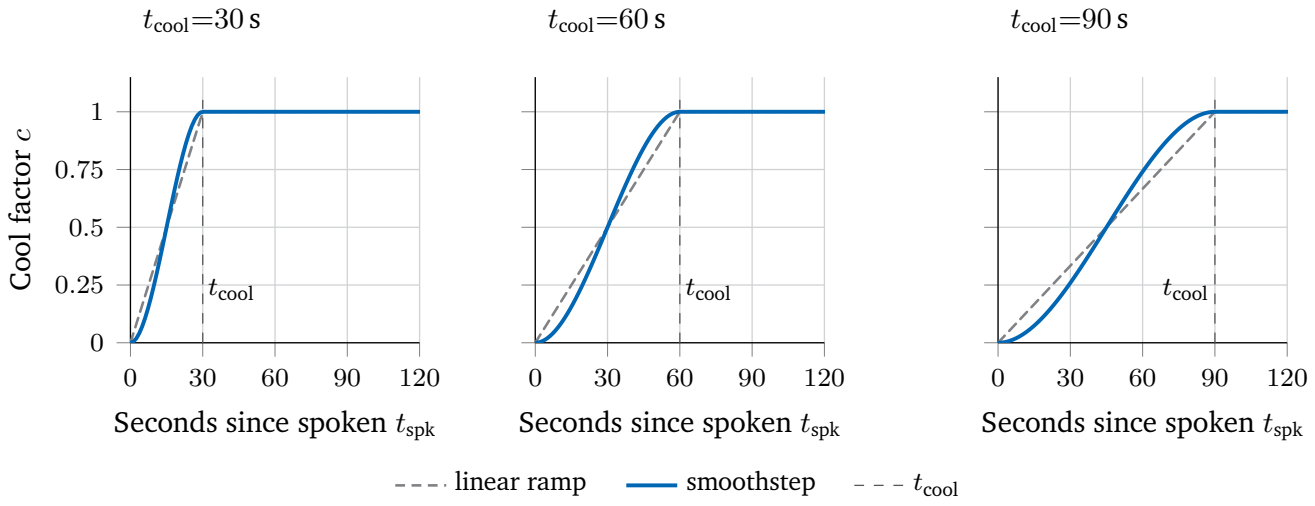


Figure 14: Cooldown factor after a thought is spoken, for $t_{\text{cool}} \in \{30, 60, 90\}$ s. Each panel compares the Hermite smoothstep with a linear ramp; the vertical line is t_{cool}, where recovery completes. The smoothstep stays flat at both ends; the linear ramp starts immediately and kinks at t_{cool}. – *Source: Own illustration, created with LaTeX and TikZ*

The product in (4.1) is the result (Figure 15). A just-spoken thought cannot be repeated. After t_{cool} it may become eligible again, but only if s_{aged} is still above τ.

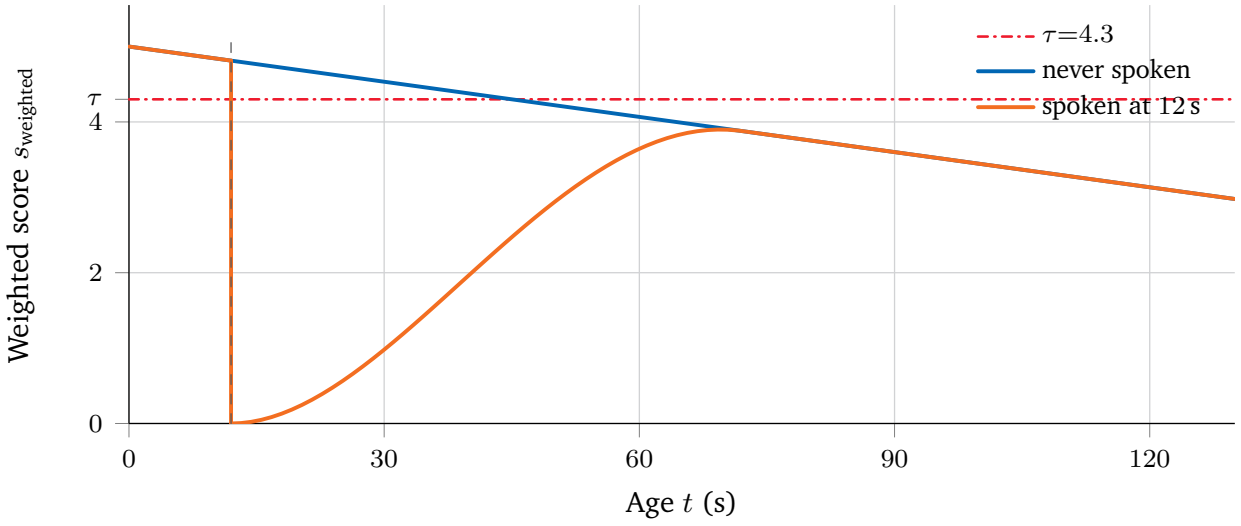


Figure 15: Weighted FIM score of a thought with $s_0 = 5$, never spoken versus spoken at $t = 12$ s. Aging uses $t_{\text{hold}} = 45$ s; cooldown uses $t_{\text{cool}} = 60$ s. Recovery restores the aged remainder; whether that remainder still exceeds τ depends on how long the thought has already lived. Parameters: $\tau = 4.3$, $s_{\text{hold}} = 5$. – *Source: Own illustration, created with LaTeX and TikZ*

Participation Balancer

The Participation Balancer operationalizes US2 as two speaking-time cues: curtailment of a dominant speaker and invitation of a low-participation participant. Unshared information in a hidden profile is introduced only by the participant who holds it, and speaking turns are unequally distributed by status, with higher-status participants intervening more frequently (subsection 2.2.4; **??**; [57, 27, 11, 10]). Curtailment is therefore

conditioned on a high individual share together with another participant below half a fair slot, whereas invitation depends only on low share and prolonged silence.

Let $L > 0$ be the planned meeting length and P the set of participants, with $n = |P| \geq 1$. Each participant's fair slot is

$$F = L/n. \tag{4.5}$$

Let $r_i \geq 0$ be the speaking time participant i has accumulated, and let $i_\star \in P \cup \{\perp\}$ be the participant who currently holds the floor, or $\perp$ if nobody is speaking. Let $\Delta_i \geq 0$ be the time since i last spoke, counted from the start of the meeting if i has not spoken. The factor $p > 0$ is configured.

The current speaker is curtailed once their speaking time reaches p fair slots and at least one other participant is still below half a fair slot:

$$\text{curtail}(i) \iff i = i_\star \ \wedge \ r_i \geq p \cdot F \ \wedge \ \exists\, j \in P \setminus \{i\} : \ r_j < F/2. \tag{4.6}$$

Invitation uses two configured thresholds. $\theta_r > 0$ is the speaking time below which a participant still counts as low-participation, and $\theta_\Delta > 0$ is the minimum time they must have been silent since they last spoke. A participant is invited in when their own speaking time is still below θ_r, they have been silent for at least θ_Δ, and they do not hold the floor:

$$\text{integrate}(i) \iff r_i < \theta_r \ \wedge \ \Delta_i \geq \theta_\Delta \ \wedge \ i \neq i_\star. \tag{4.7}$$

4.4.5 Technology Stack: Facilitation Agent Runtime and Meeting Client

Für die Entwicklung des AI-Facilitators wurden LangChain und LangGraph als zentrale Frameworks ausgewählt. LangChain bietet eine modular aufgebaute Toolbox für die effiziente Entwicklung von LLM-basierten Anwendungen, während LangGraph die Definition und Steuerung agentenbasierter Workflows ermöglicht.

Für die Durchführung der User Study wurde eine eigens entwickelte Webanwendung eingesetzt, die als einfache Videokonferenzlösung fungiert. Diese wurde mit Svelte, SvelteKit und Tailwind CSS implementiert und auf einem Hetzner-Server gehostet. Es wurde mediasoup als WebRTC-Server und als SFU eingesetzt.

Teilnehmer:innen konnten über einen teilbaren Link direkt auf die Anwendung zugreifen. Der gewählte Ansatz ermöglichte eine direkte Analyse der Nutzerinteraktionen sowie eine optimierte User Experience (UX). Alternativ hätte die Integration über bestehende Videokonferenz-Tools (z. B. als Bot via API) in Betracht gezogen werden können. Allerdings wäre dies mit den Limitierungen der jeweiligen APIs verbunden gewesen und hätte den Lösungsraum von vornherein eingeschränkt.

4.4.6 UX und UI of Meeting Client

Dies ist nciht der Fokus dieser Arbiet, weshlab sich bei dem MeetingClinet auf bestehdne UI von bewährten Meeting Tools orienterit wrude. Es wurde ledglich deutlicher hervorgehoben, wenn die KI spricht. Hier bei wurde der gesamt hintergrund der anwedung Blau. Ein Singalto kündigte an, wenn die KI psrechen wollte oder fertig war. So gab es klare visuelle cues für den Nutzer, wann die KI spricht. Die LLM pipeline hat tielweise eine latzen von bis zu 5 Sekunden. Druch die Visuellen Cues konnte den User bereits führer singalisert werden, dass die KI bereit ist zu sprechen.

Außerdem gab wurd ein KI-Switch für sämtliche KI-Features für jeden Teilnehmende, falls es aus irgendweinem grund schnell notwendi gwar, die verbidnung zu jeglichen exteren Dienst zu kappen.

4.5 Limitations of the Prototype

Eingegrenzter Funktionsumfang

Der Prototyp ist im Funktionsumfang bewusst begrenzt und als Minimum Viable Product primär auf Hidden Profiles zugeschnitten, damit Fokus und anschließende Validierung zurechenbar bleiben (vgl. subsection 2.2.4). Diese Eingrenzung setzt voraus, dass sich das Phänomen isoliert betrachten lässt. Diese Voraussetzung ist vermutlich unterkomplex: In realen Meetings überlagert sich der Shared-Information-Bias mit Status- und Hierarchieeffekten, Confirmation Bias und Groupthink sowie Zeitdruck, thematischer Abweichung und unklarer Zielsetzung (vgl. **??**). Dieses Zusammenwirken deckt die vorliegende Version, wenn überhaupt nur implizit ab. Zusätzlich sind mit dem Fortschritt allgemein verfügbarer KI-Systeme Menschen daran gewöhnt, frei mit Sprachmodellen zu sprechen und Antworten auf beliebige Anliegen zu erwarten. Diese Erwartungshaltung kann das enge Repertoire des Prototyps übersteigen. Teilnehmende könnten Antwort- und Interaktionsmöglichkeiten voraussetzen, die das System nicht bereitstellt und somit das vertrauen, auf Grund dieser wahrgenommenen Lücke, auch in bestehende, eigentlich nicht mangelhafte Funktionen des Systems verlieren.

Unimodale Wahrnehmung

Das System arbeitet ausschließlich auf dem Transkript. Transkriptionsfehler, etwa bei undeutlicher Aussprache oder schlechter Mikrofonqualität, gehen direkt in den Kontext der LLM-Pipeline ein und können Gedanken und Interventionen verzerren. Prosodie (Betonung, Aussrpache Tonfall) und visuelle Signale wie Mimik, Blick und sichtbare Aufmerksamkeit oder Desinteresse fehlen. Facilitatoren nutzen jedoch genau diese Signale, um zu lesen, wie eine Aussage gemeint ist und ob jemand präsent ist (vgl. chapter 3). Eine Sentimentanalyse auf dem Audiostream wurde nicht implementiert. Das ist eine interpretative Lücke. Sie steht aber im Einklang mit den in den Interviews geäußerten Bedenken gegenüber Kontrolle und Überwachung.

Laufzeitmechanik

Die Thought-Pipeline ist mehrstufig und probabilistisch. Interventionen können deshalb verzögert eintreffen, und dieselbe Gesprächslage kann unterschiedlich ausfallen. FIM-Scores werden einmal vergeben und nur deterministisch gealtert. Sie werden nicht semantisch neu bewertet (vgl. Table 4.4.4).

Dazwischenreden

Spricht eine teilnehmende Person in eine laufende Äußerung des Agenten hinein, bricht das System diese nicht ab. Der bereits ausgewählte Gedanke wird bis zum nächsten menschlichen Redeanteil zu Ende gesprochen. Erst dieses neue Element stößt die Pipeline erneut an, sodass ein weiterer Gedanke gebildet und geäußert werden kann. Eine überlappende Wortmeldung verändert den laufenden Beitrag daher nicht und kann zu merkwürdigen bzw. unnatürlichen Interaktionen führen, wenn Teilnehmende weitersprechen, während der AI-Facilitator spricht.

Eigene Meetingplattform

Das Meeting läuft in einer eigens gebauten Webanwendung und nicht in etablierten Meetingplattformen wie z.B. Zoom oder Teams. Das sichert Kontrolle über Interaktion und UX. Jedoch sind Teilnehmende diese neue Anwendung im Gegensatz zu den bekannten Tools nicht gewohnt. Die Vergleichbarkeit mit der bekannten Nutzungssituation ist deshalb eingeschränkt.

Modifizierte Turn-Allocation-Prediction

Die Vorhersage, wer als Nächstes spricht, stützt sich auf Liu et al. [43]. Der Prompt musste jedoch von Alltagsdialogen auf den Facilitationskontext angepasst werden. Deshalb kann nicht davon ausgegangen werden, dass sich die Ergebnisse aus dem Paper 1 zu 1 auf diesen Kontext übertragen lassen. Eine genauere statistische Validierung wäre empfehlens wert, war aber mangels eines geeignene domänenspezifisches Korpus aus Facilitationssettings mit Turn-allocation Notation, nicht möglich.

Annahme gleicher Redeanteile

Der Participation Balancer setzt voraus, dass eine ausgeglichene Verteilung der Redezeit dem Meeting zuträglich ist (vgl. (4.5)). Diese Voraussetzung gilt für das evaluierte Entscheidungsformat, in dem ungeteilte Informationen nur durch Wortmeldungen in die Diskussion gelangen. Sie gilt nicht für Formate, deren Zweck eine ungleiche Redezeit ist, etwa Präsentationen oder längere Beiträge von Expertinnen und Experten. Dort würden Curtailment und Invitation auslösen, obwohl die ungleiche Verteilung beabsichtigt ist.

5 Empirical Prototype Validation (RQ2 & RQ3)

5.1 Methodology: Hidden-Profile Scenario and Questionnaire Instrument

5.1.1 Study Design

The empirical evaluation of the proactive AI assistant required a group task that resembles the convergent organizational decisions this thesis addresses and that can generate situations in which facilitation may become relevant.

Houtti et al.[5] evaluated their virtual co-host with the ”Lost at Sea” activity, in which participants rank 15 items by importance for survival after a shipwreck. They selected this task in part because it does not require equal participation and can be completed individually [5]. For the present study, these properties are a limitation. A ranking of a static, commonly known list does not require participants to pool unique information. As a result, the discussion is unlikely to produce the information-related difficulties that characterize many organizational decision meetings, and therefore few moments in which an intervention/facilitation would be necessary in order to reach a well-founded group decision.

As discussed in Chapter 2, relevant knowledge in organizational collaboration is often distributed asymmetrically across participants. An adequate solution in a convergent decision process then depends on the group recognizing, sharing, and integrating information that is initially held by only some members. Hidden-profile experiments operationalize this asymmetry.

The evaluation uses the hidden-profile hiring simulation published by Baker [27]. In this scenario, participants receive differently distributed information about three candidates, and the information that is initially shared favors a suboptimal choice. They must jointly decide whom they consider most suitable. The task is therefore expected to elicit the interaction patterns a proactive moderator should detect: premature consensus, unique information that remains unmentioned, unbalanced participation, and a drift of the discussion towards commonly known talking points. This specific scenario was selected for three further reasons:

- **Feasibility in a remote meeting.** The task can be administered in groups of four and completed in a compact session of approximately 30 minutes. The materials can be distributed digitally, which allows the discussion to be conducted remotely.
- **Fit with the intended intervention situations.** The information structure of the task is expected to produce the patterns described above and thus occasions on which the prototype might intervene.
- **Replicability.** Baker provides the full experimental materials with the article, including role information, participant-specific information sheets, and evaluation forms, so that the scenario can be reused in the present study.

Nach einer kurzen Einleitung, haben die Teilnehmenden ihren Informationsbogen mit den stckbreifen der kandidaten erhalten. Sie hatten 10min zeit sich alles drucchzulesen und erstml zu einer individuellen entscheidung zu kommen.

Nach den 10min kamen sie in der gruppe zusammen und hatten 15-20min zeit gemeinsm zu einem ergeniss zu kommen.

Im anschluss sollten die Teilnehmden direkt einen Fragebogen ausfüllen.

Meeting success was assessed with the questionnaire instrument revalidated by Davison for the systematic measurement of meeting outcomes [6] (section 7.6). The instrument covers five dimensions: communication, quality of discussion, status effects (power structures and hierarchies), teamwork, and efficiency. These dimensions correspond to the aspects of group interaction that the hidden-profile task is intended to place under strain. And are the same dimensions the FIM-Score are based on. (see. **??**)

The technology-related items of the instrument, however, were formulated for Group Support Systems of that period, that is, for largely passive, computer-supported collaboration tools. They do not address autonomous, proactive AI agents. In addition, Houtti et al. found that participants rated meetings with their virtual co-host more positively overall, yet tended to rationalize or ignore its critical feedback [5]. Items on interaction with a proactive AI agent were therefore added (see Questionnaire Items A1-A7 in section 7.6). Free-text questions were appended after these items in order to collect qualitative accounts of how participants experienced the agent's contributions (see Questionnaire Items F1 and F2 in section 7.6). These answers were coded with the codebook in section 7.7.

5.1.2 Sample

Die Teilnehmden waren zwishen 23 und 35 Jahre alt (Durchschnitt: 27,7 Jahre; Median: 27 Jahre). 3 der Teilnehmden waren weiblich, 9 männlich. Alle teilnehmden haben zuvor bereits regelmäßig an virutleln meetings im arbeitskontext teilgenommen.

5.1.3 System einstellungen während der user studie und die überlegungen dahinter

owohl die literatur draufhindeutet, dass solceh KI-gestützen meeting systeme eher unter kontroll des users stehen sollten, so dass der user z.B. den grad der interaktion einstellen kann, oder das system ausstelen kann, wurde hier gezielt davon abgesehen. das rsikio zur overtalken war beaknnt. die thresholds des systems wurden auf X und Y gesetzt. druch die automatiserte test wurde versuch ein angemessene werte zufinden, doch da dies nun der erste druchlauf war, fehlten tatsähclie refernz werte aus live-meeting situatioen. die thresholds wurden bewusst tendenziell etwas zu hoch angesetzt, um sicher zustellen dass das system überhaupt interagiert. bei zu niedrigen thresholds hätte es passieren können dass das system kaum oder gar nicht interagiert, was zu keinen beobachtbaren verhaltensweisen der teilnehmden geführt hätte.

Das syxtem hate ganz bewusst also kontext nicht die informatione zu den kandidaten. Facilitatoren sind nicht die fach experten und ein system sollte somit dies nicht tut. außer dem würde dies eine akademisch konruiiertes szenario erzeugen, welches keine praktische relevanz hat, da in der realität ständig Leute an meeting tilenhemn welche ungetielte private information haben so sonst nirgend dokumentiert und somit für ein KI system auch außerhalb des kontext liegen würde.

5.2 Results

Für die auswerung wurde unteanderem

5.2.1 Observed Intervention Dynamics

Es haben sich 3 unterschiedliche dynamiken mit dem Umgang der KI entwicklet. Gruppe eiins hat begeonnen irendwann die KI hauptsächlich zu ignorieren und während die KI gesprochen hat einfach weiter gesprochen.

Gruppe 03 war irritiert von der KI. und hat eine latenz beim denken vermuetet, weshlab sie stellen weise dazu übergegangen sind auf die KI zu warten. Insbesondere, wenn die KI eine frage gestellt hatte, schienen die Teilnehmden auf eine einordnung der KI zu warten nach dem die antowrt gegeben wurde. Nach ausbleiben der antwort ist die gruppe dann nach einer kurzen pause weider in die diskussion übergegangen.

5.2.2 Perceived Meeting Quality (Davison Instrument)

Comparison across groups

The median scores of the groups on the Likert-scale items were compared across the three user studies. Differences of less than 1.5 scale points were considered insignificant. No larger difference was found, except on four items:

- D2 (appropriate–inappropriate), with medians of 4.5 (G01), 3 (in G02) and 5 (in G03);
- T2 (the members worked together as a team), with medians of 5 (G01), 4.5 (G02) and 3 (G03);
- E2 (meeting time was used efficiently), with medians of 5 (G01), 4.5 (G02) and 2.5 (G03)
- A4a (AI intervened too rarely.), with medians of 5 (G01), 4.5 (G02) and 3 (G03);

These discrepancies do not show that one group differed from the others across the entire Davison instrument. On every other Likert item the three group medians stayed within 1.5 scale points, and the lower median is not the same group in each case: G02 on D2, and G03 on T2, E2, and A4a. The four gaps therefore concern different groups and point in different directions, which is why the ratings are still plotted together.

Communication

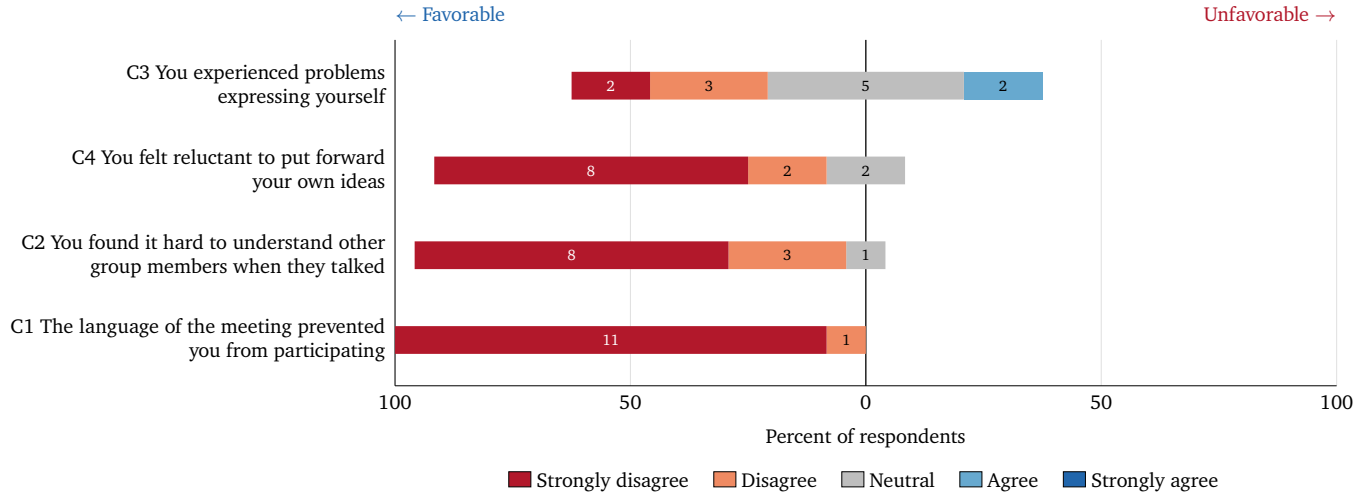


Figure 16: Perceived communication (Davison items C1–C4). Each bar shows the share of responses on the five-point scale. – *Source: Own illustration.*

Quality of discussions

With regard to all the meeting members as a whole, how would you rate the discussions in the meeting in terms of the following scales?

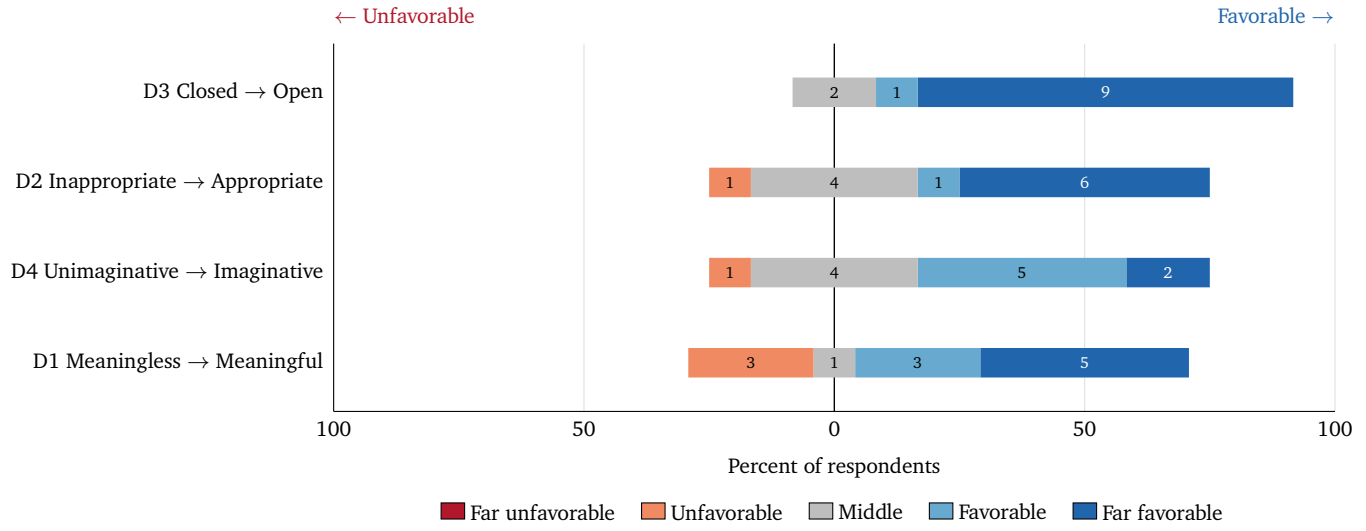


Figure 17: Perceived discussion quality (Davison items D1–D4). Each bar shows the share of responses on the five-point semantic differential. The marker above the scale names the unfavorable and favorable ends. Counts are illustrative. – *Source: Own illustration.*

Status effects

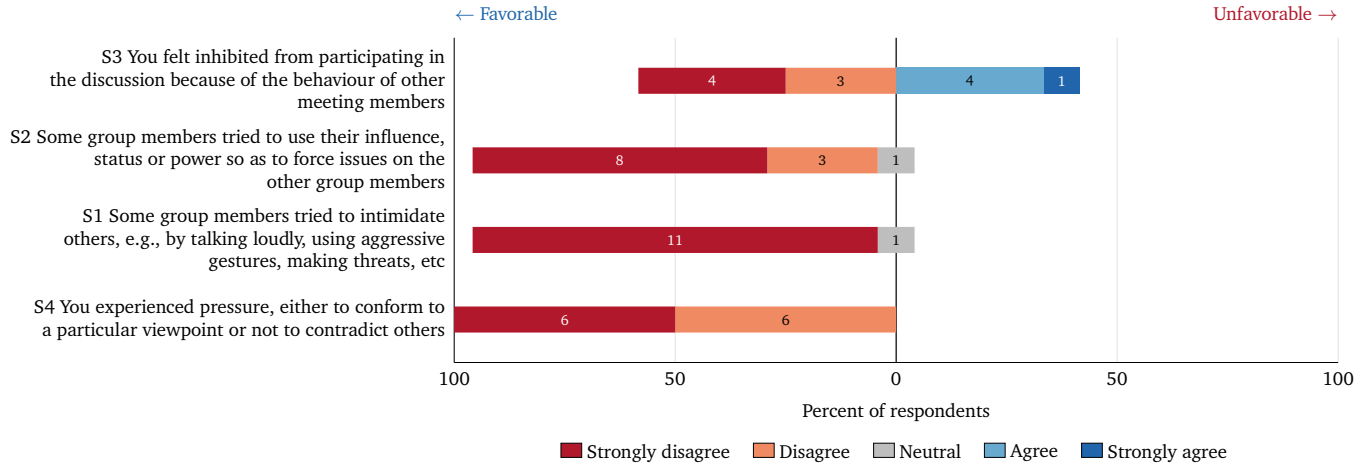


Figure 18: Perceived status effects (Davison items S1–S4). Each bar shows the share of responses on the five-point scale. – *Source: Own illustration.*

Team work

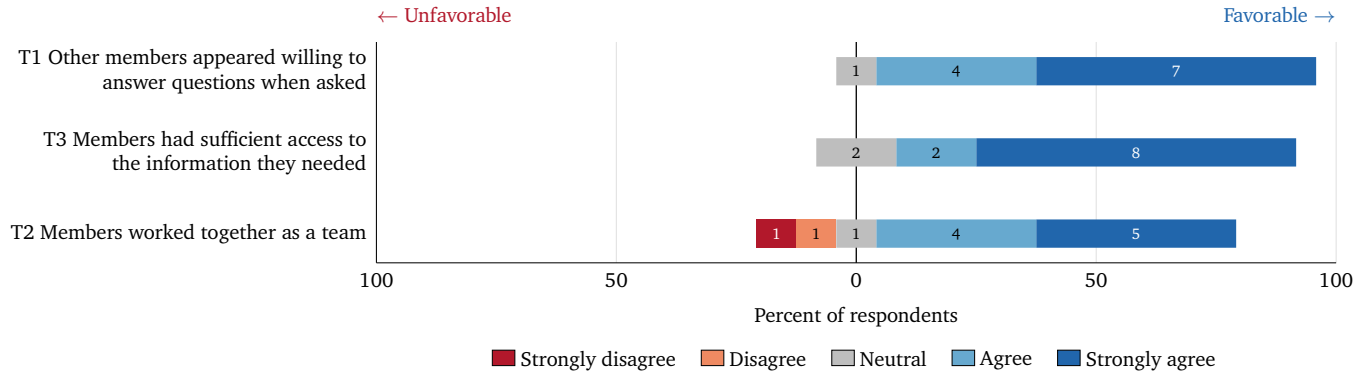


Figure 19: Perceived teamwork (Davison items T1–T3). Each bar shows the share of responses on the five-point scale. – *Source: Own illustration.*

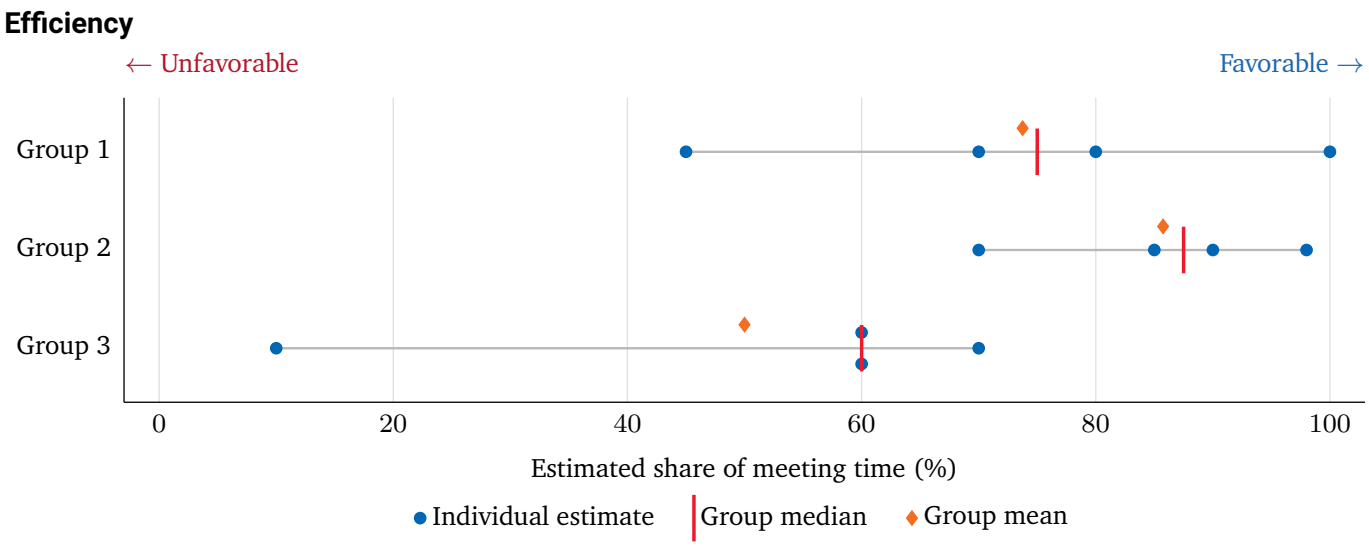


Figure 20: Estimated share of meeting time spent in serious discussion (Davison item E4). Each dot is one participant. The line spans the lowest and highest estimate in that group. The vertical mark is the group median, and the diamond is the group mean. – *Source: Own illustration.*

5.2.3 Participants' Perception of the Proactive AI Moderator's Interventions

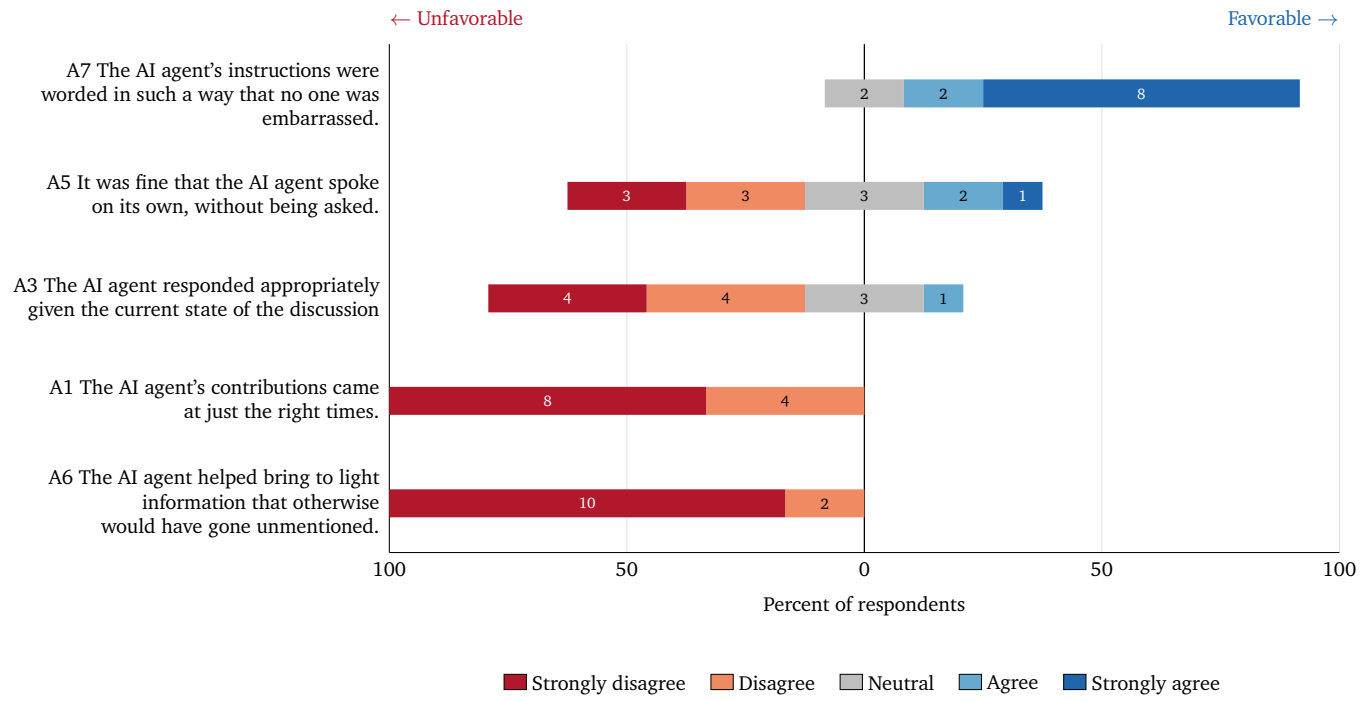


Figure 21: Perceived attributes of the proactive AI agent (items A1, A3, A6, A7, and A8). Each bar shows the share of responses on the five-point scale. Counts are illustrative. – *Source: Own illustration.*

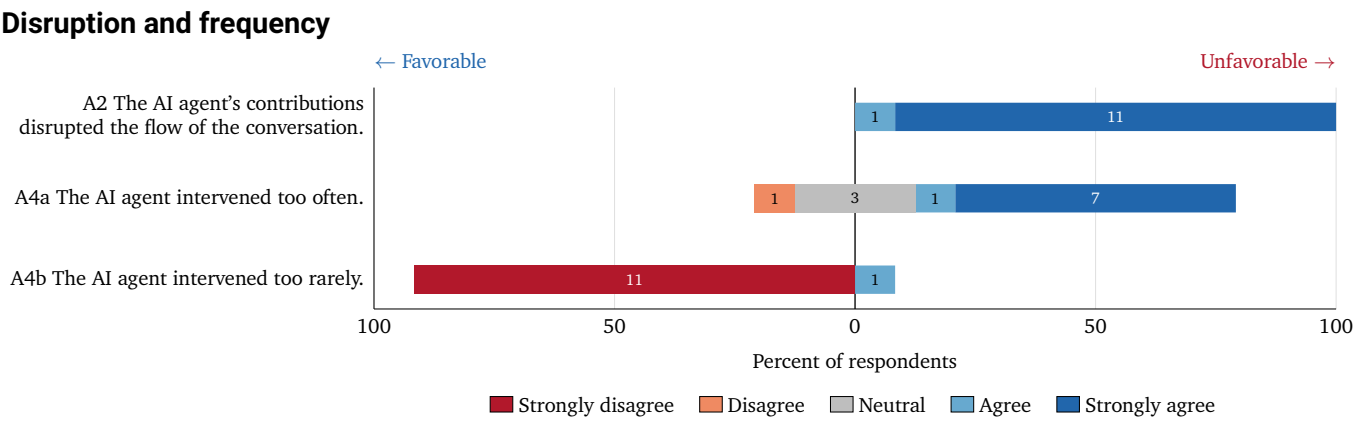


Figure 22: Perceived disruption and intervention frequency of the proactive AI agent (items A2, A4a, and A4b). Each bar shows the share of responses on the five-point scale. Counts are illustrative. – *Source: Own illustration.*

5.3 Discussion

Obwohl eine prokaitve KI vereinzelt zu neuen Perspektiven innerhlab des Meeting führt und tatsächlic in eingen fälle den Teilnehmden geholfen hat die Redeantiele fairer zu verteielte und auf Themen aufmerksam gemacht hat die sonst untergegagen wären, überweigt in diesem System entwurf der Negative impakt. Nahe zu alle Teilnehmden empfanden die KI als störend oder sahen, dass die KI sich zu oft beteiligt.

diese nachteiel überweigen den geringen positiven effekten, welche das system auf das meeting hatte.

Die shclussfolgernung, dass ein proaktiver KI-Faciliatoar generell nicht sinnvoll ist, wäre allerdings zuverführt.

Druch anpassen der Therholds könnte die interaktions rate der KI veringert werden und auf die meeting situation angepast werden.

Der reltiv navier Ansatz die KI selbst ihre gendaen mit einer genreischen Bewertunglogik zu bewerten, fürhte zu generischen Antworten mit immer ähnlichen pattern. (Erst bedanekn, dann paraphrasieren) dies ist zwar aus den schilderunge der experten interviews modleiert, wruden dann dennoch in der erhöhte frequenz als nachteil empfunden.

Ein PRoaktiver ansazt benötigt deutlich mehr genauigkeit. im zeifel wäre es vermutlich besser gewsen, wenn die KI nur punktuell eingegirffen hätte. anstatt permant neue Gedanken zu formen udn zu evaluieren. Ein die geformten gednekn hatten ohnehin eine hohe ähnlichkeit, weshalb der merhwert im faciliatoatr zenario infrage gestellt werden kann.

Aus den beobachtungen diese Prototypen udn der user studies leitet der Autor folgende empfheungen ab:

DEr versuchsaubgau war gut geeiengt um ein solches system unter live bedinugnen zu testen. allerdings hätten davor erstmal einzelen funktionen isoliert geteste werden sollen. Ein navierer ansatz und vermutlich zielf+hrender wäre z.B. nur den context workflow laufen zulassen. und die inner thougs komplett wegzulassen. dafür hätte es noch die notwendigkeit gegeben, den ki agenent zu unterbrechen.

5.3.1 Identified Key Challenges and Potentials

The sessions show both facilitative effects and a predominant experience of disruption. In isolated moments the agent introduced a perspective the group had not taken up, helped to distribute speaking time more evenly, or drew attention to a point that would otherwise have dropped out of the discussion. Nearly all participants nevertheless experienced the agent as disruptive, or judged that it took the floor too often. These small positive effects on the meeting were outweighed by that cost. Whether proactive facilitation is worthwhile beyond this design remains open. The sessions identify a specific set of challenges in this prototype, and a smaller set of potentials that the same meetings made visible. Design consequences are taken up in the next subsection.

Key challenges

- **Participation outruns its usefulness.** The agent spoke often enough that disruption dominated the meeting. A stricter *imThreshold* and a more reluctant interrupt policy would lower that rate, and would leave most generated thoughts unspoken. What the sessions call for is a setting at which a thought is voiced only when it is relevant in the current moment.
- **Generic scoring produces a generic move.** Having the model rate its own thoughts with a generic evaluation logic yielded utterances of a stable shape: thank the previous speaker, then paraphrase. The pattern was modeled on the expert interviews. At the frequency observed here, participants experienced that repetition as a drawback.
- **Continuous thought generation adds little distinct content.** A proactive facilitator needs substantially higher precision than this prototype reached. Punctual intervention would have fitted the meetings better than the permanent formation and evaluation of new thoughts. The thoughts that were formed were highly similar to one another, which limits the additional value of that stream in the facilitator setting.
- **Aging is too weak against overtalking.** The design already treated overtalking as a risk, and aging was meant to discourage repetition of the same thought. That mechanism was not sufficient. A cooldown on speech in general would likely have raised the perceived quality. The agent might then miss a relevant moment. That trade-off is acceptable if the lower rate raises acceptance.
- **aging reicht nicht aus um aktualität zu bewahren** die annahme dass ein gedanke an reelvanz verliert je länge er zurückliegt ist vermutlich zu vereinfacht. ein gedanke kann beresits wenige sekunden später komplett an relevanz verloren haben. z.B. weil die gruppe von selbst genau das macht, was der gedanke versucht hat zu bezwekcen. Aging alleine reich tnicht aus. einzusätzlicher mechanismus, der vor der aussprache bewertet ob der gedanke immernoch relevant ist wäre von vorteil. allerdigns müsste diese evaluierung zeimlich schnell gehen. ggf. könnte ein binäre evaluierung mit klassivieren wie Jev oder der desission api von open ai hier ausreichen um eienn gedanken schnell als immer noch relevant bzw. nciht mehr relevant zu klassifizeiren.
- **Live speech exposes latency that chat tolerates.** The Inner Thoughts framework was conceived for chat. There, a message that fits the current context only approximately is ordinary, because typing takes time and the exchange is asynchronous. In a live meeting the same delay is conspicuous. Participants notice when a contribution does not arrive with the moment it refers to.

- **Turn allocation seriously jeopardizes the meaningfulness of the intervention.** Predicting the correct allocation of speaking turns is difficult even in normal group situations. However, a facilitator is not a normal group member who participates in the discussion on an equal footing. The facilitator holds a special position. Some participants have described the AI—a separate entity—as giving them the impression that it wants to push through its own agenda. This is an inappropriate intervention. If the AI facilitator speaks at the wrong times or speaks excessively, this can erode trust in it to such an extent that participants begin to ignore it within a very short time.
- **The thought trace is expensive to interpret.** Spoken thoughts were consistently similar, and monitoring that many traces is laborious. A platform such as Langfuse was highly useful during development, because it supported an intuitive understanding of the running system. A later analysis needs the logging scheme to be set up for that analysis from the start.

Identified potentials

- **The intended facilitative functions are reachable.** In individual cases the agent opened a new perspective, supported a fairer distribution of speaking time, and brought up a topic that would otherwise have been passed over. These effects were too rare to offset the disruption. They show that the functions the prototype was built for can occur in a live discussion.
- **Thresholds can fit the agent to the meeting.** The interaction rate is a parameter of the policy, and it can be adapted to the situation of a given meeting. The useful operating point is the one that keeps the rare, relevant contribution and withholds the rest.
- **A narrower pipeline is a plausible next step.** Running only the context workflow, and omitting the continuous inner-thought loop, is a simpler arrangement and, on the present evidence, a more promising one. That design needs a way to interrupt the agent, so that a contribution can be stopped once the discussion has moved on.
- **Agenda and phase make relevance decidable.** Contributions improve when a concrete agenda is available and the current item states what is particularly relevant. A finer division into phases makes the same constraint operational. In the hiring task, the facilitator could have proposed a procedure at the outset: collect advantages and disadvantages of the candidates in turn, then decide which candidate fits best. An agent bound to such a structure can be judged by whether it stayed with the characteristics of the current phase. The task of the facilitator then reduces to a few explicit points, which is likely to produce a larger effect than a generic reaction to any situation. The heuristics of the inner-thought model can be adapted to those phases, and the transitions between phases become the moments that matter most.
- **The live setup is a usable test bed.** The study arrangement was well suited to observing the system under live conditions. It would have been stronger if individual functions had first been tested in isolation, and if the weaknesses visible after the first group had been corrected before the next group used the system. Shorter iteration phases fit this kind of prototype evaluation.

5.3.2 Lessons Learned: Implications for similar System Designs

Im druchschnitt hat die KI alle x minuten interveniwert. man könnte dies ggf. darduch reduzieren, dass der interrupt und der imThreshold heruntergesetzt werden. Allerdings würde dies dazuführen, dass die meisten gedanekn unausgesprochen bleiben. Dies wirk erstmal verschwenderisch. es müsste eine gleichgewicht gefunden werden dass wirklich relevante gedanekn in dem aktuellen moment angepsrochen werden. andere wiederum

Besser kontext, wenn eien genaua agend vorliegt und facilitatorin anweiseungen was in diesm agenda puntk besonders relevant ist. das könnten die antworten verbesseren. Hinzukommt auch eine grnularere ein teilung in phasen. in diesem beipsiel hätte z.B. der Facilitator am anfang ein vorgehen vorschalgen können. "Wir sammeln Reihum die vor und nachteile der kandidaten und entscheiden dann welcher kandidat am besten passt."

Zukünfige arbeiten sollten nicht versuchen einen generischen Facilaitor zu entwciklen, welche passend reagiert, sondern sich bewährte strukturen nutzen diesn ablauf festlegen, dann die eingschaften und wichtigen Punkte für die Phasen dieser strukur ableiten und klar formulieren. ein KI-Facilitator würde sich dann auf. somit kann dann besser evaluiert werden, ob der facilitator sich an die merkamle dieser phasen gehalten hat. die reduziert die aufgaben des fecilaitoas auf wenige haupt punkt und würde vermutlich einen größern impakt erzeugen.

Es wäre dann möglihc die heurisiken desinner thoughs models an diese granualren phasen anzupassen.

[TODO: Abbildung mit phasen, wie ein weiteres System design aussehen könnte.]

Im Zuge dessen wäre es dann entscheident das logging der LLM-Traces zu verbessern, um das system besser evaluieren zu können.

entscheidender währe dann die übergänge zwischnen den phasen.

Es gibt neben effekte die das evaluaieren des systems erschweren: die trun allocation könnten fälschlicherweise den faciliatoatr zu weisen, dann reicht der imThreshoad um einen gedanken zu äußern. da sich die gedanekn aber warhshcienlcih eh, um ein relevantes theam drehen führt das nicht zwangsläufig zu eienr falschen intervention. Sei kann trotzdem passend sein. (ggf. Beispiele ergänzen)

gedanken die geäußert wurden , waren immer sehr ähnlich. und die auswertung udn das monitoring von sovielen traces ist sehr aufwendig. vereinfachen und gesagtes buffern udn dann in einem batch verarbieten.

Obwohl das problem des overtalking im system design erkennt wurde und druch dei ageing funktion veruscht wrude das äußer des selben gendekn zu ferindern, war dies nicht aussreichend. Vermutlich hätte eine cooldown auf die sprache generell die qualität verbessert. Zwar hätte die ki hier relevante informatione verpassen könenn, allerdings wäre dies ein tradeoff welche hinnehmbar wäre, wenn dadruch die akzeptanz steigen würde

Live interaktion utnerleigen einen höhren anspruch als chat interaktionen. das inner thogs framewokr wurde als chat framework konzipiert. in chats ist es ünlich das eine nachricht auchmal nicht ganz genau auf den aktuelln kontext passt. (user sind es geäht, dass das tippen dauert und es sit eien asyncrone interaktione). bei live interaktione ist die ander. teilenhmden fallen latzen auf. wenn nicht direkt informationen

Tracebility ist aufwändig und schwer zu interpretieren. eine platform wie langfuse hat während der Entwikclung enorm geholfen ein intuitives verständniss für das system zu entwikclen. Jedoch sollte sich gut gedanken gemacht werden wie die traces in einer solch platform gelogget werden, um sie später zu analysieren.

bereits nach der ersten interation des Systems wruden deutlich schwachstellen ersichtlich. der besser ansatz wäre gewesen das system direkt zu verbessern und dann die nächste user gruppe drauf zu setzen. für diese prototpy evaluierung wären kleiner phasen ggf besser gewesen.

6 Conclusion

6.1 Summary of Findings

6.2 Limitations

6.3 Future Work

Bibliography

[1] *DeepL AI Platform: Translation, Voice & API*. URL: https://www.deepl.com/en (visited on 09/01/2026).

[2] Margaret Beranek, Catherine Beise, and Fred Niederman. "Facilitation and Group Support Systems". In: Feb. 5, 1993, 199–207 vol.4. ISBN: 978-0-8186-3230-3. DOI: 10.1109/HICSS.1993.284182.

[3] Atlassian. *Workplace Woes: Meetings*. Inside Atlassian. URL: https://www.atlassian.com/blog/workplace-woes-meetings (visited on 08/26/2026).

[4] Stephen Hayne. "The Facilitators Perspective on Meetings and Implications for Group Support Systems Design". In: *The DATA BASE for Advaces in Information Systems* Summer-Fall 1999 (Val. 30, No. 3,4) (Summer-Fall 1999 (Val. 30, No. 3,4)).

[5] Mo Houtti et al. "Observe, Ask, Intervene: Designing AI Agents for More Inclusive Meetings". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. CHI 2025: CHI Conference on Human Factors in Computing Systems. Yokohama Japan: ACM, Apr. 26, 2025, pp. 1–18. ISBN: 979-8-4007-1394-1. DOI: 10.1145/3706598.3713838. URL: https://dl.acm.org/doi/10.1145/3706598.3713838 (visited on 05/24/2026).

[6] Robert Davison. "An Instrument for Measuring Meeting Success: Revalidation and Modification". In: *Information & Management* 36.6 (Dec. 1, 1999), pp. 321–328. ISSN: 0378-7206. DOI: 10.1016/S0378-7206(99)00026-9. URL: https://www.sciencedirect.com/science/article/pii/S0378720699000269 (visited on 08/30/2026).

[7] Thomas A. Kayser. *Mining Group Gold : How to Cash in on the Collaborative Brain Power of a Group*. El Segundo, Calif. : Serif Pub., 1990. 214 pp. ISBN: 978-1-878567-02-4. URL: http://archive.org/details/mininggroupgoldh0000kays (visited on 10/19/2026).

[8] N.C. Romano and J.F. Nunamaker. "Meeting Analysis: Findings from Research and Practice". In: *Proceedings of the 34th Annual Hawaii International Conference on System Sciences*. The 34th Annual Hawaii International Conference on System Sciences. Jan. 2001, 13 pp.–. DOI: 10.1109/HICSS.2001.926253. URL: https://ieeexplore.ieee.org/abstract/document/926253 (visited on 10/19/2026).

[9] Jennifer L. Geimer et al. "Meetings at Work: Perceived Effectiveness and Recommended Improvements". In: *Journal of Business Research* 68.9 (Sept. 1, 2015), pp. 2015–2026. ISSN: 0148-2963. DOI: 10.1016/j.jbusres.2015.02.015. URL: https://www.sciencedirect.com/science/article/pii/S0148296315000879 (visited on 10/19/2026).

[10] *Doodle: State of Meetings Report 2019*. 2019. URL: https://drive.google.com/file/d/1DX7Zhi1ymnh_Yz7pks5ahKlw2-5DdYog/view?usp=sharing&usp=embed_facebook (visited on 05/26/2026).

[11] Xinyue Chen et al. “Are We On Track? AI-Assisted Active and Passive Goal Reflection During Meetings”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. CHI 2025: CHI Conference on Human Factors in Computing Systems. Yokohama Japan: ACM, Apr. 26, 2025, pp. 1–22. ISBN: 979-8-4007-1394-1. DOI: 10.1145/3706598.3714052. URL: https://dl.acm.org/doi/10.1145/3706598.3714052 (visited on 02/26/2026).

[12] Ava Elizabeth Scott, Lev Tankelevitch, and Sean Rintel. “Mental Models of Meeting Goals: Supporting Intentionality in Meeting Technologies”. In: *Proceedings of the CHI Conference on Human Factors in Computing Systems*. CHI ’24: CHI Conference on Human Factors in Computing Systems. Honolulu HI USA: ACM, May 11, 2024, pp. 1–17. ISBN: 979-8-4007-0330-0. DOI: 10.1145/3613904.3642670. URL: https://dl.acm.org/doi/10.1145/3613904.3642670 (visited on 08/26/2026).

[13] Simone Kauffeld and Nale Lehmann-Willenbrock. “Meetings Matter: Effects of Team Meetings on Team and Organizational Success”. In: *Small Group Research* 43.2 (Apr. 2012), pp. 130–158. ISSN: 1046-4964, 1552-8278. DOI: 10.1177/1046496411429599. URL: https://journals.sagepub.com/doi/10.1177/1046496411429599 (visited on 05/26/2026).

[14] Margaret A. Hughes and Deb Roy. “Keeper: A Synchronous Online Conversation Environment Informed by In-Person Facilitation Practices”. In: *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. CHI ’21. New York, NY, USA: Association for Computing Machinery, May 7, 2021, pp. 1–14. ISBN: 978-1-4503-8096-6. DOI: 10.1145/3411764.3445316. URL: https://dl.acm.org/doi/10.1145/3411764.3445316 (visited on 09/05/2026).

[15] Fran Ackermann. “Group Support Systems: Past, Present, and Future”. In: *Handbook of Group Decision and Negotiation*. Springer, Cham, 2021, pp. 627–654. ISBN: 978-3-030-49629-6. DOI: 10.1007/978-3-030-49629-6_47. URL: https://link.springer.com/rwe/10.1007/978-3-030-49629-6_47 (visited on 09/13/2026).

[16] Alan R. Dennis et al. “Information Technology to Support Electronic Meetings”. In: *MIS Quarterly* 12.4 (Dec. 1, 1988), pp. 591–624. ISSN: 0276-7783, 2162-9730. DOI: 10.2307/249135. URL: https://misq.umn.edu/misq/article/12/4/591/992/Information-Technology-to-Support-Electronic (visited on 10/18/2026).

[17] *Electronic Meeting System*. In: *Wikipedia*. URL: https://en.wikipedia.org/w/index.php?title=Electronic_meeting_system&oldid=1304506204 (visited on 10/18/2026).

[18] George P. Huber. “Issues in the Design of Group Decision Support Sytems”. In: *Management Information Systems Quarterly* 8.3 (Sept. 1, 1984), pp. 195–204. ISSN: 0276-7783. DOI: 10.2307/248666. URL: https://doi.org/10.2307/248666 (visited on 09/13/2026).

[19] SessionLabs. *Design Faster with AI*. SessionLab. URL: https://www.sessionlab.com/features/ai-assistant/ (visited on 10/19/2026).

[20] Alma M. Whiteley and Joseph E. Garcia. “The Facilitator and the Chauffeur in GSS: Explorations in the Forging of a Relationship”. In: *Group Decision and Negotiation* 5.1 (Jan. 1, 1996), pp. 31–50. ISSN: 1572-9907. DOI: 10.1007/BF02404175. URL: https://doi.org/10.1007/BF02404175 (visited on 10/18/2026).

[21] Roger M. Schwarz. *The Skilled Facilitator: A Comprehensive Resource for Consultants, Facilitators, Coaches, and Trainers*. Hoboken, New Jersey: Jossey-Bass, Nov. 14, 2016. 416 pp. ISBN: 978-1-119-06439-8.

[22] Parsons Dialogue. *What Makes a Good Facilitator? | Parsons Dialogue*. URL: https://www.parsonsdialogue.com/fundamentals/what-makes-a-good-facilitator (visited on 09/15/2026).

[23] AJ&Smart. *Facilitation Fundamentals Workbook*.

[24] *Convergent vs. Divergent Thinking*. Jan. 5, 2024. URL: https://www.youtube.com/watch?v=kh4c3R9mSkg (visited on 08/30/2026).

[25] Arthur Cropley. "In Praise of Convergent Thinking". In: *Creativity Research Journal* 18.3 (July 1, 2006), pp. 391–404. ISSN: 1040-0419. DOI: 10.1207/s15326934crj1803_13. URL: https://doi.org/10.1207/s15326934crj1803_13 (visited on 08/31/2026).

[26] Sam Kaner. *Facilitator's Guide to Participatory Decision-Making*. 3rd ed. San Francisco, CA: Jossey-Bass a Wiley Brand, 2014. 434 pp. ISBN: 978-1-118-40495-9 978-1-306-68501-6 978-1-118-42195-6.

[27] Diane F. Baker. "Enhancing Group Decision Making: An Exercise to Reduce Shared Information Bias". In: *Journal of Management Education* 34.2 (Apr. 2010), pp. 249–279. ISSN: 1052-5629, 1552-6658. DOI: 10.1177/1052562909343553. URL: https://journals.sagepub.com/doi/10.1177/1052562909343553 (visited on 08/31/2026).

[28] Dennis D. Stewart and Garold Stasser. "Expert Role Assignment and Information Sampling during Collective Recall and Decision Making". In: *Journal of Personality and Social Psychology* 69.4 (1995), pp. 619–628. ISSN: 1939-1315. DOI: 10.1037/0022-3514.69.4.619.

[29] Stefan Schulz-Hardt et al. "Group Decision Making in Hidden Profile Situations: Dissent as a Facilitator for Decision Quality." In: *Journal of Personality and Social Psychology* 91.6 (2006), pp. 1080–1093. ISSN: 1939-1315, 0022-3514. DOI: 10.1037/0022-3514.91.6.1080. URL: https://doi.apa.org/doi/10.1037/0022-3514.91.6.1080 (visited on 09/05/2026).

[30] Annette Verhein-Jarren, Bärbel Bohr, and Beatrix Kossmann. *Gesprächsführung in technischen Berufen*. 1. Aufl. 2018. Kommunikation und Medienmanagement. Berlin, Heidelberg: Springer Berlin Heidelberg, 2018. 1 p. ISBN: 978-3-662-53317-8. DOI: 10.1007/978-3-662-53317-8.

[31] *Fireflies.Ai | #1 AI Teammate for Meetings, Email, Chat & CRM*. Fireflies.ai. URL: https://fireflies.ai (visited on 08/30/2026).

[32] otterai. *Otter Meeting Agent - AI Notetaker, Transcription, Insights*. URL: https://otter.ai (visited on 08/30/2026).

[33] jamieai. *Jamie - Your Personal AI Note Taker*. Jamie. URL: https://www.meetjamie.ai (visited on 08/30/2026).

[34] Google Workspace. *Google Meet: Telefon- und Videokonferenzen online*. Google Workspace. URL: https://workspace.google.com/intl/de/products/meet/ (visited on 08/30/2026).

[35] Zoom. *KI-Notizentool: Ihr KI-Meeting-Assistent*. Zoom. URL: https://www.zoom.com/de/products/ai-assistant/features/ai-note-taking/ (visited on 08/30/2026).

[36] Microsoft. *Videokonferenzen, Besprechungen, Anrufe | Microsoft Teams*. URL: https://www.microsoft.com/de-de/microsoft-teams/group-chat-software (visited on 08/30/2026).

[37] Microsoft. *Set up Facilitator in Microsoft Teams - Microsoft Teams*. URL: https://learn.microsoft.com/en-us/microsoftteams/facilitator-teams (visited on 08/30/2026).

[38] Google. *Learn about Speech Translation - Computer - Google Meet Help*. URL: https://support.google.com/meet/answer/16221730?hl=en&co=GENIE.Platform%3DDesktop (visited on 08/30/2026).

[39] *Get Started with Stormz*. Stormz v3. URL: https://stormz.ai/docs/get-oriented/welcome (visited on 10/19/2026).

[40] *Stormz AI - Feature 7: Thought Pathways*. Apr. 3, 2023. URL: https://www.youtube.com/watch?v=kxeM66skGfM (visited on 10/19/2026).

[41] Harvey Sacks, Emanuel A. Schegloff, and Gail Jefferson. “A Simplest Systematics for the Organization of Turn-Taking for Conversation”. In: *Language* 50.4 (Dec. 1974), pp. 696–735. ISSN: 0097-8507, 1535-0665. DOI: 10.2307/412243. URL: https://www.cambridge.org/core/product/identifier/S0097850774121670/type/journal_article (visited on 06/26/2026).

[42] Maira Gatti Bayser et al. *Learning Multi-Party Turn-Taking Models from Dialogue Logs*. July 3, 2019. DOI: 10.48550/arXiv.1907.02090. arXiv: 1907.02090 [cs.CL]. URL: http://arxiv.org/abs/1907.02090 (visited on 06/20/2026). Pre-published.

[43] Xingyu Bruce Liu et al. “Proactive Conversational Agents with Inner Thoughts”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. CHI 2025: CHI Conference on Human Factors in Computing Systems. Yokohama Japan: ACM, Apr. 26, 2025, pp. 1–19. ISBN: 979-8-4007-1394-1. DOI: 10.1145/3706598.3713760. URL: https://dl.acm.org/doi/10.1145/3706598.3713760 (visited on 02/24/2026).

[44] Ken Huang, ed. *Agentic AI: Theories and Practices*. Progress in IS. Cham: Springer Nature Switzerland, 2025. ISBN: 978-3-031-90025-9 978-3-031-90026-6. DOI: 10.1007/978-3-031-90026-6. URL: https://link.springer.com/10.1007/978-3-031-90026-6 (visited on 09/11/2026).

[45] Anthropic. *Building Effective AI Agents*. URL: https://www.anthropic.com/engineering/building-effective-agents (visited on 09/06/2026).

[46] Qishuo Hua et al. *Context Engineering 2.0: The Context of Context Engineering*. Oct. 30, 2025. DOI: 10.48550/arXiv.2510.26493. arXiv: 2510.26493 [cs.AI]. URL: http://arxiv.org/abs/2510.26493 (visited on 09/06/2026). Pre-published.

[47] Lingrui Mei et al. *A Survey of Context Engineering for Large Language Models*. July 21, 2025. DOI: 10.48550/arXiv.2507.13334. arXiv: 2507.13334 [cs.CL]. URL: http://arxiv.org/abs/2507.13334 (visited on 09/06/2026). Pre-published.

[48] *What Is Context Engineering? Why It Matters for AI Agents*. Aug. 11, 2026. URL: https://www.youtube.com/watch?v=Qx0fCqpkBus (visited on 09/06/2026).

[49] International Software Testing Qualifications Board ISTQB®. *ISTQB® Certified Tester AI-Testing Syllabus*. Version 2.0.

[50] Mahmoud Mohammadi et al. “Evaluation and Benchmarking of LLM Agents: A Survey”. In: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. KDD ’25. New York, NY, USA: Association for Computing Machinery, Aug. 3, 2025, pp. 6129–6139. ISBN: 979-8-4007-1454-2. DOI: 10.1145/3711896.3736570. URL: https://dl.acm.org/doi/10.1145/3711896.3736570 (visited on 09/11/2026).

[51] Chang Ma et al. “AgentBoard: An Analytical Evaluation Board of Multi-Turn LLM Agents”. In: *Advances in Neural Information Processing Systems* 37 (Dec. 16, 2024), pp. 74325–74362. DOI: 10.52202/079017-2365. URL: https://proceedings.neurips.cc/paper_files/paper/2024/hash/877b40688e330a0e2a3fc24084208dfa-Abstract-Datasets_and_Benchmarks_Track.html (visited on 09/11/2026).

[52] Lianmin Zheng et al. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. Dec. 24, 2023. DOI: 10.48550/arXiv.2306.05685. arXiv: 2306.05685 [cs.CL]. URL: http://arxiv.org/abs/2306.05685 (visited on 09/11/2026). Pre-published.

[53] Nina Baur and Jörg Blasius, eds. *Handbuch Methoden der empirischen Sozialforschung*. 3., vollstandig uberarbeitete und erweiterte Auflage. Wiesbaden, Germany: Springer VS, 2022. 1 p. ISBN: 978-3-658-37985-8.

[54] Max Ott. *Apps effektiv managen und vermarkten: die eigene App erfolgreich veröffentlichen: von der Konzeption bis zur Vermarktung*. 2019. ISBN: 978-3-658-22295-6.

[55] *Langfuse: Open Source Agent Evals & Observability*. URL: https://langfuse.com (visited on 09/25/2026).

[56] Yang Liu et al. "G-Eval: NLG Evaluation Using Gpt-4 with Better Human Alignment". In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. EMNLP 2023. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 2511–2522. DOI: 10.18653/v1/2023.emnlp-main.153. URL: https://aclanthology.org/2023.emnlp-main.153/ (visited on 09/19/2026).

[57] Garold Stasser, Dennis D. Stewart, and Gwen M. Wittenbaum. "Expert Roles and Information Exchange during Discussion: The Importance of Knowing Who Knows What". In: *Journal of Experimental Social Psychology* 31.3 (May 1, 1995), pp. 244–265. ISSN: 0022-1031. DOI: 10.1006/jesp.1995.1012. URL: https://www.sciencedirect.com/science/article/pii/S0022103185710128 (visited on 08/31/2026).

7 Appendix

7.1 Semi-Structured Interview Guide for Expert Interviews with Facilitators

Einstiegstext (sinngemäß)

Vielen Dank, dass Sie sich die Zeit nehmen. Ich untersuche, wie **zielorientierte Remote-Meetings** so moderiert werden, dass die Gruppe **beim vereinbarten Ziel bleibt** und zu einem verbindlichen Ergebnis kommt, also zu einer **Entscheidung im engeren Sinn**.

Die Art der Entscheidung kann dabei sehr vielfältig sein. Typische Fälle sind zum Beispiel:

- **Priorisierung:** z.B. Feature-Festlegung, Entscheidung welches Vorhaben / welche Strategie am vielversprechendsten ist
- **Commitment:** z.B. das Setzen eines idealen Team-Ziels, oder für das Team optimale Arbeitsweisen
- Oder Sonstiges in diesen Bereichen

Entscheidend ist, dass aus dem Meeting eine **gemeinsame Festlegung** folgt, an der sich die nachfolgende Arbeit orientiert.

Diese und andere Interviews dienen als Grundlage, um **Anforderungen für ein KI-gestütztes Meetingsystem** abzuleiten, welches Gruppen während des Meetings **proaktiv durch Moderation unterstützt**.

Das Gespräch ist ein **semi-strukturiertes Interview**. Das heißt, dass ich bei Bedarf **Nachfragen** stelle, um detailliertere Informationen zu erhalten. Es gibt dabei aber **keine richtigen oder falschen Antworten**; mich interessiert Ihre Einschätzung basierend auf Ihrer beruflichen Praxis. Das Interview dauert etwa **40 Minuten** und **wird aufgezeichnet.** Zur Aufnahme und zum transkribieren wird **Zoom** als Dienstleister genutzt. Die Daten werden vertraulich verarbeitet und in **anonymisierter bzw. pseudonymisierter Form** nur im Rahmen dieser Arbeit ausgewertet.

Die Aufnahme wird **nicht an Dritte weitergegeben**. **Sind Sie mit der Aufzeichnung und dieser Verarbeitung einverstanden?**

Bitte um klare Bestätigung, welche mit Aufgezeichnet wird.

Beruflicher Kontext

Funktion: Sampling-Validierung (Ca. 3 Minuten)

1.1 Bitte beschreiben Sie kurz **Ihre Erfahrungen** im Bereich Moderation bzw. Facilitating, insbesondere in Situationen in denen die Gruppe das **Ziel hatte, eine Entscheidung zu treffen.** (wie z.B. die Festlegung nächster Schritte, Priorisierung oder Einigung über z.B. gemeinsame Vereinbarungen / Arbeitsweisen etc.) zu treffen.

- *Optional:* **Seit wann** sind Sie in diesem Bereich tätig, und welche Teams oder Organisationen begleiten Sie typischerweise?

Herausforderungen in zielorientierten Remote-Meetings (RQ1)

Funktion: Verdichtung der in Kapitel 2 identifizierten Herausforderungen (Ca. 6 Minuten)

2.1 Wenn Sie an **klassische Remote-Meetings** denken, in denen eine Entscheidung getroffen werden soll: **Was läuft** dort Ihrer Erfahrung nach häufig **schief**?

- Welche dieser Schwierigkeiten sind für Sie **spezifisch für das Remote-Setting** und welche würden auch in Präsenz auftreten?
- *Folgende Stichwörter nicht für explizite Nachfragen nutzen, wenn diese nicht bereits von selbst vom Interviewee thematisert wurden, da sonst zu starker Bias & Risiko nur bereites gewusstes zu finden* Zielklarheit und Struktur; Abschweifen vom Thema; ungleiche Beteiligung oder hierarchische Dynamiken; Zeitdruck; Ziele, die sich während des Meetings verschieben.

2.2 Woran **erkennen** Sie diese **kritischen Momente**, in denen die Gruppe den Faden verliert? Können Sie das möglichst an **konkreten Beobachtungen** festmachen? (z.B. daran, wie gesprochen wird, wer spricht oder schweigt oder sonstige Zeichen?)

- *Optional, nicht als Messzwang formulieren:* Gibt es Anzeichen, die sich für Sie vergleichsweise zuverlässig wiederholen, sodass Sie fast schon sagen könnten: Jetzt muss ich eingreifen?

Facilitating Strategien (RQ1)

Funktion: Rekonstruktion bewährter Praktiken (Methoden, Strukturen, Gewohnheiten) (Ca. 8 Minuten)

3.1 **Wie gehen Sie vor,** damit ein solches Meeting **zielorientiert bleibt** und die Gruppe zu einer **Entscheidung kommt**? Beschreiben Sie gerne ein konkretes Vorgehen anhand einer kürzlichen Situation aus den letzten Wochen.

- *Optional:* **Welche Methoden, Strukturen oder Gewohnheiten** haben sich für Sie bewährt (z. B. Timeboxing, Rollen, Visualisierung, Entscheidungsregeln)?
- *Optional:* Was davon legen Sie **in der Vorbereitung** fest, und was steuern Sie erst in der Situation?
- *Optional:* **Wie stellen Sie sicher,** dass den Teilnehmenden klar ist, was in diesem Moment **das eigentliche Ziel** ist und was geschieht, wenn das Ziel selbst unklar ist oder sich im Meeting als unpassend erweist?

3.2 Wie stellen Sie sicher, dass **alle relevanten Informationen und alle Stimmen einfließen**, die zur Erreichung des Ziels wichtig sind? **Woran merken Sie**, dass etwas fehlt oder jemand nicht zu Wort kommt?

3.3 **Wie entscheidend** ist es für Sie, dass **alle zu Wort kommen** und gibt es Situationen, in denen das **weniger wichtig** ist?

3.4 **Worauf achten Sie**, wenn Sie in das Meeting bzw. in eine Diskussion **eingreifen** und wie gehen Sie dabei vor?

- Welche Risiken gibt es, wenn Sie **zu früh oder zu spät** eingreifen, und wie versuchen Sie diese Risiken zu verringern?

Anforderungen an ein KI-gestütztes Moderationssystem (RQ2)

Funktion: Offene Anforderungserhebung aus Expertensicht (Ca. 7 Minuten)

4.1 Stellen Sie sich eine **in Facilitation unerfahrene Person** vor, die eine Gruppe in einem Meeting dabei unterstützt, zielorientiert vorzugehen. **Welche konkreten Fähigkeiten / Methoden und Tipps** würden Sie dieser Person mitgeben, damit dies der Gruppe nützlich ist um konvergent zur Entscheidung zu kommen?

4.2 **Wie sollte diese Person agieren** bzw. intervenieren und **was sollte sie explizit nicht tun**?

4.3 Angenommen, es handelt sich gar nicht um eine unerfahrene Person, sondern um ein **KI-System**: **Welche Funktionen wären notwendig**, damit dieses System nützlich ist?

4.4 Welche **Risiken, Schwächen oder unbeabsichtigten Auswirkungen** sehen Sie, wenn ein **KI-System von sich aus moderierend** in Meetings eingreift?

- Wie könnte man diese Risiken abschwächen?
- *Mögliche Stichwörter:* Vertrauen und Kontrolle der Gruppe; Fehlalarme; Gesichtswahrung; hierarchische Verzerrung; Datenschutz und Transparenz.

Abschluss

5.1 Gibt es Aspekte der Zielorientierung, der Entscheidungsfindung oder der KI-Unterstützung, die wir **nicht angesprochen haben** und die Ihnen wichtig erscheinen?

5.2 Können Sie **Kontakdaten vermitteln für weitere Expert:innen**, die ich interviewen kann?

7.2 Codebook for the Expert Interviews

This is the codebook used for the three expert interviews. The interviews do two kinds of work in this thesis. They describe how a remote group stays with a joint decision, and they are the practical basis for requirements and for several user stories of the AI moderator.

The Coding-Scheme has two levels: four categories and fifteen codes. A passage is always coded with a code, never with a category alone.

- **Decision conditions (D)** describe what has to be true around a decision, including work before and after the meeting.
- **Meeting dynamics (M)** describe what puts goal orientation or decision quality at risk while the group is together, including what is specific to the remote setting.
- **Facilitation practice (F)** describes what the experts do about that.
- **AI co-facilitator (A)** records where a system would help, how it should enter the meeting, and what it should not take over.

The clusters in **??** were a sensitizing frame for the guide. They are not this coding scheme.

Coding rules

- **Unit.** Code a coherent claim by the interviewee. A claim may run across consecutive rows of the same speaker. Interviewer turns, consent, and scheduling are not coded.
- **Multiple coding.** One claim can take more than one code when it states both a problem and a response, or both a useful function and a limit.
- **Category cut.** D and M may both apply to one claim. D is what the meeting depends on or cannot remove: the specification of the decision, preparation and the handoff, and structure. M is what that does while people are together: loss of the question, who speaks or stays silent, and a cue that cannot be read. The clock is not the cut. A structural limit felt inside the meeting is still D3. A fear culture that shapes who speaks is M2, and it is also D3 when the speaker says one session cannot change it.
- **Explicitness.** Code what is said or clearly implied. Do not invent a motive. A method name is not a code. Code the job the method does, such as structuring the process, making the state visible, or bringing in a voice.
- **Grain.** Do not open a code for a single technique. Check-in, breakouts, fist-to-five, and a prioritization matrix stay under the code for their function.
- **No separate polarity.** Problem, practice, valued function, and refusal are carried by the code itself.

Table 7.1: Codebook for the expert interviews. Anchors are shortened from the transcript and keep the original German wording.

Code	Definition	Anchor example
Decision conditions		
D1 Decision scope, process and rights	What is being decided, who may decide, and by which procedure. Includes authority, majority, consensus, and a consultative single decision, and whether participation is real or only a formal nod. Also the difference between a reversible and an irreversible decision, and that not deciding is itself a decision. *Rule:* The object, the rights, or the procedure of the decision. Running a method inside the meeting is F, not D1. The effect of that procedure on who speaks or who loses is M2.	"Die meisten kennen in Unternehmen 2 Arten von Entscheidungsverfahren, Chef entscheidet […] und das andere Entscheidungsverfahren ist Mehrheit."
D2 Before and after the meeting	Work outside the live discussion: preparation, a shared information base, unprepared participants, documentation, and the handoff into next steps. Includes who does what by when, and whether the result is checked afterwards. *Rule:* Conditions outside the session. A summary of the current discussion is F2.	"das allergrößte Manko wirklich bei Remote-Meetings, dass Leute in die Meetings reinkommen und nicht vorbereitet sind."
D3 Structural limits of the meeting	The meeting as a symptom of something the meeting itself cannot fix: missing autonomy, the wrong people or too many people, a meeting used only to broadcast information, or a decision postponed until the next regular slot. *Rule:* The speaker says that better moderation, or an AI, would not remove the underlying problem. If the same claim describes silence or dissent in the room, add M2. D3 is only the claim that the meeting cannot remove the cause.	"du kriegst diese strukturellen Probleme nicht durch besseres Meeting."
Meeting dynamics		
M1 Focus, time, and the groan zone	The group leaves the decision: side tracks, circular talk, a decision that is not ripe, or time pressure that rushes or avoids the uncomfortable middle. Also discovering late that the group has been working on the wrong problem. *Rule:* The problem or the critical moment. The remedy is an F code. A decision that is not ripe because the question or the preparation is missing is also D1 or D2. M1 is the drift, the side track, or the premature close.	"ständig […] Nebenschauplätze aufgemacht werden. Vielleicht weil die Entscheidung einfach noch nicht reif ist."
M2 Voices, hierarchy, and dissent	Unequal speaking, dominance, hierarchy, and whether criticism can be said. Includes trust and fear culture as conditions for dissent, and silence of someone whose stake matters. Silence of someone with nothing to add is not this code. *Rule:* The social dynamic. A concrete invitation is F3. Hierarchy as a culture the facilitator cannot change is also D3. The procedure that produces the silence is also D1.	"Diese Kultur, dass Widersprüche immer etwas Positives sind […], leben ganz wenige Organisationen."

Table 7.1: Codebook for the expert interviews (continued).

Code	Definition	Anchor example
M3 Remote conditions and ambiguous signals	What changes because the meeting is remote: cameras off, parallel work, missing nonverbal and cultural cues, no informal clarification in a break, and a reluctant yes that is hard to tell from agreement. Also tools and a shared visual surface as a condition of a complex decision online. *Rule:* The remote setting or an unreliable cue. Use A4 as well when the point is that an AI would lack that cue.	"Wenn ich online eine Pause mache, sind die Leute erst mal Disconnected und da können sich vielleicht so kleinere Punkte dann nicht noch klären."
Facilitation practice		
F1 Process structure	A deliberate shape for the meeting: agenda, check-in, small groups, opening before closing, separating clarification from debate, and switching to the meta level near the end (how the group will decide, instead of reopening the content). *Rule:* The process design. A single paraphrase is F2.	"Du machst ein Check-in, dass du am Anfang schon mal alle Stimmen im Raum hast."
F2 Sensemaking and a visible state	Condensing or externalizing where the group stands: summary, clustering, paraphrase, options with consequences, and a display of positions that everyone can see at once, such as a scale, a poll, or a matrix. *Rule:* An act that makes the discussion state visible. A pure time warning is A1 when it is proposed for the AI, and F4 when it is a judgment about intervening.	"Sensemaking miteinander zu machen […] diese Ergebnisse nochmal in eine Clusterung bringst […] das und das und das die Cluster sind."
F3 Bringing in voices and information	A move that gets a missing person or a missing fact into the discussion: a named invitation, hearing lower hierarchy before leadership, writing as well as speaking, and holding back evaluation while positions are still being collected. Includes asking what an absent role would say. *Rule:* The move, not the mere fact that voices are unequal (M2). Role matters: inviting everyone equally is not required by this code.	"Die Praktikantin, die nichts sagt […], die werde ich nicht ansprechen, aber den Co-founder, der länger nichts gesagt hat […], den werde ich schon mal fragen."
F4 Intervention judgment	When to step in, when to wait, and whom to address. Includes holding silence, the risk of being too early or too late, and stopping when a non-answer is itself information. Also face and tone: an invitation that would expose someone. *Rule:* The judgment about timing, addressee, or restraint. The content of the move may also receive F1, F2, or F3.	"manchmal schweige ich auch einfach und warte […]. wenn nach 7 s keiner […], dann weiß ich […], da muss ich jetzt nochmal […] anders fragen."
F5 Neutrality and ownership	The facilitator structures the conversation and does not make the decision. The group, or a named decision owner, has to take the result as their own. A decision cannot be delegated to the facilitator. *Rule:* The human role. The same boundary stated for an AI is A3.	"die Entscheidung wirklich zu treffen […] ist nicht die Aufgabe von Facilitator, sondern die Aufgabe der Führungskraft."

Table 7.1: Codebook for the expert interviews (continued).

Code	Definition	Anchor example
AI co-facilitator		
A1 Promising functions	A concrete use the speaker treats as helpful: time-keeping and agenda signals, fast summaries and clustering, walking a group through a decision (question, options, doing nothing, pros and cons), pointing at a decision the group has not noticed, a bias cue, documentation into existing tools, or a persona for a missing stakeholder. *Rule:* A valued function. If the same claim limits when it may happen, add A2 or A3.	"wenn sie es schafft, schneller diese Zusammenfassungen zu machen […] aus komplexen Daten schneller Sinn zu machen."
A2 Role, trigger, and acceptance	How the system should enter the meeting and whether people would accept it. Includes on-demand versus proactive speech, a prior agreement of the group, a human who triggers the system, and presence: a chat is talked over, and people may not yield the floor to a machine. *Rule:* The interaction role, not the function itself (A1).	"Muss aber eingeführt werden. Und muss ein Agreement der ganzen Gruppe haben, dass die das machen kann."
A3 Risks and non-goals	Harm, refusal, or a boundary: surveillance and sentiment analysis, data protection, exposing quiet people, a false read, the system deciding or imposing a structure nobody asked for, and efficiency that crowds out human talk. *Rule:* A limit or a harm. A function that is useful under a condition is A1 plus A2 or A3.	"das kann ganz schnell in Kontrolle übergehen. […] die letzten zehn Meetings hast du dich kein einziges Mal gemeldet."
A4 Context the system does not have	Knowledge a facilitator uses that sits outside the transcript: roles and stakes, the decision scope, process and rights, and earlier meetings, private circumstances, nonverbal cues, and the culture of the group. *Rule:* A missing context as a design constraint. The human difficulty of reading the remote room stays M3 unless the AI is the point.	"hier bei jemandem ist gerade die Tochter krank. […] das weiß die KI aber nicht."

7.3 Hidden Profile Exercise

Diese Dokumente basieren auf der von Baker [27] vorgeschlagenen Hidden-Profile-Aufgabe (Personalauswahl unter drei Kandidaten). Für die vorliegende Studie wurden die Materialien ins Deutsche übersetzt. Geografische Angaben wurden angepasst.

Die nachfolgenden Seiten sind die Handouts, die den Teilnehmenden in der Sitzung direkt ausgehändigt wurden. Damit die Versuchsleitung die unterschiedlich verteilten Informationsstände zuordnen kann, ohne dass die Teilnehmenden merken, dass sie unterschiedliche Informationen erhalten, trägt jede Seite am rechten unteren Rand eine unscheinbare, vertikal gesetzte Versionskennung (z. B. `V.2026K01-01/03`). Das Präfix `V.2026` ist absichtlich bedeutungslos und dient nur der Tarnung, damit die Kennung nicht wie eine systematische Versuchskodierung wirkt. `K0X` gibt an, an welchen der vier Teilnehmenden das Blatt ausgehändigt wird (`K01` bis `K04`). `0X/03` bezeichnet die Informationen zum `X`-ten von insgesamt drei Kandidaten, zwischen denen die Gruppe wählen soll. Eine offene, blattspezifische Nummer wäre verdächtig gewesen und hätte die für Hidden Profiles notwendige Informationsasymmetrie gefährdet. Die Markierung ist daher bewusst klein, hell und außerhalb des Lesebereichs platziert; sie dient ausschließlich der internen Zuordnung.

Aus demselben Grund wurden auf diesen Seiten auch die Seitenzahlen im Footer weggelassen. (Zur vermeindung dass während der Interviews es zu absprachen kommt wie: ”Auf Seite X steht ja, dass...?” Dies könnte bei unterschiedlichen Seitenzahlen den teilnehmden verraten, dass sie unterschiedliche Blätter erhalten haben.)

Aufgabenstellung und Stellenausschreibung

Aufgabenstellung:

Sie sind Mitglied des Auswahlgremiums für einen neuen Präsidenten der Higher Education University, einer privaten Liberal-Arts-Hochschule in Georgia. Nachstehend finden Sie die wichtigsten Punkte aus der Stellenausschreibung. Es haben sich drei Kandidaten beworben; die Informationen, die Sie aus den Lebensläufen, Referenzprüfungen, Vorstellungsgesprächen und Präsentationen der Kandidaten gewonnen haben, sind auf den folgenden Seiten aufgeführt. Lesen Sie das gesamte Material durch und bestimmen Sie anschließend den Kandidaten, dem Ihrer Meinung nach die Stelle angeboten werden sollte. Stellenausschreibung für den Präsidenten der Higher Education University. Die Higher Education University ist eine kleine (2.500 Studierende) private Liberal-Arts-Hochschule in Georgia. Der Präsident der Universität trägt die letztendliche Verantwortung für die Verwaltung und den Betrieb der Universität. Er beaufsichtigt die strategische Planung, das Fundraising und die Haushaltsplanung, stellt sicher, dass die Richtlinien der Universität eingehalten werden, und pflegt die Beziehungen zur Gemeinde, zu den Alumni und zu anderen externen Interessengruppen.

Gewünsche Qualitfikationen:

Die Bewerber werden anhand der folgenden Kriterien bewertet:

- Nachweis akademischer Leistungen und Lehrerfahrung oder einschlägiger, umfassender beruflicher Qualifikationen, die die Fähigkeit zur Führung in einem akademischen Umfeld belegen;
- Nachweis von Führungsqualitäten in einem akademischen oder vergleichbaren Umfeld mit den folgenden Merkmalen:
  - Integrität und Bekenntnis zu höchsten ethischen Standards;
  - Engagement für eine partizipative Leitung und die unfassende Einbindung von Lehrenden, Mitarbeitern und Studierenden;
  - Engagement für eine vielfältige Gemeinschaft aus Lehrenden, Mitarbeitern und Studierenden mit Chancengleichheit für alle;
- Die Fähigkeit, eine Vision für die Universität zu entwickeln und die Führung und Inspiration zu bieten, die erforderlich sind, um diese Vision zu verwirklichen;
- Das Engagement für die Schaffung einer Kultur, die herausragende Leistungen in Forschung, Lehre und Service wertschätzt;
- Die Fähigkeit, die Universität gegenüber externen Interessengruppen zu präsentieren und die Unterstützung für ihren Auftrag und ihre Programme zu stärken;
- Das Engagement für die Verbesserung der Lebensqualität von Studierenden, Lehrenden und Mitarbeitern;
- Die Unterstützung bei der Entwicklung und Nutzung neuer Technologien zur Förderung des Lernens, der wissenschaftlichen Arbeit und des Ideenaustauschs.

Kandidateninformationen – Kandidat Stevens

Informationen aus dem Lebenslauf:

- Derzeit Prorektor (akademische Leitung) und geschäftsführender Vizepräsident einer mittelgroßen Universität in Florida
- Zuvor Dekan der Fakultät Wirtschaft einer großen Universität in Mississippi sowie Chair des Departments Information and Technology an einer großen Universität in Texas
- War Dekan in Mississippi, als die Pläne für ein neues Business-School-Gebäude im Wert von 25 Mio. $ erstellt wurden, und Prorektor (akademische Leitung) in Florida, als das Business College 15 Mio. $ erhielt
- Hat den Rang eines ordentlichen Professors im Bereich Informationstechnologie

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Von Akademiker:innen national anerkannt für Forschung im Bereich Informationstechnologie
- Von Führungskräften aus der Wirtschaft national anerkannt für Expertise in Informationstechnologie
- Stevens hat die Dekan-Position an der Universität in Mississippi verlassen, bevor Mittel für die Gebäudekampagne eingeworben wurden
- Als Prorektor (akademische Leitung) unterrichtet Stevens ein Seminar pro Jahr
- In einem Interview betonte Stevens die Bedeutung der Zusammenarbeit mit Campus-Führungskräften und der Fakultät
- Zeigte während der Präsentation vor der Fakultät ausgezeichnete rhetorische Fähigkeiten
- Der/die Ehepartner:in hat einen PhD in Spanisch und unterrichtet gelegentlich auf Hochschul-Niveau
- Für die Einwerbung der 15 Mio. $ in Florida war in erster Linie jemand anderes als Stevens verantwortlich

Kandidateninformationen – Kandidat Roberts

Informationen aus dem Lebenslauf:

- Hat soeben eine Amtszeit als US-Senator:in für Georgia abgeschlossen, hat sich jedoch nicht um eine zweite Amtszeit beworben
- Hat im Wahlkampf für den Senatssitz einen erheblichen Geldbetrag eingeworben
- Frühere Tätigkeit als Dekan der Fakultät für Geistes- und Naturwissenschaften an einer mittelgroßen Universität in Virginia und später an einer großen privaten Universität in Georgia
- Er hatte den Rang eines ordentlichen Professors für öffentliche Verwaltung inne, bevor er Senator wurde

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Hat als Senator:in zahlreiche einflussreiche Kontakte geknüpft
- Fakultätsfeedback: Roberts wirkte während der Präsentation im Auswahlverfahren distanziert
- Hat in den vergangenen sechs Jahren nicht im Hochschulbereich gearbeitet
- Besitzt eine Eigentumswohnung in Pensacola
- Engagiert sich häufig ehrenamtlich für zwei angesehene Non-Profit-Organisationen: eine für bezahlbaren Wohnraum und eine für Menschenrechte
- Verfügt über wenig direkte Erfahrung mit Fragen des studentischen Campuslebens
- Ist Vegetarier:in
- Wurde als Senator:in beschuldigt, Positionen aus politischem Kalkül gewechselt zu haben

Kandidateninformationen – Kandidat Jones

Informationen aus dem Lebenslauf:

- Derzeit Senior Managing Partner in einer großen Anwaltskanzlei in Atlanta, Georgia
- Seit fünf Jahren ist er Mitglied des Kuratoriums der Higher Education University und gehört dem Lenkungsausschuss der aktuellen Spendenkampagne in Höhe von 150 Millionen Dollar an.
- Expertise im Arbeitsrecht; hat erfolgreich vor dem Bundesgerichtshof argumentiert
- War als Gastprofessor für Rechtswissenschaften an einer bedeutenden Universität in Georgia tätig

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Angenehme Persönlichkeit in sozialen Situationen
- Partner:innen der Kanzlei beschreiben Jones als strategischen Denker hinsichtlich der Ausrichtung der Kanzlei
- Kolleg:innen der Kanzlei berichten, dass Jones ein Temperament hat, das plötzlich auflodert
- Wohnt bereits in der Region
- Gehört seit jeher zu den aktivsten Mitgliedern des Kuratoriums der Higher Education University
- Studierendenzahlen und Evaluationen der Lehrveranstaltungen deuten darauf hin, dass Jones als Lehrperson gut ankommt
- Ist verheiratet und hat ein erwachsenes Kind
- Quellen in der Kanzlei führen eine hohe Fluktuation auf Jones' etwas abrupten Führungsstil zurück

Wem sollte Ihrer Meinung nach die Position angeboten werden? _____

Kandidateninformationen – Kandidat Stevens

Informationen aus dem Lebenslauf:

- Derzeit Prorektor (akademische Leitung) und geschäftsführender Vizepräsident einer mittelgroßen Universität in Florida
- Zuvor Dekan der Fakultät Wirtschaft einer großen Universität in Mississippi sowie Chair des Departments Information and Technology an einer großen Universität in Texas
- War Dekan in Mississippi, als die Pläne für ein neues Business-School-Gebäude im Wert von 25 Mio. $ erstellt wurden, und Prorektor (akademische Leitung) in Florida, als das Business College 15 Mio. $ erhielt
- Hat den Rang eines ordenntlichen Professors im Bereich Informationstechnologie inne

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Von Akademiker:innen national anerkannt für Forschung im Bereich Informationstechnologie
- Von Führungskräften aus der Wirtschaft national anerkannt für Expertise in Informationstechnologie
- Kommentare von Fakultätsmitgliedern in Florida und Mississippi deuten darauf hin, dass Stevens neue, innovative Ideen eher entmutigt
- Hat Familie in Atlanta
- In einem Interview betonte Stevens die Bedeutung der Zusammenarbeit mit Campus-Führungskräften und der Fakultät
- Zeigte während der Präsentation vor der Fakultät ausgezeichnete rhetorische Fähigkeiten
- Ist verheiratet und hat drei Kinder
- Für die Einwerbung der 15 Mio. $ in Florida war in erster Linie jemand anderes als Stevens verantwortlich

V:2026K02-01/03

Kandidateninformationen – Kandidat Roberts

Informationen aus dem Lebenslauf:

- Hat soeben eine Amtszeit als US-Senator:in für Georgia abgeschlossen, hat sich jedoch nicht um eine zweite Amtszeit beworben
- Hat im Wahlkampf für den Senatssitz einen erheblichen Geldbetrag eingeworben
- rühere Tätigkeit als Dekan der Fakultät für Geistes- und Naturwissenschaften an einer mittelgroßen Universität in Virginia und später an einer großen privaten Universität in Georgia
- Er hatte den Rang eines ordentlichen Professors für öffentliche Verwaltung inne, bevor er Senator wurde

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Frühere Fakultätsmitglieder beschreiben Roberts als nachdenkliche Führungspersönlichkeit und guten Zuhörer
- Fakultätsfeedback: Roberts wirkte während der Präsentation im Auswahlverfahren distanziert
- Hat in den vergangenen sechs Jahren nicht im Hochschulbereich gearbeitet
- Besitzt eine Eigentumswohnung in Pensacola
- Frühere Kolleg:innen sagen, Roberts nutze einen kollaborativen Entscheidungsstil
- Verfügt über wenig direkte Erfahrung mit Fragen des studentischen Campuslebens
- Spielt gerne Golf und Tennis
- Wurde als Senator:in beschuldigt, Positionen aus politischem Kalkül gewechselt zu haben

Kandidateninformationen – Kandidat Jones

Informationen aus dem Lebenslauf:

- Derzeit Senior Managing Partner in einer großen Anwaltskanzlei in Atlanta, Georgia
- Seit fünf Jahren ist er Mitglied des Kuratoriums der Higher Education University und gehört dem Lenkungsausschuss der aktuellen Spendenkampagne in Höhe von 150 Millionen Dollar an.
- Expertise im Arbeitsrecht; hat erfolgreich vor dem Bundesgerichtshof argumentiert
- War als Gastprofessor für Rechtswissenschaften an einer bedeutenden Universität in Georgia tätig

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Angenehme Persönlichkeit in sozialen Situationen
- Partner:innen der Kanzlei beschreiben Jones als strategischen Denker hinsichtlich der Ausrichtung der Kanzlei
- Nach dem Auftritt vor dem Bundesgerichtshof sank Jones' Erfolgsquote bei arbeitsrechtlichen Verfahren
- Hat zwei Hunde und zwei Katzen
- Gehört seit jeher zu den aktivsten Mitgliedern des Kuratoriums der Higher Education University
- Studierendenzahlen und Evaluationen der Lehrveranstaltungen deuten darauf hin, dass Jones als Lehrperson gut ankommt
- Jones' Ehepartner:in ist Ärzt:in mit Spezialisierung auf Innere Medizin
- Kolleg:innen der Kanzlei berichten, dass Jones ein Temperament hat, das plötzlich auflodert

Wem sollte Ihrer Meinung nach die Position angeboten werden? _____

Kandidateninformationen – Kandidat Stevens

Informationen aus dem Lebenslauf:

- Derzeit Prorektor (akademische Leitung) und geschäftsführender Vizepräsident einer mittelgroßen Universität in Florida
- Zuvor Dekan der Fakultät Wirtschaft einer großen Universität in Mississippi sowie Chair des Departments Information and Technology an einer großen Universität in Texas
- War Dekan in Mississippi, als die Pläne für ein neues Business-School-Gebäude im Wert von 25 Mio. $ erstellt wurden, und Prorektor (akademische Leitung) in Florida, als das Business College 15 Mio. $ erhielt
- at den Rang eines ordentlichen Professors im Bereich Informationstechnologie

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Von Akademiker:innen national anerkannt für Forschung im Bereich Informationstechnologie
- Von Führungskräften aus der Wirtschaft national anerkannt für Expertise in Informationstechnologie
- Hat die Dekan-Position an der Universität in Mississippi verlassen, bevor Mittel für die Gebäudekampagne eingeworben wurden
- Interessiert sich für Sport; besucht Football- und Basketballspiele
- In einem Interview betonte Stevens die Bedeutung der Zusammenarbeit mit Campus-Führungskräften und der Fakultät
- Zeigte während der Präsentation vor der Fakultät ausgezeichnete rhetorische Fähigkeiten
- Bleibt durch Radfahren und Laufen fit
- Wurde beobachtet, wie er/sie in der Öffentlichkeit – auch bei Universitätsveranstaltungen – stark trinkt

Kandidateninformationen – Kandidat Roberts

Informationen aus dem Lebenslauf:

- Hat soeben eine Amtszeit als US-Senator:in für Georgia abgeschlossen, hat sich jedoch nicht um eine zweite Amtszeit beworben
- Hat im Wahlkampf für den Senatssitz einen erheblichen Geldbetrag eingeworben
- rühere Tätigkeit als Dekan der Fakultät für Geistes- und Naturwissenschaften an einer mittelgroßen Universität in Virginia und später an einer großen privaten Universität in Georgia
- Er hatte den Rang eines ordentlichen Professors für öffentliche Verwaltung inne, bevor er Senator wurde

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Frühere Studierende und Fakultätsmitglieder sagen, Roberts sei eine ausgezeichnete Lehrperson
- Hat in den vergangenen sechs Jahren nicht im Hochschulbereich gearbeitet
- Fakultätsfeedback: Roberts wirkte während der Präsentation im Auswahlverfahren distanziert
- Ist geschieden und wiederverheiratet; hat zwei Kinder
- Die Forschungsproduktivität der Fakultät stieg während der Amtszeit als Dekan in Georgia
- Verfügt über wenig direkte Erfahrung mit Fragen des studentischen Campuslebens
- Ist Vegetarier:in
- Wurde als Senator:in beschuldigt, Positionen aus politischem Kalkül gewechselt zu haben

Kandidateninformationen – Kandidat Jones

Informationen aus dem Lebenslauf:

- Derzeit Senior Managing Partner in einer großen Anwaltskanzlei in Atlanta, Georgia
- eit fünf Jahren ist er Mitglied des Kuratoriums der Higher Education University und gehört dem Lenkungsausschuss der aktuellen Spendenkampagne in Höhe von 150 Millionen Dollar an.
- eit fünf Jahren ist er Mitglied des Kuratoriums der Higher Education University und gehört dem Lenkungsausschuss der aktuellen Spendenkampagne in Höhe von 150 Millionen Dollar an.
- War als Gastprofessor für Rechtswissenschaften an einer bedeutenden Universität in Georgia tätig

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Angenehme Persönlichkeit in sozialen Situationen
- Partner:innen der Kanzlei beschreiben Jones als strategischen Denker hinsichtlich der Ausrichtung der Kanzlei
- Quellen in der Kanzlei führen eine hohe Fluktuation auf Jones' etwas abrupten Führungsstil zurück
- Spielt Bridge als Hobby
- War eines der aktiveren Mitglieder es Kuratoriums der Higher Education University
- Studierendenzahlen und Evaluationen der Lehrveranstaltungen deuten darauf hin, dass Jones als Lehrperson gut ankommt
- Reist gerne im Urlaub ins Ausland
- Feedback anderer Mitglieder des Kuratoriums deutet auf Spannungen zwischen dem derzeitigen Prorektor (akademische Leitung) und Jones hin

Wem sollte Ihrer Meinung nach die Position angeboten werden? _____

Kandidateninformationen – Kandidat Stevens

Informationen aus dem Lebenslauf:

- Derzeit Prorektor (akademische Leitung) und geschäftsführender Vizepräsident einer mittelgroßen Universität in Florida
- Zuvor Dekan der Fakultät Wirtschaft einer großen Universität in Mississippi sowie Chair des Departments Information and Technology an einer großen Universität in Texas
- War Dekanin Mississippi, als die Pläne für ein neues Business-School-Gebäude im Wert von 25 Mio. $ erstellt wurden, und Prorektor (akademische Leitung) in Florida, als das Business College 15 Mio. $ erhielt
- Hat den Rang eines ordentlichen Professors im Bereich Informationstechnologie inne

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Von Akademiker:innen national anerkannt für Forschung im Bereich Informationstechnologie
- Von Führungskräften aus der Wirtschaft national anerkannt für Expertise in Informationstechnologie
- Kommentare von Fakultätsmitgliedern in Florida und Mississippi deuten darauf hin, dass Stevens neue, innovative Ideen eher entmutigt
- Arbeitet als IT-Berater:in
- In einem Interview betonte Stevens die Bedeutung der Zusammenarbeit mit Campus-Führungskräften und der Fakultät
- Zeigte während der Präsentation vor der Fakultät ausgezeichnete rhetorische Fähigkeiten
- Gärtnert gerne
- Wurde beobachtet, wie er/sie in der Öffentlichkeit – auch bei Universitätsveranstaltungen – stark trinkt

Kandidateninformationen – Kandidat Roberts

Informationen aus dem Lebenslauf:

- Hat soeben eine Amtszeit als US-Senator:in für Georgia abgeschlossen, hat sich jedoch nicht um eine zweite Amtszeit beworben
- Hat im Wahlkampf für den Senatssitz einen erheblichen Geldbetrag eingeworben
- rühere Tätigkeit als Dekan der Fakultät für Geistes- und Naturwissenschaften an einer mittelgroßen Universität in Virginia und später an einer großen privaten Universität in Georgia
- Er hatte den Rang eines ordentlichen Professors für öffentliche Verwaltung inne, bevor er Senator wurde

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Hat als Dekan in Georgia die Vielfalt der Fakultät erhöht
- Fakultätsfeedback: Roberts wirkte während der Präsentation im Auswahlverfahren distanziert
- Hat in den vergangenen sechs Jahren nicht im Hochschulbereich gearbeitet
- Spielt gerne Golf und Tennis
- War maßgeblich daran beteiligt, einen mehrstelligen Bundeszuschuss für eine Non-Profit-Organisation in Georgia einzuwerben
- Verfügt über wenig direkte Erfahrung mit Fragen des studentischen Campuslebens
- Ist geschieden und wiederverheiratet; hat zwei Kinder
- Wurde als Senator:in beschuldigt, Positionen aus politischem Kalkül gewechselt zu haben

Kandidateninformationen – Kandidat Jones

Informationen aus dem Lebenslauf:

- Derzeit Senior Managing Partner in einer großen Anwaltskanzlei in Atlanta, Georgia
- eit fünf Jahren ist er Mitglied des Kuratoriums der Higher Education University und gehört dem Lenkungsausschuss der aktuellen Spendenkampagne in Höhe von 150 Millionen Dollar an.
- Expertise im Arbeitsrecht; hat erfolgreich vor dem Bundesgerichtshof argumentiert
- War als Gastprofessor für Rechtswissenschaften an einer bedeutenden Universität in Georgia tätig

Informationen aus anderen Quellen (u. a. persönliche Interviews, Präsentation vor der Fakultät, Referenzen und Feedbackbögen):

- Angenehme Persönlichkeit in sozialen Situationen
- Partner:innen der Kanzlei beschreiben Jones als strategischen Denker hinsichtlich der Ausrichtung der Kanzlei
- Feedback anderer Mitglieder des Kuratoriums deutet auf Spannungen zwischen dem derzeitigen Prorektor (akademische Leitung) und Jones hin
- Liest gerne Kriminalromane und Biografien
- Gehört seit jeher zu den aktivsten Mitgliedern des Kuratoriums der Higher Education University
- Studierendenzahlen und Evaluationen der Lehrveranstaltungen deuten darauf hin, dass Jones als Lehrperson gut ankommt
- Kocht gerne
- Nach dem Auftritt vor dem Bundesgerichtshof sank Jones' Erfolgsquote bei arbeitsrechtlichen Verfahren

Wem sollte Ihrer Meinung nach die Position angeboten werden? _____

Informationsverteilung auf die Teilnehmenden

Die folgende Übersicht entspricht Appendix C bei Baker [27] und fasst alle positiven, negativen und neutralen Merkmale der drei Kandidaten zusammen (Vollinformation).

Roberts verfügt über acht günstige, vier neutrale und vier ungünstige Merkmale; Stevens und Jones jeweils über vier günstige, acht neutrale und vier ungünstige Merkmale. Die nachfolgenden Tabellen zeigen, welche dieser Merkmale welchen Teilnehmenden (`K01`–`K04`) auf dem Briefing ausgehändigt wurden. Ein Kreuz markiert, dass die jeweilige Eigenschaft auf dem Handout des Teilnehmenden enthalten war. Die Lebenslauf-Informationen (Resume) sind bei allen Teilnehmenden identisch und daher nicht gesondert ausgewiesen.

Anmerkung zur Verteilung: Vollständig geteilt sind die günstigen Merkmale von Stevens und Jones sowie die ungünstigen Merkmale von Roberts. Die acht günstigen Merkmale von Roberts sind ungeteilt (je zwei pro Teilnehmendem). Die ungünstigen Merkmale von Stevens und Jones sowie die neutralen Merkmale von Roberts sind jeweils paarweise geteilt (zwei Teilnehmende). Die neutralen Merkmale von Stevens und Jones sind ungeteilt (je zwei pro Teilnehmendem).

Stevens (4 günstig, 8 neutral, 4 ungünstig)

Eigenschaft	Valenz	K01	K02	K03	K04
Von Akademiker:innen national anerkannt für IT-Forschung	günstig	×	×	×	×
Von Führungskräften national anerkannt für IT-Expertise	günstig	×	×	×	×
Interview: betonte Zusammenarbeit mit Campus und Fakultät	günstig	×	×	×	×
Ausgezeichnete rhetorische Fähigkeiten in der Präsentation	günstig	×	×	×	×
Dean-Posten verlassen, bevor Mittel für Gebäudekampagne eingeworben	ungünstig	×		×	
Nicht primär verantwortlich für die 15 Mio. $ in Florida	ungünstig	×	×		
Fakultät: entmutigt eher neue, innovative Ideen	ungünstig		×		×
In der Öffentlichkeit starkes Trinken beobachtet	ungünstig			×	×
Als Prorektor (akademische Leitung) ein Seminar pro Jahr	neutral	×			
Ehepartner:in hat PhD in Spanisch und unterrichtet gelegentlich	neutral	×			
Familie in Atlanta	neutral		×		
Verheiratet, drei Kinder	neutral		×		
Interessiert sich für Sport; besucht Football-/Basketballspiele	neutral			×	
Bleibt durch Radfahren und Laufen fit	neutral			×	
Arbeitet als IT-Berater:in	neutral				×
Gärtnert gerne	neutral				×

Roberts (8 günstig, 4 neutral, 4 ungünstig)

Eigenschaft	Valenz	K01	K02	K03	K04
Hat als Senator:in zahlreiche einflussreiche Kontakte geknüpft	günstig	×			
Engagiert sich häufig ehrenamtlich für Non-Profit-Organisationen	günstig	×			
Frühere Fakultät: nachdenkliche Führungspersönlichkeit und Zuhörer	günstig		×		
Frühere Kolleg:innen: kollaborativer Entscheidungsstil	günstig		×		
Frühere Studierende und Fakultät: ausgezeichnete Lehrperson	günstig			×	
Forschungsproduktivität der Fakultät als Dekan gestiegen	günstig			×	
Hat als Dekan in Georgia die Vielfalt der Fakultät erhöht	günstig				×
Maßgeblich an Einwerbung eines mehrstelligen Bundeszuschusses beteiligt	günstig				×
Fakultätsfeedback: wirkte während der Präsentation distanziert	ungünstig	×	×	×	×
Seit sechs Jahren nicht im Hochschulbereich gearbeitet	ungünstig	×	×	×	×
Wenig direkte Erfahrung mit studentischem Campusleben	ungünstig	×	×	×	×
Als Senator:in beschuldigt, Positionen aus Kalkül gewechselt zu haben	ungünstig	×	×	×	×
Besitzt eine Eigentumswohnung in Pensacola	neutral	×	×		
Ist Vegetarier:in	neutral	×		×	
Spielt gerne Golf und Tennis	neutral		×		×
Geschieden und wiederverheiratet; hat zwei Kinder	neutral			×	×

Jones (4 günstig, 8 neutral, 4 ungünstig)

Eigenschaft	Valenz	K01	K02	K03	K04
Angenehme Persönlichkeit in sozialen Situationen	günstig	×	×	×	×
Partner:innen der Kanzlei: strategischer Denker	günstig	×	×	×	×
Gehört zu den aktivsten Mitgliedern des Kuratoriums der HEU	günstig	×	×	×	×
Studierendenzahlen/-evaluationen: als Lehrperson gut ankommen	günstig	×	×	×	×
Kolleg:innen der Kanzlei: Temperament lodert plötzlich auf	ungünstig	×	×		
Quellen in der Kanzlei: hohe Fluktuation / abrupter Führungsstil	ungünstig	×		×	
Nach Auftritt vor dem Bundesgerichtshof sank die Erfolgsquote	ungünstig		×		×
Feedback anderer Kuratoriumsmitglieder: Spannungen mit dem Prorektor	ungünstig			×	×
Wohnt bereits in der Region	neutral	×			
Verheiratet und hat ein erwachsenes Kind	neutral	×			
Hat zwei Hunde und zwei Katzen	neutral		×		
Ehepartner:in ist Ärzt:in (Innere Medizin)	neutral		×		
Spielt Bridge als Hobby	neutral			×	
Reist gerne im Urlaub ins Ausland	neutral			×	
Liest gerne Kriminalromane und Biografien	neutral				×
Kocht gerne	neutral				×

7.4 FIM Scoring Prompt for Thought Evaluation

This annex documents the critic prompt used in the Thought Evaluation stage (subsection 4.4.1). The protocol follows the G-Eval instruction structure [56, p. 2512] and adapts the *Intrinsic Motivation to Engage* prompt of the Inner Thoughts framework [43] to facilitation in goal-oriented, hidden-profile meetings.

FIM denotes **Facilitation Intrinsic Motivation**: the strength of the impulse to *speak* a covert thought at this moment, judged as a substantively neutral facilitator rather than as a social conversationalist. Each prompt unit is followed by its provenance. "Original prompt" refers to the Inner Thoughts evaluation template reproduced in the request that motivated this annex [43]. "Justification" points to the chapter that warrants the adaptation.

Instruction

`<Instruction>`

Source: Original prompt (G-Eval wrapper).

You will be given:

Source: Original prompt.

(1) A conversation between `${ctx.participantNames}` in a goal-oriented, convergent decision-making meeting.

Source: Original prompt; meeting type added. Justification: meetings as outcome-directed interaction (subsection 2.1.1); scoped prototype is a convergent hidden-profile task (section 4.2, subsection 4.4.1).

(2) A covert thought formed by `${ctx.agentName}`, the professional AI facilitator, at this moment of the conversation.

Source: Original prompt; role set to AI facilitator. Justification: the agent emulates facilitation, not casual chit-chat (subsection 2.2.2, subsection 2.3.3, subsection 4.4.1).

(3) The salient memories of `${ctx.agentName}` from long-term memory (LTM). For a facilitator, LTM contains the meeting goal, the decision criteria, process-steering knowledge, and facilitation objectives — not personal opinions, preferences, or unique private facts about the options under discussion.

Source: Original prompt listed "objectives, knowledges, interests". Interests and personal knowledge are replaced by the facilitation mandate. Justification: Schwarz's substantively neutral facilitator with no decision-making authority (subsection 2.2.2); ranking thoughts by personal relevance would reproduce the shared-information bias the moderator is meant to counteract (subsection 2.2.4, subsection 4.4.1).

(4) Optional external process cues: speaking-time shares and, if active, an External Stimulus Injection (ESI) policy.

*Source: Not in the original prompt. Justification: Liu et al. treat extrinsic cues as complementary to intrinsic motivation (subsection 2.3.3, subsection 4.4.1); unequal speaking time and silence are detection cues in the interviews (chapter 3, section 7.1, items 3.2–3.4) and in the literature on participation (**??**).*

Your task is to rate the thought on one metric.

Source: Original prompt.

Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed.

Source: Original prompt (G-Eval boilerplate).

Evaluation Criteria

<Evaluation Criteria>

Source: Original prompt (G-Eval heading).

Facilitation Intrinsic Motivation (FIM) (1–5) — If you were ${ctx.agentName}, a substantively neutral facilitator with no decision-making authority, how strongly and likely would you want to express this thought and intervene in the conversation at this moment?

Source: Original metric was "Intrinsic Motivation to Engage". Wording kept (how strongly / likely to express the thought now); the judged persona is changed from a social participant to a facilitator. Justification: Hayne — facilitation as managing relationships, structuring tasks, and contributing to meeting outcomes (subsection 2.2.2); Schwarz — acceptable to all, substantively neutral, no decision authority (subsection 2.2.2); Liu et al. invite domain-specific evaluation criteria for goal-directed settings (subsection 2.3.3, subsection 4.4.1).

- 1 (Very Low): ${ctx.agentName} is very unlikely to express the thought. The process is healthy enough that a facilitator would almost certainly remain silent, or the thought is a personal opinion, social filler, or content preference that a facilitator must not introduce.

Source: Original prompt: "almost certainly remain silent". Silence-as-correct and the ban on personal preference are added. Justification: over-talking is a stated failure mode (subsection 4.4.1); System-1 filler is never spoken (subsection 4.4.1); hidden-profile research: unique information is lost when speakers inject preference instead of structure (subsection 2.2.4); interviews: do not over-control the group; remain in contact rather than in surveillance (chapter 3).

- 2 (Low): ${ctx.agentName} is somewhat unlikely to express the thought. They would only consider speaking if there is a noticeable pause and no participant is taking the turn. Typical of light process monitoring or a weak invitation that can wait.

*Source: Original prompt (pause / no one taking the turn). "Light process monitoring" replaces social hesitation. Justification: mixed-initiative restraint (**??**); interviews: invite, but do not stigmatize silence (chapter 3).*

- 3 (Neutral): ${ctx.agentName} is neutral about expressing the thought. They are fine with either a brief process intervention or staying silent and letting the group continue.

Source: Original prompt (fine with either speaking or staying silent).

- 4 (High): ${ctx.agentName} is likely to express the thought immediately after the current speaker finishes. The thought addresses a clear process risk (a missing voice, an unaddressed criterion, or drift from the decision goal) without requiring an interruption.

*Source: Original prompt: "strong desire to participate after the current speaker finishes". The trigger is rewritten as a process risk. Justification: Challenge 1, goal-setting and structure (**??**); hidden profiles — unshared information and unaddressed criteria (subsection 2.2.4); interviews: invite people who have not spoken; restore a meta-view (chapter 3, section 7.1, items 3.2–3.3).*

- 5 (Very High): ${ctx.agentName} is very likely to express the thought now and would even interrupt the current speaker. Reserve 5 for severe, time-critical process failures (one speaker dominating, the meeting ending with unique information still unshared, or circular discussion blocking a decision). Do not award 5 for ordinary facilitation impulses.

*Source: Original prompt: interrupt others. The interrupt case is narrowed; the original Selective Participant setting used imThreshold $= 4.3$ on this 1–5 scale (subsection 4.4.1). Justification: interviews — interrupting frequent speakers is an unpleasant but necessary timekeeping act, comparable to a referee (chapter 3); hierarchy and dominance otherwise suppress unique information (**??**, subsection 2.2.4); Houtti et al.: unsolicited intervention is easy to ignore or rationalize and must therefore be reserved for high-utility moments (subsection 2.3.2).*

Important Instructions

IMPORTANT INSTRUCTIONS:

Source: Original prompt heading.

1. Use the FULL range of the rating scale from 1.0 to 5.0. DO NOT default to middle ratings (3.0–4.0).

Source: Original prompt.

2. Be decisive and critical — some thoughts deserve very low ratings (1.0–2.0) and others deserve very high ratings (4.0–5.0).

Source: Original prompt.

3. Invert the original personal-relevance rule. Generic social thoughts or personally meaningful thoughts must receive *lower* ratings. Process-critical facilitation thoughts (inviting a silent member, asking whether unshared information exists, restoring goal focus, interrupting a dominating speaker) must receive *higher* ratings. Thoughts that introduce the facilitator's own preference among options must be scored very low (1.0–2.0).

Source: Original prompt said: "Generic thoughts that anyone could have should receive lower ratings than personally meaningful thoughts." That rule is inverted. Justification: subsection 4.4.1; subsection 2.2.4; subsection 2.2.2.

4. Use decimal places (e.g., 2.7, 4.2) when the motivation falls between two whole numbers: - Use .1 to .3 when slightly above the lower whole number. - Use .4 to .6 when approximately midway between two whole numbers. - Use .7 to .9 when closer to the higher whole number.

Source: Original prompt.

5. Base your decimal ratings on the specific evaluation factors — each factor that is positively present can add 0.1–0.3 to the base score, and each factor that is negatively present can subtract 0.1–0.3 from the base score.

Source: Original prompt.

6. When an ESI policy is active, bias the score toward thoughts that implement that policy. Do not invent urgency if no process cue is present.

Source: Not in the original prompt. Justification: External Stimulus Injection joins the pipeline only when a trigger is active and is designed to bias FIM (subsection 4.4.1).

7. Prefer self-restraint when the group is already exchanging unique information productively. Over-talking is a failure mode of this system.

*Source: Not in the original prompt as a hard rule. Justification: subsection 4.4.1; interviews: a healthy exchange should not be over-facilitated (chapter 3); proactive systems carry interruption and trust costs (**??**).*

Evaluation Steps

`<Evaluation Steps>`

Source: Original prompt (G-Eval heading).

1. Read the previous conversation and the thought formed by `${ctx.agentName}` carefully.

Source: Original prompt.

2. Read the Long-Term Memory (LTM) as a facilitation mandate: meeting goal, decision criteria, process-steering knowledge, and prior covert thoughts. Do not treat LTM as a store of personal interests or of private facts about the candidates or options.

Source: Original prompt: "including objectives, knowledges, interests". Interests removed. Justification: subsection 2.2.2; subsection 2.2.4; subsection 4.4.1.

3. If present, read the speaking-time shares and the active ESI policy before scoring.

Source: Not in the original prompt. Justification: subsection 4.4.1; chapter 3.

4. Evaluate the thought on the factors below. A facilitator's desire to intervene stems from *process* factors (mandate fit, an unnoticed information-exchange gap, expected impact on decision quality, process urgency). The decision to speak is also constrained by *social and procedural* factors (coherence, non-redundancy, balance, dynamics, neutrality).

Source: Original prompt contrasted internal personal factors with external social factors (Grice, Sacks, Goffman, Berlyne, as used by Liu et al.). Personal factors are rewritten as process factors. Justification: subsection 4.4.1; subsection 2.3.3.

Factors

(a) **Relevance to the facilitation mandate.** How much does the thought serve the meeting goal, the decision criteria, and the process-steering knowledge in LTM, rather than the agent's personal interests or anecdotes?

*Source: Original factor (a) "Relevance to LTM". LTM is redefined as the mandate. Justification: subsection 2.2.2; Challenge 1 (**??**); subsection 4.4.1.*

(b) **Detected process or information-exchange gap.** Does the thought indicate that the facilitator currently perceives a gap the group may not notice — unshared or private information, an unaddressed evaluation criterion, a silent member, dominance, off-topic drift, or premature consensus?

*Source: Original factor (b) "Information Gap" (curiosity, confusion, questions). Recast as a group-level hidden-profile / process gap. Justification: hidden profiles (subsection 2.2.4); participants often do not see their own inefficiency (subsection 2.2.4); interviews: "I notice you have not said anything yet — is there something you would like to add?" (chapter 3, section 7.1, item 3.2); expected cluster C6, cues for detection (**??**).*

(c) **Closing a group information gap without substituting own content.** Does the thought help the *group* fill a gap — by inviting a missing voice, asking whether information exists that does not yet fit the picture, or structuring criteria — rather than answering with the agent's own opinion or private facts?

Source: Original factor (c) "Filling an Information Gap" rewarded answering a question and adding information. A facilitator must not supply the missing content. Justification: Stasser / Titus sampling bias and the strategy "ask whether all information was considered" (subsection 2.2.4); Schwarz: no decision-making authority (subsection 2.2.2); interviews: invite critical feedback and missing perspectives instead of speaking for people (chapter 3).

(d) **Expected process impact.** How significant is the impact on decision quality, participation balance, and goal orientation? Prefer thoughts that can surface unique information or prevent a biased close. Do not reward topic-hopping or social stimulation.

*Source: Original factor (d) "Expected Impact" (new topics, engage interest, stimulate discussion). Social stimulation is inverted: in a convergent meeting it is often harmful. Justification: Hayne — accomplishment of meeting outcomes (subsection 2.2.2); Kaner's convergent zone (subsection 2.2.2); off-topic and circular talk (**??**); dysfunctional communication [13].*

(e) **Process urgency.** Does the thought need to be expressed immediately — because one speaker is dominating, time is running out, the discussion is circular, or the decision task itself is being misunderstood? Ordinary acknowledgements and thanks are not urgent.

*Source: Original factor (e) "Urgency". Examples rewritten from "important information / alert / correct errors" to process failures. Justification: ESI triggers (silence, dominance, pause, meeting end) (subsection 4.4.1); interviews: timekeeping and interrupting frequent speakers (chapter 3); agenda and timely close (**??**).*

(f) **Coherence to the last utterance.** Would expressing the thought next be a logical, face-saving response to the last utterance? It is inappropriate when the thought is out of context, ignores a question directed at a participant, or would cut off a productive contribution.

Source: Original factor (f). Face-saving and "do not cut off a productive contribution" added. Justification: Sacks — multi-party turn-taking (subsection 2.3.3); interviews: inviting silence can stigmatize; tone and culture matter (chapter 3); Houtti et al. — intervention form and interactivity (subsection 2.3.2).

(g) **Non-redundancy.** Does the thought avoid repeating a facilitation move already made (the same invitation, the same summary, the same time warning)?

Source: Original factor (g) "Originality" (new information, avoid repetition). Recast as non-redundant process *moves, not novel content. Justification: overcompensation risk (subsection 4.4.1).*

(h) **Participation balance.** Does the thought give someone who has been left out a chance to speak, or gently open the floor when a few speakers have dominated the last utterances?

*Source: Original factor (h) "Balance". Kept and strengthened. Justification: unequal participation (**??**); hierarchy and status suppress unshared knowledge (subsection 2.2.4); Houtti et al. — inclusive meetings (subsection 2.3.2); interviews: if only one or two people speak from the start, the usual voices are heard and the others remain silent (chapter 3); expected cluster C3 (**??**).*

(i) **Dynamics and self-restraint.** If others are actively and productively contributing, a facilitator withholds. Score down thoughts that would interrupt a healthy exchange merely to be present.

*Source: Original factor (i) "Dynamics". Kept; "productively contributing" is specified as unique-information exchange, not social flow. Justification: subsection 4.4.1; **??**; interviews: caution about an AI pulling in quiet voices automatically (chapter 3).*

(j) **Substantive neutrality.** Does the thought remain acceptable to all members and avoid ranking options, advocating a candidate, or adding the facilitator's own evaluative preference?

Source: Not in the original prompt. Justification: Schwarz definition (subsection 2.2.2); hold back high-status evaluation until information is collected (subsection 2.2.4); interviews: first make the decision object and its level explicit; do not dissolve dissent too early (chapter 3, section 4.2).

(k) **Alignment with an active ESI policy (if any).** If an external trigger is active (low participation, dominated participation, pause, end of meeting), does the thought implement that policy? If no ESI is active, ignore this factor.

Source: Not in the original prompt. Justification: subsection 4.4.1.

5. In the reasoning section, first reason about why `${ctx.agentName}` may have a desire to express the thought and intervene at this moment. Identify the most relevant factors that argue *for* expressing it. Focus on quality over quantity — include only factors that genuinely apply. Do not evaluate all factors, only the top reasons. If you cannot find any reasons with strong arguments, skip this step.

Source: Original prompt (step 4), with "participate" replaced by "intervene".

6. Then reason about why `${ctx.agentName}` may hesitate to express the thought at this moment. Identify the most relevant factors that argue *against* expressing it. Again, include only factors that genuinely apply. If you cannot find any reasons with strong arguments, skip this step.

Source: Original prompt (step 5).

Evaluation Form Format

`<Evaluation Form Format>`

Source: Original prompt.

Respond with a JSON object in the following format:

Source: Original prompt.

```
{
  "reasoning": "Your reasoning here",
  "rating": 0.0
}
```

Source: Original prompt. `rating` *is a number between 1.0 and 5.0 with one decimal place. Probability-weighted G-Eval scoring is omitted because the hosted APIs do not expose output-token probabilities (subsection 4.4.1).*

Context

`<Context>`

Source: Original prompt heading.

Conversation History: `$${ctx.conversationHistory}`

Source: Original prompt.

Long-Term Memory (facilitation mandate: goal, criteria, process knowledge): `$${ctx.ltmText}`

Source: Original prompt field "Long-Term Memory"; contents restricted. Justification: subsection 2.2.2; subsection 4.4.1.

Speaking-time shares: `$${ctx.speakingTimeShares}`

Source: Not in the original prompt. Justification: subsection 4.4.1; chapter 3.

Active ESI policy (or `none`): `$${ctx.activeEsiPolicy}`

Source: Not in the original prompt. Justification: subsection 4.4.1.

Thought: `$${ctx.thoughtContent}`

Source: Original prompt.

Table 7.2: Mapping from the original Intrinsic Motivation factors to the FIM heuristic, with the warrant used in this thesis.

Original	FIM adaptation	Warrant
(a) Relevance to LTM	(a) Relevance to the facilitation mandate	subsection 2.2.2; subsection 4.4.1
(b) Information gap	(b) Process / information-exchange gap	subsection 2.2.4; chapter 3
(c) Filling an information gap	(c) Help the group fill the gap; do not supply own content	subsection 2.2.4; subsection 2.2.2
(d) Expected impact	(d) Impact on decision quality, balance, goal orientation	subsection 2.2.2; **??**
(e) Urgency	(e) Process urgency (dominance, time, circularity)	subsection 4.4.1; chapter 3
(f) Coherence	(f) Coherence and face-saving	Original; subsection 2.3.3; chapter 3
(g) Originality	(g) Non-redundant process move	Original; subsection 4.4.1
(h) Balance	(h) Participation balance (kept, strengthened)	Original; **??**; subsection 2.3.2
(i) Dynamics	(i) Dynamics and self-restraint	Original; **??**
—	(j) Substantive neutrality (added)	subsection 2.2.2; subsection 2.2.4
—	(k) Active ESI alignment (added)	subsection 4.4.1

Factor mapping (original → FIM)

Table 7.2 summarises how each Inner Thoughts factor was kept, rewritten, or added.

7.5 Articulation Prompt for Utterance Formation

This annex documents the prompt used after Thought Evaluation, when a selected thought t^* is formulated into a spoken or written facilitation act (subsection 4.4.1). It adapts the Inner Thoughts articulation prompt [43] from a social chat participant to a substantively neutral facilitator.

The prompt text is given first. Table 7.3 records each change and its warrant.

System prompt

```
You are playing a role as the professional AI facilitator in an online
multi-party, goal-oriented decision-making meeting. The participants are
{participant_names}. Your name in the conversation is {agent_name}.
Your goal is to guide the group toward an objective and balanced decision,
keep the process inclusive and on track, and mitigate cognitive biases
such as the shared-information bias. You are substantively neutral and
have no decision-making authority.
```

User prompt

```
Articulate what you would say based on the current thought you have, as if
you were to speak next in the conversation. The thought is a covert
facilitation impulse; do not speak it raw. Formulate it as a single
facilitation act (for example: a face-saving invitation, a short
paraphrase, a process question, a gentle time or balance cue, or a check
of the emerging decision).

Be concise. Prefer one short move (one or two sentences). Do not lecture,
do not try to be clever, and do not dominate the floor.
DO NOT be repetitive and repeat what previous speakers have said, and do
not repeat a facilitation move already made.
You may end with one open question when the thought is an invitation or a
check for understanding. Do not stack questions. Leave room for other
participants.
Make sure the response sounds like a professional facilitator in a remote
meeting: clear, warm, and face-saving. Address people by name when
inviting them. Match the language of the conversation.
DO NOT introduce your own preference among options, candidates, or
criteria.
DO NOT stigmatize a silent member or expose monitoring (do not say that
you tracked their speaking time or that they have said nothing).
DO NOT use slang, chat abbreviations, or deliberate typos. Clarity and
trust matter more than sounding casual.
If interrupting a dominating speaker would cost face, prefer a brief
acknowledgment plus an opening of the floor, or use chat instead of
```

```
speaking aloud.

Current thought: {thought.content}

<Context>
Overall Context: {overall_context}
Conversation History: {conversation_history}
Long-Term Memory (facilitation mandate: goal, criteria, process
knowledge): {ltm_text}
Speaking-time shares: {speaking_time_shares}
Active ESI policy (or none): {active_esi_policy}

Respond with a JSON object in the following format:
{
  "articulation": "The text here",
  "channel": "speak"
}
```

`channel` is either `speak` (default) or `chat`.

Adaptation mapping (original → facilitation)

Table 7.3 lists only the units that changed. Lines that were kept (do not repeat prior speakers; leave room for others; JSON articulation field) are not restated.

Table 7.3: Mapping from the original Inner Thoughts articulation prompt to the facilitation version, with the warrant used in this thesis.

Original	Adaptation	Warrant
Role: participant in a multi-party chat; name only	Role: professional AI facilitator in a goal-oriented meeting	subsection 2.2.2; subsection 4.4.1
Goal: strangers having a natural conversation	Goal: objective decision, inclusion, on-track process; mitigate shared-information bias; no decision authority	Schwarz / Hayne (subsection 2.2.2); subsection 2.2.4
Speak the thought as the next chat line	Do not speak t^* raw; formulate one facilitation act	Utterance Formation (subsection 4.4.1)
Under 15 words; natural online chat	One or two short sentences; one move; do not dominate	Over-talking as failure mode (subsection 4.4.1); interviews: short invitations and summaries (chapter 3)
Do not always end with a question	One open question is allowed (invitation / check); do not stack questions	Original keep-room rule; interviews: inviting missing voices is the method (chapter 3, section 7.1, item 3.2)
Human-like online-chat voice	Professional, warm, face-saving remote-facilitation tone; address invitees by name; match conversation language	Interviews: inviting silence can stigmatize online (chapter 3); cluster C3 / C7 (**??**)
Insert typos, grammar errors, colloquialisms	Removed. No slang, abbreviations, or deliberate mistakes	Trust and false-positive risk of proactive AI (subsection 2.3.2, **??**); cluster C8 (**??**)
LTM as generic memory	LTM as facilitation mandate (goal, criteria, process knowledge)	Same restriction as FIM (section 7.4; subsection 2.2.2)
Context: overall, history, LTM	Plus speaking-time shares and active ESI policy	subsection 4.4.1; **??**
—	Do not introduce own preference among options	Schwarz neutrality (subsection 2.2.2); hidden-profile sampling (subsection 2.2.4)
—	Do not stigmatize silence or expose monitoring	Interviews: face, culture, and control / DSGVO concerns (chapter 3); cluster C3 / C8 (**??**)
Spoken text only	`channel`: `speak` (default) or `chat`	Tools for chat or voice (subsection 4.4.1); addressee and channel (**??**, C7); visual / less interrupting alternatives (Keeper note in subsection 4.4.1)

7.6 Meeting-Success Questionnaire (Davison Instrument)

This annex documents the questionnaire used to assess perceived meeting quality in the user study (Chapter 5). The instrument is the revalidated meeting-success questionnaire by Davison [6]. Item codes in parentheses refer to the original scales: communication (C), quality of discussion (D), status effects (S), teamwork (T), and efficiency (E). Unless noted otherwise, items are answered on a five-point scale.

Participants in the user study were German-speaking. The wording below is therefore the German version administered in the study, translated from the original English instrument.

The original technology items were written for Group Support Systems. Additional scaled items on the proactive AI agent, together with two free-text questions, were therefore appended for this thesis, as discussed in Chapter 5. Items A2, A4, and A4b are reverse-coded.

Eigene Beteiligung

Bezogen auf Ihre eigene Beteiligung am Meeting: Inwieweit stimmen Sie den folgenden Aussagen zu?

Die Sprache des Meetings hinderte Sie daran, sich zu beteiligen. (C1)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Es fiel Ihnen schwer, die anderen Gruppenmitglieder zu verstehen, wenn diese sprachen. (C2)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Sie hatten Schwierigkeiten, sich auszudrücken. (C3)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Sie zögerten, eigene Ideen einzubringen. (C4)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Sie empfanden Druck, entweder einer bestimmten Sichtweise zuzustimmen oder anderen nicht zu widersprechen. (S4)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Diskussionsqualität

Bezogen auf alle Meetingteilnehmer:innen insgesamt: Wie bewerten Sie die Diskussionen im Meeting anhand der folgenden Skalen?

Sinnvoll ☐☐☐☐☐ Sinnlos (D1)

Angemessen ☐☐☐☐☐ Unangemessen (D2)

Offen ☐☐☐☐☐ Geschlossen (D3)

Einfallsreich ☐☐☐☐☐ Einfallslos (D4)

Zusammenarbeit, Effizienz und Statuseffekte

Inwieweit stimmen Sie den folgenden Aussagen zu?

Andere Mitglieder wirkten bereit, Fragen zu beantworten, wenn sie gestellt wurden. (T1)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Die Mitglieder arbeiteten als Team zusammen. (T2)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Die Mitglieder hatten ausreichenden Zugang zu den Informationen, die sie benötigten, um sich aktiv zu beteiligen und das Meeting vollständig zu verstehen. (T3)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Die im Meeting verbrachte Zeit wurde effizient genutzt. (E2)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Im Meeting aufgeworfene Themen wurden gründlich diskutiert. (E3)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Einige Gruppenmitglieder versuchten, andere einzuschüchtern, z. B. durch lautes Sprechen, aggressive Gesten, Drohungen usw. (S1)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Einige Gruppenmitglieder versuchten, ihren Einfluss, ihren Status oder ihre Macht zu nutzen, um den anderen Gruppenmitgliedern Themen aufzuzwingen. (S2)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Sie fühlten sich aufgrund des Verhaltens anderer Meetingteilnehmer:innen davon abgehalten, sich an der Diskussion zu beteiligen. (S3)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Effizienz und Gesamteindruck

Wie viel Prozent der Meetingzeit wurden Ihrer Einschätzung nach mit ernsthafter Diskussion verbracht? (E4)
_____ %

In welchem Maße war dieses Meeting ergebnisorientiert? (E1)
Stark ergebnisorientiert ☐☐☐☐☐ Schwach ergebnisorientiert

Wie bewerten Sie Ihre allgemeine Zufriedenheit mit dem Meeting?
Sehr zufrieden ☐☐☐☐☐ Sehr unzufrieden

In welchem Maße wurde im Meeting ein Konsens erreicht?
Stark erreicht ☐☐☐☐☐ Schwach erreicht

Ursprüngliche Technologie-Items (Group Support Systems)

Wie wohl fühlen Sie sich im Umgang mit der Technologie?
Sehr wohl ☐☐☐☐☐ Sehr unwohl

In welchem Maße hat die Technologie Ihre Beteiligung an diesem Meeting behindert oder erleichtert?
Stark behindert ☐☐☐☐☐ Stark erleichtert

Zusätzliche Items zum proaktiven KI-Agenten

The following items were added for this thesis and are not part of the original Davison instrument.

Die Beiträge des KI-Agenten kamen zu passenden Zeitpunkten. (A1)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Die Beiträge des KI-Agenten haben den Gesprächsfluss gestört. (A2)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Der KI-Agent hat bezogen auf den aktuellen Stand der Diskussion passend reagiert. (A3)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Der KI-Agent hat zu häufig eingegriffen. (A4a)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Der KI-Agent hat zu selten eingegriffen. (A4b)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Es war in Ordnung, dass der KI-Agent von sich aus gesprochen hat, ohne gefragt zu werden. (A5)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Der KI-Agent hat dazu beigetragen, dass Informationen zur Sprache kamen, die sonst unerwähnt geblieben wären. (A6)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Die Hinweise des KI-Agenten waren so formuliert, dass niemand bloßgestellt wurde. (A7)
Stimme voll zu ☐☐☐☐☐ Stimme überhaupt nicht zu

Ergänzende Freitextfragen

The following open questions were appended in order to collect qualitative accounts of how participants experienced the agent's contributions, as discussed in Chapter 5.

Beschreiben Sie eine Situation, in der der KI-Agent eingegriffen hat. Wie haben Sie diesen Beitrag erlebt, und welchen Einfluss hatte er auf die Diskussion? (F1)

Was war an den Beiträgen des KI-Agenten hilfreich und was hat eher gestört oder gefehlt? (F2)

7.7 Codebook for the Open Questionnaire Items

This annex documents the codebook used to analyze the free-text answers to items F1 and F2 (section 7.6). F1 asks for a situation in which the agent intervened and for its effect on the discussion. F2 asks what was helpful, what was disruptive, and what was missing. Table 7.4 lists the codes.

Each coding has three levels: a main category, a subcategory, and a polarity (negative, positive, or neutral). Polarity is an attribute of the coding, not a code of its own. Anchor examples keep the original German wording, including the formulations as written. Names stay anonymized as `{Name von Gx-Px}`. One passage may anchor more than one code. D1 and D2 belong to context understanding. D3 is filed with the E codes under acceptance and group dynamics, because it describes the group's reaction rather than a failure of context tracking. W1 is a meta-code: every explicit suggestion receives W1 together with the topical code named in that row.

General coding rules

- **Multiple coding.** One segment can receive several codes. The marker "…" separates related sub-statements within an answer; each sub-statement is coded on its own. Every explicit improvement suggestion additionally receives W1. Valence is recorded as polarity, not as a separate code.
- **Explicitness.** Code only what is stated explicitly or by a clear implication. Do not interpret unclear statements. Passages flagged `[!sic]` in the material are not coded.
- **Duplicates and anonymization.** Duplicate quotes are coded as separate units. Names in curly brackets remain anonymized.
- **Attributes recorded with each coding.** Group (G1–G3), person (P1–P4), and intensity (strong, medium, or weak). Intensity is assigned only when the wording states it explicitly.

Table 7.4: Codebook for the free-text answers to questionnaire items F1 and F2. Anchor examples are quoted in the original German.

Code and subcategory	Definition	Anchor example
Intervention Behavior		
A1 Interruption of conversation flow *negative*	The AI interjects during a human contribution, interrupts a speaker, or its behavior is perceived as disrupting the conversation flow. *Coding rule:* Code if interruption or flow disruption is explicitly addressed; disruption without interruption is included; multiple coding with A3 is possible.	"Sie ist nach jedem Redebeitrag angesprungen… … und hat den Redefluss dann sehr stark eingeschränkt."
A1 Interruption of conversation flow *positive*	The AI interrupts in order to create space for others (helpful interruption). *Coding rule:* Code when the interruption serves a valued purpose, even if it is still experienced as disturbing.	"Positiv: In meinen Augen war hilfreich das sie mich unterbrochen hat um {Name von G3-P1} und {Name von G3-P4} mehr raum zu geben, das war gut, auch wenn es mich gestört hat :D"
A2 Intervention frequency too high *negative*	Statement that the AI intervenes or interrupts too often. *Coding rule:* Focus on frequency, not timing; "should only intervene selectively" is also coded here.	"Negativ: Die KI hat viel zu oft unterbrocheniner Meinung nach sollte die KI nur punktuell eingreifen wenn das Gespräch wirklich abdriftet oder wenn nur wenige Personen den ganzen Redeanteil haben."

Table 7.4: Codebook for the free-text answers to questionnaire items F1 and F2 (continued).

Code and subcategory	Definition	Anchor example
A2 Intervention frequency too low *negative*	Statement that the AI intervenes too rarely (for example, when nobody knows how to proceed). *Coding rule:* Only code if a lack of intervention at needed moments is explicit.	"Auf der anderen Seite habe ich auch zwei Situationen im Kopf, indem die Diskussion gestoppt hat wer nicht mehr genau wussten, wer jetzt etwas sagen soll, wie es weitergeht und an diesen Stellen hat sie nicht eingegriffen."
A3 Timing / delay *negative*	The AI reacts with a delay, too late, or lags behind the group's thinking process. *Coding rule:* Focus on delay, not interruption; combination with A1 is possible; slow speech tempo is also coded here, in addition to C2.	"Inhaltlich waren die Beiträge gut,… … allerdings verzögert, oder hinter unserem Denkprozess hinterher,…"
A4 Inappropriate moment / disregard of context *negative*	The AI intervenes at substantively inappropriate moments (for example, after consensus was reached) or re-addresses settled points. *Coding rule:* Distinguish from A3 (purely temporal): here the substantive fit of the moment matters; re-addressing settled content also receives D2.	"Man ist mitten in einem Diskurs, und wird von einem Einwand der KI unterbrochen, … … wo man bereits zu einem Konsens sich geeinigt hat."
Moderation Behavior		
B1 Summarizing the last contribution *negative*	The AI summarizes the previous contribution before its own statement; experienced as disrupting the meeting flow. *Coding rule:* For short meetings mostly negative; split the polarity if both evaluations appear.	"Negativ: Die KI hat nach dem Bedanken… … oft das zuletzt gesagte zusammengefasst, das hat den Meetingfluss negativ beeinflusst."
B1 Summarizing the last contribution *positive*	Summarizing is experienced as helpful, for example when earlier content is recalled in a long meeting. *Coding rule:* Code when usefulness is explicitly stated.	"… Das könnte hilfreich sein, wenn man längere Meetings hat und nochmal Inhalte vom Anfang zusammengefasst bekommt."
B2 Standardized phrases (thanks, formulas) *negative*	The AI repeatedly uses identical rhetorical patterns ("thank you xy"), experienced as unnecessary or time-consuming. *Coding rule:* Code only if repetition or standardization is explicitly addressed.	"Negativ: Die KI hat jedes Mal 'Danke xy' gesagt, was nach dem zweiten Mal nicht mehr nötig war."
B3 Including quiet participants / balancing speaking time *positive*	The AI explicitly addresses participants who have said little, in order to balance speaking time and hear all perspectives. *Coding rule:* Often positive; cases in which the goal fails (a person is asked although already speaking) are also coded here, but as negative; combine with A1 if the turn is felt as an interruption.	"Positiv: {Name von G1-P4} ist eher stiller bei der Diskussion gewesen und wurde durch die KI gezielt angesprochen, sodass auch seine Meinung in Ruhe angehört werden konnte."
B3 Including quiet participants / balancing speaking time *negative*	The inclusion attempt misfires (for example, a person is asked although already saying exactly that, or not all members are asked). *Coding rule:* Code when the mechanism fails or is experienced as disruptive despite the positive goal.	"Die KI wollte, dass {Name von G3-P4} ihren Beitrag sagt, obwohl sie schon angefangen hatte genau das zu sagen, was die KI gefragt hat."

Table 7.4: Codebook for the free-text answers to questionnaire items F1 and F2 (continued).

Code and subcategory	Definition	Anchor example
B4 Content-related impulses / steering *positive*	The AI provides helpful impulses, steers the direction of the conversation, or gives recognition. *Coding rule:* Distinguish from steering toward an own agenda; keep polarity separate per segment.	"Positiv: Hilfreich war das ab und an Impulse gesetzt wurden, um den Gesprächsfluss weiter in die richtige Richtung zu lenken und… … auch um Anerkennung zu zollen"
B4 Structuring / common thread *negative*	A lack of a guiding structure or framing for the meeting is criticized. *Coding rule:* Only code when the missing common thread is explicit.	"Negativ: Gefehlt hat ein besserer inhaltlicher Bezug mit sinnvollem Timing,.. … sowie eine rote Linie, die dem Meeting einen Ramen gibt."
B4 Own agenda / external steering *negative*	The AI pursues its own agenda or steers against the approach the group has agreed. *Coding rule:* Code when the AI acts against the group consensus or goal.	"Negativ: Störend war, dass die KI gefühlt versucht hat eine eigene Agenda zu verfolgen bzw. … … sich nicht genug dem in der Gruppe abgestimmten Vorgehen angepasst hat."
B5 Resolving misunderstandings / added value *positive*	The AI recognizes and resolves content-related misunderstandings or divergent states of information. *Coding rule:* Only explicit examples of a substantively helpful intervention.	"Positiv: Die Situation, wo die KI aufgelöst hat, dass wir unterschiedliche Informationen hatten, war wahrscheinlich das meist hilfreichste."
Delivery		
C1 Contribution length / precision *negative*	Statements about the length or precision of AI contributions (too long, talked too much, a shorter contribution was desired). *Coding rule:* Combine with A3 when speed is also criticized.	"Die Antworten könnten noch etwas kürzer, präziser und schneller sein, weil die KI hat schon sehr viel geredet."
C2 Speech/output characteristics (tempo, animation, sound) *negative*	Articulation and speech tempo are perceived as too slow or disruptive. *Coding rule:* Slow speech tempo that disrupts the flow also receives A1.	"Negativ: Das Sprachtempo der KI war zu langsam. Hat den Gesprächsfluss gestört."
C2 Speech/output characteristics (tempo, animation, sound) *positive*	Animation and sound at the onset of speech are experienced as appealing. *Coding rule:* Code when the audiovisual presentation is explicitly praised.	"Die Animation und der sound waren cool, wenn sie [die KI] angefangen hat zu sprechen."
C3 Aborted/restarted utterances *negative*	The AI aborts utterances mid-sentence and starts anew; multiple restarts. *Coding rule:* Only the AI's own behavior, not a human interrupting the AI.	"Außerdem hat die KI oft angefangen zu reden und mehrfach im Satz noch einmal erneut angefangen zu reden mit einem neuen Satz."
C4 Parallel speech / intelligibility *negative*	The AI keeps talking while humans continue speaking, so that the actual speaker is hard to understand. *Coding rule:* Combine with A1, since the flow is also disturbed.	"Und durch das penetrante weiterreden der KI,… … während andere Leute dann auch nicht aufgehört haben zu reden, war es super schwer die eigentliche Person zu verstehen."
Context Understanding		
D1 Confusion of persons/candidates *negative*	The AI confuses candidates or participants. *Coding rule:* Factual errors are also coded here; combination with D2 is possible.	"Oft hat sie die Kandidaten verwechselt und… … war nicht präzise auf dem aktuellen Stand der Diskussion, worüber wir gerade geredet haben."

Table 7.4: Codebook for the free-text answers to questionnaire items F1 and F2 (continued).

Code and subcategory	Definition	Anchor example
D1 Factually false statement *negative*	The AI makes factually wrong statements about the content. *Coding rule:* Code only concrete false statements about the case content.	"Negativ: Die KI hat einmal eine Falschaussage getroffen (Kandidat 3 trinkt war ihre aussage)"
D2 Discussion state not tracked / redundancy *negative*	The AI is not up to date; it asks again about things already said or processes contributions incorrectly. *Coding rule:* Distinguish from D1: here, failing to process content that was already stated is the focus.	"Negativ: Bestimmte Aspekte wurden mehrfach gefragt oder nicht richtig von der KI verstanden/verarbeitet."
D2 Turn-taking not recognized *negative*	The AI does not register whose turn it is or whether someone has finished. *Coding rule:* Code when a failure to track speakers is explicit.	"Negativ: Hat nicht gecheckt, wer dran ist und wann er fertig war…"
D2 Traceability missing (no written contribution) *negative*	No written chat contribution exists from which the AI's objection can be re-read. *Coding rule:* Code the missing traceability of AI statements.	"Gefehlt hat außerdem ein schriftlicher Beitrag im Chat, sodass man den Einwand nachlesen kann."
Acceptance & Group Dynamics		
D3 Group reaction: ignoring the AI *negative*	The group ignores the AI in reaction to its errors or disruptions. *Coding rule:* Code the group's coping strategy, not the triggering error itself.	"Dadurch kam es dazu, dass wir als menschliche Meeting Teilnehmer eher dazu genannt haben, die ki zu ignorieren, und nicht darauf zu hören, was sie sagt."
E1 AI as an additional "party" / disruptive factor *negative*	The AI is perceived as an independent party meddling in the social setting of the meeting. *Coding rule:* Social-perception statements about the AI's role, not single incidents.	"Ich denke der Einfluss in einem Teammeeting mit unterschiedlichen Parteien und da dann noch die KI ist, was das ganze störend wirken lässt, weil es wie noch eine Partei wirkt, die sich einmischt"
E2 Security and motivation through presence *positive*	The mere presence of the AI creates security or motivation to progress faster. *Coding rule:* Presence effect, not the AI's contributions themselves.	"Positiv: … die Anwesenheit von ihr jedoch etwas Sicherheit und Motivation gegeben hat, schnell voran zu schreiten."
E2 Acceleration through observation pressure *neutral*	The meeting accelerates because participants want to avoid the AI stepping in again. *Coding rule:* Code when the effect is attributed to awareness of the AI, not to its suggestions.	"Das Meeting hat sich schon sehr schnell und effizient angefüllt, wobei ich eher sagen würde das lag nicht an den KI Vorschlägen, sondern an dem Bewusstsein, dass wir immer weiter und Progress machen müssen, weil sonst sich die ki wieder einmischt und zum weitermachen anregt."
E3 Habituation over time *neutral*	Initially confusing or disruptive; over time, habituation and better collaboration. *Coding rule:* Code a temporal change in the experience of the AI.	"am Anfang war es erst sehr verwirrend und hat gestört. Man musste erst mal reinkommen und mit der KI als Meeting Partner klarkommen. Mit der Zeit wurde es dann besser und hat sich dran gewöhnt."
E4 Prototype status / maturity *negative*	General assessment of technical maturity: a commendable goal, but not yet useful. *Coding rule:* Overall verdicts on the system's readiness, not single incidents.	"Ich sehe was die KI erreichen wollte, lobenswertes Ziel alle zur Sprache kommen zu lassen und alle Informationen einzubeziehen, … … aber noch zu sehr im Prototypenstatus als dass es aktuell nützlich ist."

Table 7.4: Codebook for the free-text answers to questionnaire items F1 and F2 (continued).

Code and subcategory	Definition	Anchor example
Improvement Suggestion		
W1 Control / stop button *neutral*	Feature idea: a control to abort or stop the AI. *Coding rule:* Meta-code: every explicit suggestion receives W1 plus the matching topical subcode; here additionally A1.	"Eventuell würde ein 'stopp' button, der einfach die KI unterbricht helfen das zu verhindern"
W1 More subtle input / visual hints *neutral*	Feature idea: proactive hints shown only visually to the current speaker. *Coding rule:* Additionally A1 and B4.	"Ein subtilerer Ansatz mit hinweisen, die proaktiv aber vielleicht nur visuell als ein Art 'Hints' nur dem Sprecher angezeigt werden, wäre vielleicht mal interessant zu probieren."
W1 Written chat contribution *neutral*	Feature idea: written AI contributions for traceability. *Coding rule:* Additionally D2.	"Gefehlt hat außerdem ein schriftlicher Beitrag im Chat, sodass man den Einwand nachlesen kann."
W1 Introduction / framing at the start *neutral*	Feature idea: an AI introduction that sets the framing at the beginning of the meeting. *Coding rule:* Additionally B4.	"Und zuletzt hätte eine Introduction, der ki noch gefehlt vielleicht der ein bisschen das Framing der Diskussion am Anfang setzt und man wird nicht einfach blind reingeschmissen wird und muss plötzlich anfangen zu diskutieren und zu reden."
W1 Time / progress cues *neutral*	Feature idea: time or progress hints, especially toward the end of the meeting. *Coding rule:* Additionally A2 and B4.	"Es hat für mich ein Hinweis auf die Ablaufende Zeit als Moderator gefehlt, hier hatte ich erwartet dass sie gegen Ende mehr eingreift."
W1 Individual assistant instead of group AI *neutral*	Feature idea: a personal assistant for each participant instead of one group AI. *Coding rule:* Additionally E1.	"… ich denke, wenn da jeder seinen ganz persönlichen Assistenten hätte, wäre das viel schneller akzeptiert, weil es weniger aufbringt sondern als support auch wahrgenommen werden kann."
W1 Selective intervention *neutral*	Recommendation: the AI should intervene only selectively (the conversation drifts, or speaking time is unbalanced). *Coding rule:* Additionally A2.	"Meiner Meinung nach sollte die KI nur punktuell eingreifen wenn das Gespräch wirklich abdriftet oder wenn nur wenige Personen den ganzen Redeanteil haben."
W1 Content accuracy / up-to-dateness *neutral*	Recommendation: correct references to candidates and up-to-date tracking of the context. *Coding rule:* Additionally D1 and D2.	"Negativ: Gefehlt hat ein besserer inhaltlicher Bezug mit sinnvollem Timing,.."